



Compressor Simulator

Technical Report

Prepared by: Anas Alanqar
Prepared for: Maikana Automation
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This report opens with a Quick Start for automation/PLC engineers new to this simulator, then has three parts: Part I is the design basis for the process model (purpose, equations, constants, limitations, verification); Part II is the operating manual for the packaged desktop application and the PLC-facing interface (installation, PLC connection, development roadmap, tag interface, operator interface, running a session, fault injection, commissioning and acceptance testing, troubleshooting); Part III is reference appendices (signal/tag list, future improvements, verification detail).

Abstract

This report covers Compressor Simulator: a design basis for the process model it implements (Part I), and an operating manual for the packaged desktop application (Part II), followed by reference appendices (Part III).

This application is a from-scratch reimplementaion of an earlier Simulink/CODESYS hardware-in-the-loop model, built to the same design point and now packaged as a standalone desktop application. Notable differences between the two are called out where relevant.

The simulator represents a generic three-stage reciprocating compressor package built to a plausible design point, but not validated against real machine performance data. It contains no control logic of its own — sequencing, permissives, and alarm evaluation are entirely the responsibility of the PLC under test, connected over OPC UA. Part II documents installing the application, making that connection, operating the interface, running a session end to end, and injecting faults to exercise a PLC's fault handling.

Contents

Abstract	i
Contents	ii
List of Figures	v
1 Purpose and Scope	1
1.1 How to Use This Report	2
2 Reader Background and Common Ground	3
3 PLC Engineer Quick Start	4
3.1 What You Are Responsible For	4
3.2 System Boundary	5
3.3 Recommended Reading Path	5
3.4 Do Not Confuse	6
Part I — Design Basis	7
4 Process Description	8
4.1 What This Means for the PLC	8
4.2 Pressure as a Consequence of Mass Accumulation	9
4.3 Worked Example — From 60 psig Source to 1149 psig Discharge	9
4.3.1 The Process Path	10
4.3.2 Source Pressure Is Not Suction Pressure	10
4.3.3 Pipeline Pressure Is a Downstream Boundary, Not a Compressor Output	11
4.3.4 Compressor Mass Flow	11
4.3.5 Three Terms That Are Easy to Conflate	11
4.3.6 Why Suction Settles Near 30 psig	12
4.3.7 Valve Mass Flow, Briefly	12
4.3.8 The Key Valves, Together	13
4.3.9 Bypass / Recycle Behaviour	13
4.3.10 Why Discharge Settles Near 1149 psig	13
4.3.11 How 117 and 377 psig Are Obtained	14
4.3.12 Temperature, Briefly	14
4.3.13 Mental Model	14
5 Governing Equations	15
5.1 Symbols	15
5.2 1. Gas density — ideal gas law	15
5.3 2. Stage pressure distribution — equal ratio	15
5.4 3. Volumetric efficiency — clearance re-expansion	16
5.5 4. Compressor mass flow — positive displacement	16
5.6 5. Valve mass flow — simplified orifice	16
5.7 6. Vessel pressure dynamics — mass balance at constant volume	17
5.8 7. Discharge temperature — polytropic compression	18
5.9 8. Valve position dynamics — rate-limited actuator	18
5.10 9. First-order lag states	19
5.11 10. Lubrication pressure target	20
5.12 11. Cooling — fan-count lookup	20
5.13 12. Engine speed model	20
6 Symbols, Constants, and Design Point	22

6.1	Design Point	22
6.2	Constants — Configured Model Parameter Set	22
6.3	Transmitter Ranges — PLC Instrumentation Interface	24
7	Assumptions and Limitations	25
8	Verification Summary	27
Part II — Using the Simulator		28
9	System Requirements and Deployment Considerations	29
9.1	Server and Virtual-Machine Deployment	29
9.2	Multiple Instances	30
10	Installation	31
10.1	Extract the Installer	31
10.2	Run the Installer	33
10.3	First Launch	35
10.4	Where Things Live Once Installed	37
11	Connecting to a PLC over OPC UA	38
11.1	The OPC UA Endpoint	38
11.2	Opening the Connect to a PLC Dialog	39
11.3	This Computer	39
11.4	Another Device on the Network	40
11.5	Confirming the Link Is Up	41
11.6	What Happens on Link Loss	42
11.7	The CODESYS Side: Bringing the Runtime Online	43
11.8	Symbol Publishing and Device Security Settings	46
11.9	Verifying the Link: Forcing a Tag from the PLC Side	49
12	PLC Development Roadmap	52
12.1	Stage 1 — Establish Communications	52
12.2	Stage 2 — Create the I/O Mapping	52
12.3	Stage 3 — Implement Equipment-Level Logic	52
12.4	Stage 4 — Implement Permissives	52
12.5	Stage 5 — Implement the Sequence	53
12.6	Stage 6 — Add Alarm and Trip Evaluation	53
12.7	Stage 7 — Fault-Test the PLC	53
12.8	Stage 8 — Validate Stop, Shutdown, and Recovery Behaviour	53
12.9	Recommended PLC Software Architecture	53
13	PLC / Simulator Interface	55
13.1	Command / Feedback Cheat Sheet	55
13.1.1	Engine	55
13.1.2	Lubrication	55
13.1.3	Process Valves	55
13.1.4	Cooling	56
13.1.5	Process Measurements	56
13.1.6	Communication	56
13.2	Important: Command Polarity	56
13.3	What Should Happen When I Issue a Command?	57
13.3.1	Auxiliary Lube On (CMD_4001)	57
13.3.2	Engine Start (CMD_4005, subject to its prerequisite commands)	57

13.3.3	Close Bypass / Load the Compressor (FC_3002 toward 100%)	57
13.3.4	Open Bypass / Unload (FC_3002 toward 0%)	57
13.3.5	Blowdown (CMD_4004 de-energised, or its permissives never asserted)	57
13.4	Normal Operating Expectations (“What Good Looks Like”)	57
13.5	Normal Stop vs. Unconditional Shutdown (USD)	58
13.6	Watchdog and Communication Supervision	58
13.7	PLC Sequence / State Reference Table	58
14	The Operator Interface	61
14.1	Overrides — Status Feedback and Always-Live Inputs	61
14.2	Faults	63
14.3	Tags	68
14.4	Engineering Trends	69
15	Running the Simulator	71
16	Fault Injection	74
16.1	Where This Differs from the Predecessor Simulink Rig	76
17	PLC Validation and Commissioning Tests	77
17.1	First PLC Commissioning Test	77
17.2	Acceptance Test Checklist	77
17.2.1	Communications	77
17.2.2	Start Sequence	78
17.2.3	Running	78
17.2.4	Normal Stop	78
17.2.5	Shutdown	79
17.2.6	Fault Injection	79
18	Troubleshooting	80
18.1	First-Time PLC Engineer: Likely Early Mistakes	80
Part III	— Appendices	82
Appendix A	— Signal / Tag List	83
18.2	Commands In (PLC → Simulator)	83
18.3	Measurements Out (Simulator → PLC)	83
18.4	Tier 2 Discrete Faults (not in the PN17481 I/O list)	84
Appendix B	— Future Improvements	85
Appendix C	— Verification Detail	86

List of Figures

Figure 1	System boundary. The dashed border marks the simulation boundary: everything inside the Compressor Simulator block is a software model of the physical plant, not real equipment. The PLC / CODESYS block is the real control logic under test and owns sequence, permissives, interlocks, alarms/trips, and package control logic; the simulator owns process physics and engine/valve/instrument response. Commands flow down and measurements/status flow back up, both over the same OPC UA link.	5
Figure 2	Process flow of the three-stage reciprocating compressor package. Stage pressures correspond to the model design point.	8
Figure 3	The process path at the model’s converged design point, source boundary to pipeline boundary. Values are the implemented model’s steady-state result, not independent inputs.	10
Figure 4	The downloaded installer archive on the desktop, before extraction.	31
Figure 5	Right-click context menu on the zip file, with Extract All... selected.	32
Figure 6	The zip file and the resulting extracted folder shown together on the desktop.	32
Figure 7	Inside the extracted folder: the installer executable, CompressorSim-Setup-.exe.	33
Figure 8	SmartScreen’s initial warning. Click “More info” to reveal the run-anyway option.	33
Figure 9	After clicking “More info,” SmartScreen identifies the app and publisher and reveals the “Run anyway” button.	34
Figure 10	The installer wizard’s destination-folder step. Defaults are appropriate for almost all installs; click Next through the remaining pages to finish.	35
Figure 11	The desktop after installation completes, showing the new Compressor Simulator shortcut alongside the original installer files.	36
Figure 12	Compressor Simulator on first launch: the ARIEL JGH/4 P&ID with the compressor train stopped and blown down, and the Overrides tool dock open on the right.	37
Figure 13	Compressor Simulator with no PLC connected: the compressor train stopped and blown down, all suction/discharge ESD and blowdown valves in their fail-safe position, and the Overrides dock open and editable on the right.	38
Figure 14	The Connect to a PLC dialog in its default, not-connected state.	39
Figure 15	“This computer” selected as the PLC location.	39
Figure 16	Connected to a local CODESYS runtime. The dialog shows “Connected — this computer”; the header’s connection dot and endpoint reflect the same state.	40
Figure 17	Results of Scan my network: OPC UA servers discovered on the local subnet, each listed with its hostname and IP address.	40
Figure 18	A discovered server’s address populated into the connection field after selecting it from the scan results.	41
Figure 19	Connected to a remote CODESYS runtime at 172.20.10.2 — the dialog and header both reflect the non-localhost endpoint.	41
Figure 20	Overrides dock once connected: “OPC UA connected — the PLC is driving. These are read-only indicators,” with every tile reflecting the PLC’s commanded value (Speed command 75%, Bypass command 100%, etc.) instead of accepting input.	42
Figure 21	Launching the CODESYS Control Win V3 - x64 runtime from Windows search.	43
Figure 22	The CODESYS IDE’s Communication Settings tab, showing the gateway (Gateway-1, localhost:1217) and the active target device, DESKTOP-KC0FH83.	44

Figure 23	The Login (Alt+F8) toolbar button, which connects the IDE to the running application on the target device.	44
Figure 24	CODESYS Control Win V3 console log at startup: “OPC UA Server Started,” listening at <code>opc.tcp://DESKTOP-KC0FH83:4840</code>	45
Figure 25	CODESYS’s Device User Login prompt, requesting credentials with sufficient rights to view the Device object.	45
Figure 26	CODESYS IDE online: “Device [connected]” and “Application [run]” both shown in green in the Devices tree.	46
Figure 27	CODESYS Devices tree for a project with symbol publishing configured, and the Communication Settings tab showing the active target device.	47
Figure 28	Menu location for Device Security Settings, under the Device dropdown.	47
Figure 29	Device Security Settings dialog with the two required non-default values: <code>CommunicationMode = ALL</code> , <code>Activation = ACTIVATED</code> , under <code>CmpOPCUAServer</code>	48
Figure 30	Menu location for the Runtime Security Policy, under the Device dropdown.	48
Figure 31	Runtime Security Policy dialog: Optional encryption, Code Signing = All, Optional user management, Allow anonymous login enabled.	49
Figure 32	GVL_PLC in the CODESYS IDE with <code>CMD_4005</code> (“CAT engine start command”) selected, currently <code>FALSE</code>	50
Figure 33	Simulator HMI immediately before the force: <code>CMD_4005</code> is still <code>FALSE</code> on the PLC side, and the header shows the engine <code>STOPPED</code>	50
Figure 34	<code>CMD_4005</code> forced to <code>TRUE</code> in the CODESYS GVL_PLC watch view.	51
Figure 35	The simulator HMI shortly after the force: <code>CMD_4005</code> is now <code>TRUE</code> , sourced entirely from the PLC-side change — nothing was clicked in the simulator itself. The engine still reports <code>STOPPED</code> here, since the start command alone doesn’t spin the engine without the rest of the sequence (ESD healthy, driven-equipment ready, idle/rated-speed selection) also being asserted.	51
Figure 36	Compressor Simulator in dark theme, unit running under PLC control: all three stages <code>RUN</code> , coolers <code>ON</code> , discharge pressures building stage over stage (<code>ST1</code> 117 psig, <code>ST2</code> 364 psig, <code>ST3</code> 1071 psig), engine at 850 rpm.	61
Figure 37	Status feedback (read-only): cooler run feedback (<code>RS_4011</code> , <code>RS_4012</code>), engine jacket-water temperature (<code>TT_2014</code>), and engine oil pressure (<code>PT_1007</code>).	62
Figure 38	Operator / ECU inputs (always live): unit shutdown (<code>PB_5001</code>), local stop (<code>PB_5003</code>), remote stop (<code>PB_5004</code>), remote ESD (<code>ESD_5002</code>), CAT alarm (<code>XA_6002</code>), CAT fail SD (<code>XS_6003</code>).	63
Figure 39	Engine faults — low lube oil pressure, slow lube build, engine fails to start, mag pickup (speed signal) fault, and an overspeed bias slider — followed by the start of the Process group (blocked discharge, valve stuck).	64
Figure 40	Cylinder temperature bias per cylinder (1-4), cooler motor trip switches, and Tier 2 scrubber-level switches (suction, <code>ST2</code> , <code>ST3</code> , fuel-gas).	65
Figure 41	Further Tier 2 discrete faults: vibration trips (compressor frame, engine, skid/piping), oil/jacket-water low-level switches, fuel-gas pressure low, and cylinder lubricator no-flow (banks 1-2).	66
Figure 42	Signal freeze/invalid controls for every analog tag, a Link drop switch (suspends OPC UA writes to exercise the PLC’s own watchdog), and the start of the Instrumentation group (signal lag, signal noise).	67

Figure 43	Instrumentation faults (signal lag, signal noise) and the Boundary Conditions panel — source pressure, pipeline pressure, suction temperature, ambient temperature — for changing the process’s external conditions.	68
Figure 44	Tags tab: a flat, alphabetically-ordered list of tag names and live values (discrete switches as true/false, analogs with engineering units).	69
Figure 45	Engineering Trends with four pressure pens selected (suction, ST1 discharge, ST2 discharge, final discharge) over a 30-minute rolling window, ramping up during a startup.	70
Figure 46	The predecessor Simulink/CODESYS rig’s reference sequencer state diagram (READY – PURGE – BLOWDOWN – PRELUBE – CAT_START – WARMUP – LOADING – RUNNING, with COOLDOWN / NORMAL_STOP_POSTLUBE and SHUTDOWN / USD exit paths). Shown for context only – this application implements none of these states, timers, or transitions itself; every step below is something a connected PLC’s own sequencer (which may follow this same state machine, or a different one) must decide and command.	72

1 Purpose and Scope

WHAT THIS IS. This application simulates a **generic three-stage reciprocating gas compressor package** — a positive-displacement machine, not a centrifugal, screw, or other rotodynamic compressor, and not a performance-sizing tool. Section 2 and Section 3 establish, in a page or two, everything a first-time reader needs to know about that machine before going further.

BASIS AND VALIDATION STATUS. This report documents the current implementation and intended use of the Compressor Simulator. Unless explicitly stated otherwise, references to pressures, temperatures, flows, timing, and transient behaviour describe the behaviour produced by the implemented simulation model. Automated software tests (Section 8, Appendix C) verify selected model equations, configured design-point conditions, interface behaviour, and software invariants; they do not constitute validation against measured performance of a physical compressor. No vendor performance data, site acceptance data, or machine-specific performance test data were available for such validation during development. Project-specific PLC sequences, permissives, alarm/trip thresholds, timers, reset philosophy, shutdown actions, and safety requirements remain subject to the applicable project documentation and engineering review.

This simulator exists to exercise PLC control and sequencing logic against repeatable simulated process behaviour, to support operator and engineer familiarisation, and to allow fault conditions — loss of lube oil, cooler trips, blocked discharge, sensor failures — to be exercised within the simulator without requiring those conditions to be deliberately created on the physical compressor. Sequences that would otherwise require real thermal and pressure time constants to play out (minutes to tens of minutes) can be exercised over the same simulated duration, because the simulator executes in real time using the configured model time constants. These time constants are simulation parameters and should not be interpreted as validated measurements of the physical compressor unless explicitly supported by machine data.

This is not a performance-prediction tool, a rod-load or mechanical design tool, or a pulsation study, and it is not validated against real machine performance data — none was supplied during development, so it should not be used to guarantee machine performance.

The simulator represents a generic three-stage reciprocating compressor package built to a plausible design point rather than measured from a specific real machine. Several parameters — most importantly the clearance fraction (Section 5, eq. 4) — were back-solved to hit that design point in the absence of real performance data. This is the basis for the “generic, not validated” framing used throughout the rest of this document.

The application contains no control logic of its own: no timers, permissives, state machines, PID loops, or alarm evaluation. It produces a simulated process response based on the implemented process model and configured parameters — the pressures, temperatures, flows, and speeds that follow from the commands it receives; deciding what those values mean is the job of the PLC under test. This application, unlike its predecessor, does not bundle a reference PLC sequencer — Section 15 explains what that means in practice for operating it.

1.1 How to Use This Report

This document serves two different readers, and neither needs to read it cover to cover before starting work. Pick the path below that matches the job at hand; each links back to the other where the detail actually lives.

If you are writing the PLC program	If you are reviewing/verifying the simulator's process model
1. Section 1 — Purpose and Scope	1. Section 1 — Purpose and Scope
2. Section 2 — Reader Background / Common Ground	2. Section 4 — Process Description
3. Section 3 — PLC Engineer Quick Start	3. Section 5 — Governing Equations
4. Section 4 — Process Description	4. Section 6 — Constants and Design Point
5. Section 11 — OPC UA / CODESYS Connection	5. Section 7 — Assumptions and Limitations
6. Section 12 — PLC Development Roadmap	6. Section 8 — Verification Summary
7. Section 13 — PLC/Simulator Interface	7. Appendix C — Verification Detail
8. Appendix A — Signal / Tag List	
9. Section 15 — Operating Walkthrough	
10. Section 16 — Fault Injection	
11. Section 17 — PLC Validation / Commissioning Tests	
12. Section 18 — Troubleshooting	
13. (refer to Section 5 only when detailed process behaviour needs explaining)	

Either way, Section 5 through Section 8 (Part I) remain the design-basis reference — detailed, and deliberately not a prerequisite for starting PLC development.

2 Reader Background and Common Ground

This report assumes the reader is a technically competent automation/controls engineer: fluent in PLC programming, digital and analog I/O, permissives, interlocks, alarms, trips, state machines/sequencers, timers, industrial instrumentation, OPC UA, and general process-control terminology. None of that is re-taught here.

What this report does **not** assume is that the reader has worked on this specific compressor package before. This section is the minimum, compressor-specific common ground needed to interpret everything that follows — not a course in reciprocating-compressor engineering.

- This is a **three-stage reciprocating gas compressor** (Section 4) — a positive-displacement machine, not a curve-lookup rotodynamic one.
- Gas enters at **suction**, and pressure rises across **three compression stages** in series.
- **Intercooling** occurs between stages, and an **aftercooler** conditions the final discharge.
- Final compressed gas leaves toward the **process/pipeline**.
- The **bypass (recycle) valve** routes discharge gas back toward suction, unloading the compressor without stopping it.
- The **blowdown valve** depressurises the modelled **suction-side** volume — not the discharge side (Section 4 explains why that distinction matters).
- **ESD valves** (suction and discharge) isolate the compressor package.
- **Auxiliary lubrication / prelube** must build oil pressure before engine and compressor operation are meaningful.
- **Cooler** operation (fan count) directly affects simulated temperatures.
- An **engine** drives the compressor through a shared speed reference.
- **The PLC, not the simulator, decides how all of the above is sequenced** — permissives, timers, interlocks, and the startup/shutdown order are entirely the connected PLC's responsibility. Section 3 makes this division explicit.

3 PLC Engineer Quick Start

This section is the whole system in a few minutes: what the simulator does, what it does not do, and where the boundary between the two sits. Everything here is expanded, with the underlying physics and full procedures, later in the report — the cross-references point to where.

3.1 What You Are Responsible For

KEY MESSAGE. The simulator does not contain the PLC sequence you are supposed to develop. It has no startup/shutdown state machine, no permissives, no timers, and no alarm or trip evaluation of its own (Section 1). It produces the process response defined by the implemented simulator model for the commands it receives.

Simulator responsibilities	PLC responsibilities
• Process pressures (suction, stage, final discharge) • Compressor mass-flow response • Temperature response (cylinder, oil, aftercooler, jacket water) • Engine-speed response (crank / accelerate / decelerate / coastdown) • Valve-position dynamics (rate-limited, per-valve fail direction) • Lubrication-pressure response (prelube and running) • Simulated limit switches / status feedback • Transmitter values, including range clamping • Injected physical/instrument faults (Section 16) • OPC UA tag exchange with the PLC (Appendix A)	• Startup sequence • Normal stop sequence • Unconditional shutdown (USD) sequence • Permissives • Interlocks • Timers • State machine / sequencer • Engine-start logic • Prelube logic • Valve sequencing • Cooler sequencing • Loading / unloading logic • Alarm evaluation • Trip evaluation • Watchdog / link supervision • Reset philosophy • HMI/operator logic, where applicable

3.2 System Boundary

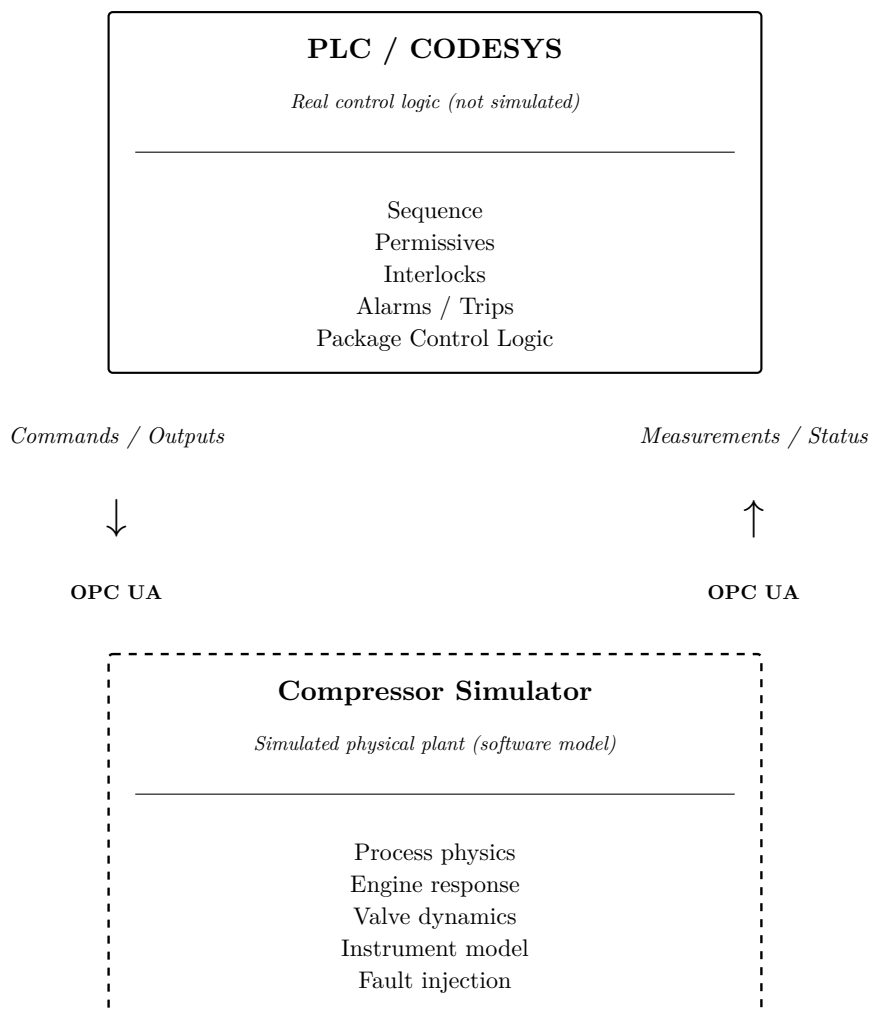


Figure 1. System boundary. The dashed border marks the simulation boundary; everything inside the Compressor Simulator block is a software model of the physical plant, not real equipment. The PLC / CODESYS block is the real control logic under test and owns sequence, permissives, interlocks, alarms/trips, and package control logic; the simulator owns process physics and engine/valve/instrument response. Commands flow down and measurements/status flow back up, both over the same OPC UA link.

Commands flow down (Appendix A, “Commands In”); measurements and status flow back up (Appendix A, “Measurements Out”). Both directions of this loop are carried over the same OPC UA link (Section 11).

3.3 Recommended Reading Path

See Section 1, “How to Use This Report,” for the full reading order for PLC development versus model review/verification. In short: read this Quick Start, Section 2, and Section 4, then go straight to Section 12 and start wiring up communications — the detailed governing equations (Section 5) are reference material, not a prerequisite.

3.4 Do Not Confuse

THE SIMULATOR IS NOT THE SEQUENCER. It supplies process physics only. The startup/shutdown state machine is entirely the connected PLC's responsibility (Section 15).

FAULTS ARE INJECTED LOCALLY. Simulator faults (Section 16) are never exposed on OPC UA as "a fault is active." The PLC sees the injected condition only through its **effect** on process/instrument tags; no separate simulator-fault indication is exposed through the normal PLC interface unless specifically documented.

SIMULATION VALUES ARE NOT VENDOR PERFORMANCE GUARANTEES. The design point (Section 6) is a plausible reference point for a generic package, not a measured or validated real-machine performance curve (Section 7).

THE PROCESS MODEL IS INTENTIONALLY SIMPLIFIED. Ideal gas, no choked flow, no cylinder thermal mass, no heat-exchanger model, no engine torque coupling (Section 7) — sufficient for PLC logic testing, not for performance, mechanical, or thermal design work.

A WORKING OPC UA LINK DOES NOT PROVE PLC LOGIC IS CORRECT. It confirms communication for the tested signal path (Section 11.9, Section 17). Sequence correctness is a separate validation step (Section 17).

A WORKING MANUAL SEQUENCE VIA OVERRIDES DOES NOT PROVE THE AUTOMATIC PLC STATE MACHINE IS CORRECT. Driving tags by hand from the Overrides dock (Section 14) only demonstrates the simulator's implemented response to that command — it says nothing about whether the PLC would have issued that command, in that order, on its own.

THE PREDECESSOR SEQUENCE IS A REFERENCE, NOT AN ENFORCEMENT MECHANISM. The state diagram in Section 15 is one valid control philosophy from an earlier rig, included for orientation. This simulator enforces none of its states or transitions.

Part I — Design Basis

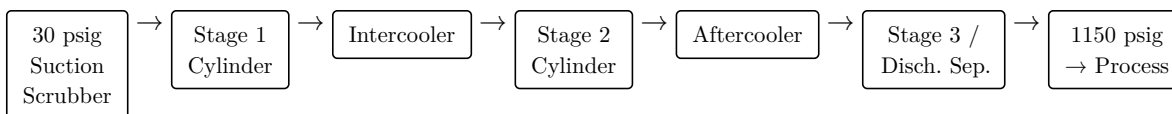
4 Process Description

A reciprocating compressor raises the pressure of a gas by drawing it into a cylinder through a suction valve, reducing the cylinder volume with a piston driven by a crankshaft, and expelling the compressed gas through a discharge valve — a positive-displacement machine, in contrast to a centrifugal machine whose flow varies continuously with pressure ratio. Gas enters through a suction scrubber, passes through the suction ESD and suction control valves into the first-stage cylinder, through an intercooler to the second-stage cylinder, through a second intercooler to the third-stage cylinder, and out through an aftercooler and the discharge ESD to the pipeline. A bypass (recycle) valve returns discharge gas to suction to unload the machine without stopping it; a blowdown valve vents the suction volume to atmosphere for depressurisation. The model lumps all discharge-side volume (second and third stage plus the discharge separator) into a single discharge volume — it does not model the discharge separator as a distinct dynamic element.

4.1 What This Means for the PLC

Being a **positive-displacement, staged, reciprocating** machine (rather than a rotodynamic one) has direct consequences for control logic:

- Compressor flow is fundamentally tied to **displacement and speed** (Section 5, eq. 5), not a pressure-ratio performance curve — there is no “compressor map” to operate against.
- **Bypass/recycle** is the mechanism for unloading and recycling flow without stopping the engine.
- **Staged compression** raises pressure progressively, stage by stage, not in one jump.
- **Intercooling** directly affects the temperature seen at each downstream stage.
- **Lubrication and prelube permissives** matter — the compressor should not be expected to run without healthy oil pressure first.
- **Engine and compressor sequencing is discrete/state-driven** (crank, accelerate, run, coast down) rather than continuous.
- **Pressure response is dynamic, not instant** (see below) — permissive checks need timeout logic, not instantaneous comparisons.
- **Blowdown/recycle path configuration matters during stopping and shutdown** — see the mass-accumulation asymmetry below, and Section 13 for the normal-stop-versus-USD consequence.



Bypass (recycle) valve returns discharge gas to suction, upstream of the suction scrubber; a blowdown valve vents the suction volume to atmosphere. Both omitted above for layout — see Figure 12 for the full P&ID.

Figure 2. Process flow of the three-stage reciprocating compressor package. Stage pressures correspond to the model design point.

Why the package is staged. As the pressure ratio across a single cylinder increases, discharge temperature rises, rod loading increases, and volumetric efficiency falls (an effect that worsens sharply with ratio — Section 5, eq. 4). Compressing this package’s full 30 → 1150 psig range in one stage would produce an impractically high discharge temperature and an impractically low volumetric efficiency. Multistage compression splits a large overall ratio across several cylinders in series with intercooling between them. For a fixed overall ratio split across k stages, distributing the ratio equally across all stages — each stage taking the k -th root of the total ratio — minimises total compression work, and

is standard multistage design practice; this is why this package's three stages each take the cube root of the total ratio rather than an arbitrary split.

4.2 Pressure as a Consequence of Mass Accumulation

This is the conceptual core of the whole model, stated here once for the rest of the document to cross-reference. Pressure is not an independent input that drives flow — it is the opposite: pressure is the integrated **result** of mass accumulating in, or draining from, a fixed volume. Gas flows in and out of a vessel for other reasons (valve positions, upstream/downstream pressure differences), and whatever net mass imbalance results is what raises or lowers the pressure. This follows from the ideal gas law: at fixed volume and temperature, $P = mR_{sp}T/V$, so pressure is directly proportional to contained mass, and $(dP)/(dt) = (R_{sp}T/V) \sum \dot{m}$ — the rate of pressure change is proportional to net mass flow rate (this is exactly Equation 8 in Section 5). If inflow exceeds outflow, pressure rises; if outflow exceeds inflow, pressure falls; if they balance, pressure holds steady regardless of how large either flow is.

Operationally, this is why blowdown and shutdown take real time rather than happening instantly, and why a PLC's pressure-permissive checks (minimum purge pressure, maximum start pressure) need timeout timers rather than instantaneous checks. It is also why the model exhibits a specific, repeatable asymmetry: **blowdown vents the suction volume, not the discharge volume**. The discharge side can only lose mass through the bypass valve into suction and then out the vent, so venting with the bypass closed collapses suction pressure toward atmospheric while discharge pressure barely moves — there is no path for discharge-side mass to leave. This single mechanism explains why an unconditional shutdown (blowdown opens immediately) and a normal stop (bypass opens first) leave the package in very different pressure states, and it applies identically wherever this document discusses blowdown, USD, or bypass behaviour below (in particular Section 15) — it is not re-derived each time it recurs.

4.3 Worked Example — From 60 psig Source to 1149 psig Discharge

The previous section stated the principle — pressure follows mass. This section walks that principle through the model's own design point (Section 6), end to end, so that Section 5 reads as the formalisation of something already understood rather than as the first encounter with it. Everything below describes **how the implemented simulator model behaves**, not a claim about what a real compressor package would do — see Section 1 for the validation status that applies to every number here.

WORKED-EXAMPLE OPERATING STATE. 1000 rpm; bypass closed (75% AO); suction control 45%; suction/discharge ESDs open; blowdown closed; both coolers running — the full driving command state is in Section 6.

4.3.1 The Process Path

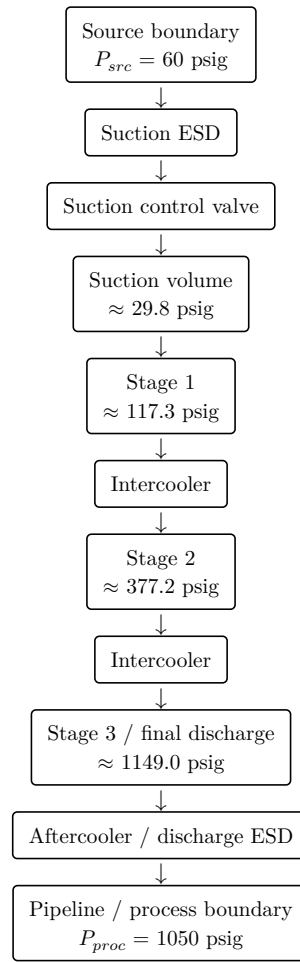


Figure 3. The process path at the model’s converged design point, source boundary to pipeline boundary. Values are the implemented model’s steady-state result, not independent inputs.

Two of these boxes are **boundary conditions** — fixed inputs to the model, defined in the simulator configuration and not themselves computed: the 60 psig source and the 1050 psig pipeline. Every other value on the path — suction, all three stage pressures, and final discharge — is a **computed result** of the mass-balance and compression equations in Section 5, evaluated at the design point’s commanded valve positions and speed (Section 6). The rest of this section explains how each of those results is arrived at.

4.3.2 Source Pressure Is Not Suction Pressure

$P_{src} = 60$ psig is the upstream boundary condition representing the gas supply available before the simulated compressor package — it is a configured input, not a measured tag. It does **not** directly set PT_1001 (suction pressure) to 60 psig, and reading it as “suction pressure” is the single most common misreading of this design point on first encounter.

Gas flows from the source toward the suction volume only when $P_{src} > P_s$ and the suction ESD and suction control valve provide an open path — the same simplified orifice relationship as every other flow in the model (Equation 7; see Section 4.3.7 below), driven by the source-to-suction pressure difference, the source gas density ρ_{src} , the suction control valve position, and the suction ESD position. At the design point that flow settles to an equilibrium far below 60 psig, for the reasons given below

(Section 4.3, “Why Suction Settles Near 30 psig”) — the 60 psig figure only ever appears as the driving pressure on the supply side of that calculation.

4.3.3 Pipeline Pressure Is a Downstream Boundary, Not a Compressor Output

The pipeline pressure is the downstream boundary pressure already present in the receiving process. In the model it represents the backpressure into which the compressor delivers gas; it is not generated by the compressor itself and is not a state the model integrates — like the source pressure, it is a fixed configured input ($P_{proc} = 1050$ psig).

For gas to flow forward into the pipeline at all, the simplified process-flow equation (Equation 7, applied to $\dot{m}_{proc}$) needs $P_d > P_{proc}$: discharge must sit above the pipeline boundary. At the design point, final discharge converges to approximately 1149 psig against a 1050 psig pipeline boundary — a driving difference of roughly $1149 - 1050 \approx 99$ psi. That 99 psi is not a value anything requires; it is what the implemented mass balance and the configured process-flow coefficient K_{proc} (Section 6) converge to once compressor mass flow and process off-take reach equilibrium (see “Why Discharge Settles Near 1149 psig” below). A different K_{proc} or a different pipeline boundary would converge to a different driving difference — nothing in the model treats 99 psi as a target.

4.3.4 Compressor Mass Flow

Mass flow rate answers a simple question: how many kilograms of gas pass through the compressor per second? At the design point, $\dot{m}_{comp} \approx 0.945$ kg/s. The implemented compressor equation (Equation 5) is:

$$\dot{m}_{comp} = V_{disp} \cdot \rho_s \cdot VE \cdot \frac{N}{60} \cdot \text{gate}$$

- V_{disp} — the configured swept displacement per revolution (stage 1, double-acting).
- ρ_s — suction gas density, computed from suction pressure and temperature (Equation 1).
- VE — volumetric efficiency (defined below).
- N — compressor/engine speed.
- **gate** — compression is enabled only when the implemented running conditions are satisfied: speed above 200 rpm and the suction ESD open beyond 2% (Equation 6).

The cause-and-effect reading: higher speed moves more mass per second; higher suction density means more mass is packed into the same swept volume; higher VE means more of that swept volume is actually fresh gas rather than re-expanded clearance gas; and a closed suction ESD or insufficient speed drives compressor mass flow to exactly zero in the implemented model — there is no partial-flow fallback. No compressor performance curve is used; flow is a direct product of displacement, density, efficiency, and speed.

4.3.5 Three Terms That Are Easy to Conflate

MASS FLOW vs. VOLUMETRIC FLOW vs. VOLUMETRIC EFFICIENCY.

Term	Units	Meaning
Mass flow rate	kg/s	Amount of gas mass passing a point per unit time. This is what Equation 5 and Equation 7 compute directly.
Volumetric flow rate	m ³ /s, m ³ /h, ...	Physical gas volume passing per unit time: $\dot{V} = \dot{m}/\rho$. Because gas density changes

Term	Units	Meaning
Volumetric efficiency (VE)	dimensionless / %	strongly with pressure and temperature, the same mass flow occupies a much smaller actual volume at discharge conditions than at suction conditions. The fraction of theoretical swept cylinder volume effectively filled with fresh suction gas after clearance re-expansion and modelled leakage (Equation 4). At the design point, $VE \approx 0.854$ (85.4%). It is a ratio, not a flow — it scales Equation 5, it does not report one.

Standard-condition volumetric flow (e.g. MMSCFD) is deliberately not reported anywhere in this document: doing so rigorously would require a defined standard condition and gas composition that were not supplied for this project (Section 7), and an invented figure would misrepresent the model as more complete than it is.

4.3.6 Why Suction Settles Near 30 psig

This is the most important individual result in this walkthrough, because it is the one first-time readers most often expect to just equal the source pressure. It does not, and the reason is the same mass-balance principle from the previous section, applied to the suction volume specifically (Equation 9):

$$\frac{dP_s}{dt} \propto \dot{m}_{sup} + \dot{m}_{byp} - \dot{m}_{comp} - \dot{m}_{bdv}$$

In plain terms: gas **enters** the suction volume from the source supply flow ($\dot{m}_{sup}$) and from bypass/recycle flow returning from discharge ($\dot{m}_{byp}$); gas **leaves** the suction volume as compressor flow ($\dot{m}_{comp}$) and blowdown flow ($\dot{m}_{bdv}$, zero when the blowdown valve is closed). If inflow exceeds outflow, suction pressure rises; if outflow exceeds inflow, it falls; when the two balance, it holds steady — regardless of how large either flow is individually.

At the design point's configured combination — 60 psig source, the design-point suction valve and ESD positions, the resulting compressor draw at 1000 rpm, and the model's configured flow coefficients (K_{suc} , Section 6) — that balance settles at an equilibrium suction pressure of approximately 29.8 psig. It is important not to describe this the other way around: **the simulator does not set suction pressure to 29.8 psig**. Nothing in the model writes that number anywhere. It is the steady-state output that this particular combination of inflows and outflows converges to; changing the suction valve position, the source pressure, or the compressor draw would converge to a different number through the same mechanism.

4.3.7 Valve Mass Flow, Briefly

Every flow referenced above — supply, bypass, process, blowdown — uses the same simplified orifice relationship (Equation 7):

$$\dot{m} = K \cdot \frac{Z}{100} \cdot \sqrt{\rho \cdot \max(\Delta P, 0)}$$

More valve opening means more available flow area and therefore more flow, all else equal. Greater upstream/downstream pressure difference means greater calculated forward flow. A fully closed valve gives zero flow regardless of pressure difference. A zero or reversed pressure difference gives zero flow too — the $\max(\Delta P, 0)$ term is what prevents the model from ever computing a reverse flow through these valves. This is explicitly the simulator’s simplified orifice implementation, not a detailed compressible-flow valve-sizing model (Section 7).

4.3.8 The Key Valves, Together

Valve	Physical/package role	Simulator role
Suction ESD	Isolation of the compressor inlet	Enables or blocks the inlet/compression path — closing it drives compressor mass flow to zero via the gate condition in Equation 5
Suction control valve	Controls/restricts gas entering the suction side	Modulates source-to-suction supply flow, $\dot{m}_{sup}$, in Equation 7
Bypass / recycle valve	Routes discharge gas back toward suction for unloading/recycle	Transfers mass directly from the discharge volume to the suction volume — $\dot{m}_{byp}$ appears with opposite signs in Equation 9 and Equation 10
Discharge ESD	Isolation between the compressor package and the downstream process	Its commanded position (Equation 13) is the Z term in the process-flow orifice equation (Equation 7) for $\dot{m}_{proc}$ — a closed discharge ESD directly zeroes process flow, not just an isolation state layered on top of it
Blowdown valve	Depressurisation path	Removes gas mass from the modelled suction volume to atmosphere — not an independently modelled discharge inventory. See the mass-accumulation asymmetry discussed above

4.3.9 Bypass / Recycle Behaviour

Bypass closed, compressor loaded: more of the compressor’s flow remains on the discharge side. Discharge pressure tends to rise, and process delivery increases according to the implemented model.

Bypass open, compressor unloaded: part of the discharge gas is transferred back toward suction. This subtracts mass from the discharge balance and adds it to the suction balance in the same step (Equation 10, Equation 9) — it tends to raise suction pressure, tends to reduce discharge pressure, reduces net process delivery, and lets the engine/compressor keep running without stopping.

The command polarity is unchanged from Section 13: a higher FC_3002 command corresponds to the bypass valve being more closed, and FC_3002 = 0% corresponds to the bypass fully open in the implemented command mapping.

4.3.10 Why Discharge Settles Near 1149 psig

The discharge-side mass balance (Equation 10) is:

$$\frac{dP_d}{dt} \propto \dot{m}_{comp} - \dot{m}_{proc} - \dot{m}_{byp}$$

The compressor adds mass to the discharge volume; process/pipeline off-take and bypass both remove it. When compressor inflow exceeds the combined outflow, discharge pressure rises; when outflow exceeds inflow, it falls; when they balance, discharge pressure stabilises. At the design point, compressor mass flow of approximately 0.945 kg/s and the configured pipeline boundary of 1050 psig converge to a final discharge pressure of approximately 1149 psig — the pressure at which $\dot{m}_{proc}$ (driven by the 99 psi difference above) removes mass from the discharge volume at the same rate the compressor is adding it. 1149 psig is not required by any real compressor; it is this configured model's equilibrium for this combination of inputs.

4.3.11 How 117 and 377 psig Are Obtained

It matters which of these numbers are **dynamically simulated** and which are **algebraically calculated** from other simulated values. The model dynamically integrates exactly two pressure inventories — suction and discharge (Equation 9, Equation 10) — via the mass-balance principle throughout this section. It does **not** separately integrate three interstage vessel mass balances.

The two intermediate stage pressures are instead derived algebraically from the current suction and discharge pressures, using the equal-stage-ratio assumption (Equation 2, Equation 3):

$$r_{tot} = \frac{P_d}{P_s}, \quad r_{stg} = r_{tot}^{\frac{1}{3}}$$

At the design point this gives the path already shown in Figure 3: 29.8 → 117.3 → 377.2 → 1149.0 psig, each stage taking the cube root of the total ratio. Stage 1 and Stage 2 pressures are therefore **derived model outputs** of the equal-ratio assumption applied to the two integrated states, not independently validated interstage pressure predictions.

4.3.12 Temperature, Briefly

Gas temperature rises through compression according to the implemented polytropic relationship (Equation 11, Section 5 item 7): a higher compression ratio produces a higher calculated discharge temperature. Intercooling reduces the inlet temperature seen by each downstream stage. Cooler behaviour itself is simplified to configured fan-count temperature targets filtered by a first-order lag, not a detailed heat-exchanger model — see Section 5 items 7 and 11 for the full treatment.

4.3.13 Mental Model

SUMMARY. The gas side of the simulator reduces to two dynamic mass inventories:

Suction: source supply + recycle – compressor draw – blowdown

Discharge: compressor draw – process delivery – recycle

Positive net accumulation raises pressure; negative net accumulation lowers it; near-zero net accumulation holds it steady. Every other equation in Section 5 exists to compute one of the terms feeding these two balances — gas density, compressor capacity, stage-pressure distribution, temperature, and valve position — or to convert the resulting suction/discharge pressures into the tags the PLC reads.

5 Governing Equations

This section states the equations implemented in the simulator process model, not what a textbook first-principles treatment would ideally use. Every simplification made relative to a rigorous treatment is stated explicitly, with its justification, in Section 7. All pressures below are absolute Pa unless noted gauge; all temperatures are Kelvin; all mass flows are kg/s, with no exceptions, throughout the internal process model (the instrumentation layer converts to psig/°F only at the boundary that crosses to the PLC).

5.1 Symbols

Symbol	Meaning	Value / units
R_{sp}	Specific gas constant	345.5 J/(kg · K)
n	Polytropic exponent	1.2693
e_T	$(n - 1)/n$	0.2122
V_{disp}	Swept volume per revolution (stage 1, double-acting)	0.023258 m ³ /rev
C	Clearance fraction	0.078 (fitted — see Section 6)
L_{slip}	Slip / leakage loss fraction	0.04
V_s, V_d	Suction, discharge vessel volumes	3.0, 4.5 m ³
N	Engine speed	rpm
Z	Valve position	0–100 %
P_{atm}	Atmospheric pressure	101325 Pa

5.2 1. Gas density — ideal gas law

$$\rho = \frac{P}{R_{sp}T} \quad (1)$$

What this means: converts the current pressure and temperature into gas density, which feeds directly into the compressor- and valve-flow calculations below (Equation 5, Equation 7).

Applied at three points: suction (ρ_s), discharge (ρ_d , using the aftercooler temperature T_{ac}), and the supply source (ρ_{src}).

Simplification: compressibility factor $Z = 1$. For illustrative comparison only, representative natural-gas compressibility factors at similar conditions (roughly 1150 psig, 245 °F) are commonly cited in the range $Z \approx 0.85$ – 0.90 ; on that general basis, densities computed by Equation 1 would be optimistic by roughly 10–15% at the discharge end relative to a real-gas treatment. No project-specific gas composition analysis was supplied to compute an actual Z -factor for this model, so this comparison is general engineering context, not project data. This simplification is acceptable because the model's purpose is PLC logic testing, not a performance guarantee.

5.3 2. Stage pressure distribution — equal ratio

$$r_{tot} = \frac{P_d}{P_s}, \quad r_{stg} = r_{tot}^{\frac{1}{3}} \quad (2)$$

$$P_1 = \min(P_s \cdot r_{stg}, P_d), \quad P_2 = \min(P_1 \cdot r_{stg}, P_d) \quad (3)$$

What this means: uses the two dynamically simulated pressures — suction and final discharge — to algebraically calculate the two intermediate stage pressures under the configured equal-ratio assumption; the interstage values are derived outputs, not independently integrated states (see Section 4.3).

At the design point: $r_{tot} = 26.1$, $r_{stg} = 2.969$, giving $30 \rightarrow 117 \rightarrow 377 \rightarrow 1149$ psig.

Guard not present in the reference model. The $\min(\cdot, P_d)$ clamp on P_1 and P_2 is a deviation this application makes from both its own internal lightweight reference model and the predecessor Simulink model, neither of which clamps the interstage pressures. It exists only to prevent interstage readings from briefly exceeding final discharge during startup or blowdown transients, and has no effect at steady state. r_{tot} is separately clamped to $[1, 30]$.

5.4 3. Volumetric efficiency — clearance re-expansion

$$VE = 1 - C \left(r_{stg}^{\frac{1}{n}} - 1 \right) - L_{slip} \quad (4)$$

What this means: reduces the theoretical swept cylinder displacement to the fraction actually filled with fresh suction gas, after accounting for clearance re-expansion and modelled leakage.

The classical reciprocating-compressor volumetric-efficiency equation: gas trapped in the clearance volume at discharge pressure must re-expand back to suction pressure before the suction valve can open, so the cylinder never fills completely, and a higher ratio means more re-expansion and less fresh gas drawn in. At the design point, $r_{stg}^{\frac{1}{n}} = 2.969^{\frac{1}{1.2693}} = 2.357$, so $VE = 1 - 0.078(1.357) - 0.04 = 0.854$. Clamped to $[0, 1]$.

$C = 0.078$ was back-solved to close the design point, because no vendor performance data was available to source it from. This is the model's single biggest fitted parameter, and it is precisely why this package is framed throughout this document as a *generic* three-stage reciprocating compressor, not a validated reproduction of a specific real machine's performance.

5.5 4. Compressor mass flow — positive displacement

$$\dot{m}_{comp} = V_{disp} \cdot \rho_s \cdot VE \cdot \frac{N}{60} \cdot \text{gate} \quad (5)$$

$$\text{gate} = \begin{cases} 1 & \text{if } N > 200 \text{ rpm and } Z_{sesd} > 2\% \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

What this means: calculates how much gas mass the running compressor transfers from suction toward discharge, per second.

Swept volume per revolution $\times$ suction density $\times$ volumetric efficiency $\times$ revolutions per second, giving kg/s.

The gate is what makes the compressor a *positive-displacement* machine rather than a curve lookup: flow is proportional to speed and independent of pressure ratio except through VE . The Z_{sesd} term means a closed suction ESD dead-heads the cylinder — no fresh gas, therefore no flow and no compression heat.

5.6 5. Valve mass flow — simplified orifice

$$\dot{m} = K \cdot \frac{Z}{100} \cdot \sqrt{\rho \cdot \max(\Delta P, 0)} \quad (7)$$

What this means: calculates flow through each simplified restriction (supply, bypass, process, blowdown) from that valve's opening, the relevant gas density, and the pressure difference across it.

Derived from the incompressible orifice equation $\dot{m} = C_d A \sqrt{2\rho\Delta P}$, with the discharge coefficient, the flow area, and the factor of 2 all folded into a single lumped constant K per valve. A linear valve characteristic is assumed (position % maps linearly to flow area). $\max(\Delta P, 0)$ prevents both reverse flow and a square root of a negative number.

Flow	Driving ΔP	Density	K
$\dot{m}_{sup}$ (supply into suction)	$P_{src} - P_s$	ρ_{src}	2.10×10^{-3}
$\dot{m}_{byp}$ (discharge $\rightarrow$ suction, recycle)	$P_d - P_s$	ρ_d	1.50×10^{-3}
$\dot{m}_{proc}$ (discharge $\rightarrow$ pipeline)	$P_d - P_{proc}$	ρ_d	1.335×10^{-4}
$\dot{m}_{bdv}$ (suction $\rightarrow$ atmosphere, vent)	$P_s - P_{atm}$	ρ_s	4.00×10^{-3}

$\dot{m}_{sup}$ is additionally gated by both valves in series: $(Z_{suc}/100) \cdot (Z_{sesd}/100)$.

Simplification: no choked-flow model. At the blowdown valve venting 1150 psig to atmosphere, the real flow would choke (reach sonic velocity at the throat, capping the mass flow rate below what the subsonic orifice equation above predicts). Vent rates during blowdown and unconditional shutdown are therefore approximate — likely faster in the model than a real choked vent, though this has not been quantified.

5.7 6. Vessel pressure dynamics — mass balance at constant volume

This is the most important equation pair in the model. **What this means:** converts net gas accumulation (inflow minus outflow) in the suction and discharge volumes into changing suction and discharge pressure — this is the mechanism worked through in full in Section 4.3. From the ideal gas law at fixed volume and temperature, differentiating gives:

$$\frac{dP}{dt} = \frac{R_{sp}T}{V} \sum \dot{m} \quad (8)$$

Suction volume:

$$\frac{dP_s}{dt} = \frac{R_{sp}T_{suc}}{V_s} (\dot{m}_{sup} + \dot{m}_{byp} - \dot{m}_{comp} - \dot{m}_{bdv}) \quad (9)$$

with $R_{sp}T_{suc}/V_s = 345.5 \times 310.93/3.0 = 35,809$ — the model computes this implicitly by evaluating the expression each step rather than as a pre-multiplied gain constant.

Discharge volume:

$$\frac{dP_d}{dt} = \frac{R_{sp}T_{ac}}{V_d} (\dot{m}_{comp} - \dot{m}_{proc} - \dot{m}_{byp}) \quad (10)$$

with $R_{sp}/V_d = 345.5/4.5 = 76.78$, matching the predecessor Simulink model's `Gain(76.78)` block exactly.

Note $\dot{m}_{byp}$ (the recycle flow) appears in **both** equations with opposite signs — it is a direct transfer of mass from the discharge volume to the suction volume, not a sink-and-independent-source pair. Both derivatives are clamped to not drive pressure below atmospheric when already at atmospheric.

Note: blowdown vents the suction volume, not the discharge volume. The discharge side can only lose mass through the bypass valve into suction and then out the vent. This is a direct, observable consequence of “pressure follows mass” (Section 5, and the process description in the previous chapter), and it is the mechanism behind the USD-versus-normal-stop asymmetry noted below.

5.8 7. Discharge temperature — polytropic compression

$$T_d = T_{in} \cdot r_{stg}^{e_T}, \quad e_T = \frac{n-1}{n} = \frac{0.2693}{1.2693} = 0.2122 \quad (11)$$

What this means: calculates the compression discharge-temperature target from the stage pressure ratio and inlet temperature.

Applied per cylinder with different inlet temperatures — the model tracks two discharge-temperature lag states, covering the first stage and lumping the second and third stages together:

State	Inlet source	Why	PLC tags fed
T_{d1}	T_{suc}	draws from the suction scrubber	TT_2004 (Cyl 1)
T_{d2}	T_{inter}	draws from the intercooler outlet	TT_2005, 2006, 2007 (Cyls 2–4)

$$T_d = \text{gate} \cdot (T_{in} r_{stg}^{e_T}) + (1 - \text{gate}) \cdot T_{amb} \quad (12)$$

When not running, both targets relax to ambient — this is why cylinder 2 (fed by the intercooler) is the one that visibly trips on cooler loss, while cylinder 1 barely moves, since only T_{inter} depends on fan count.

Simplification: these are algebraic targets, not thermal states. No cylinder metal mass or thermal inertia is modelled directly at this equation — discharge temperature responds to a ratio change algebraically. Thermal lag is added downstream (Equation 14, below) as a first-order filter on this algebraic target, not as a physically-derived thermal-mass model; a real cylinder’s discharge temperature lags a ratio change by tens of seconds due to actual metal and gas thermal mass, which this two-stage abstraction (algebraic target $\rightarrow$ generic lag) approximates but does not derive from first principles.

Cylinders 3 and 4 read identically to cylinder 2. Because the model has only two discharge-temperature lag states covering three compression stages, TT_2006 and TT_2007 would read bit-for-bit identical to TT_2005 without an added static per-cylinder calibration offset (a configurable value, currently 0, 0, +3, −2 °F for cylinders 1–4) applied at the instrumentation boundary — no two real transmitters on nominally identical cylinders agree that closely. This offset is cosmetic realism, not a modelled physical difference between the cylinders.

5.9 8. Valve position dynamics — rate-limited actuator

$$\frac{dZ}{dt} = \text{sat}(K_{valve}(Z_{target} - Z), -R_{close}, +R_{open}), \quad 0 \leq Z \leq 100 \quad (13)$$

What this means: prevents commanded valves from changing position instantaneously, applying the configured opening/closing rates below instead.

$K_{valve} = 20$ (a proportional gain on position error, matching the predecessor Simulink model’s gain(20) block) feeding a rate limiter: for any error larger than R/K_{valve} , the valve moves at the constant rate

limit — a true constant-velocity slew — while for small errors near the target, the gain term falls below the rate limit and the approach is smooth rather than a hard stop exactly at the target.

Valve	R_{open} (%/s)	R_{close} (%/s)	Initial condition	Target	Fail direction
Bypass	5	15	100 (open)	$100(1 - \text{ao_byp}/75)$	fails open
Suction control	5	5	0 (closed)	ao_suc	fails closed
Suction ESD	20	20	0 (closed)	$100 \cdot \text{CMD_sesd}$	fails closed
Discharge ESD	20	20	0 (closed)	$100 \cdot \text{CMD_desd}$	fails closed
Blowdown	50	50	100 (open unless reset “pressurised”)	$100(1 - \text{CMD_bdv})$	fails open

The simulator implements the following configured fail directions: the suction and discharge ESD valves close and the blowdown valve opens when their commands are de-energised. These configured directions determine the simulated pressure response during shutdown and link-loss conditions (Section 11, “What Happens on Link Loss”) — this is why an unconditional shutdown and a normal stop end in different simulated pressure states. Where these fail directions are also the approved project safety design is a matter for the applicable project safety documentation, not something this report establishes.

5.10 9. First-order lag states

$$\frac{dx}{dt} = \frac{x_{target} - x}{\tau} \quad (14)$$

Used for quantities that the simulator represents using a first-order lag to approximate delayed thermal or hydraulic response; the time constants below are configured simulation parameters, not measurements of physical thermal or hydraulic inertia (Section 7):

State	Target	τ (s)	Initial condition
P_{oil} (compressor lube oil)	see Equation 15	3	0 psig gauge
T_{eoil} (engine oil temperature)	190 °F running, else ambient	400	ambient
T_{ac} (aftercooler outlet)	fan-count lookup if running, else ambient	60	ambient
T_{oil} (compressor oil temperature)	fan-count lookup if running, else ambient	300	ambient

5.11 10. Lubrication pressure target

$$P_{oil,target} = \begin{cases} P_{oil,run} \cdot \min(1, \frac{N}{850}) & N > 200 \text{ rpm} \\ P_{oil,prelube} & \text{prelube pump commanded} \\ 0 & \text{otherwise} \end{cases} \quad (15)$$

120 psig at rated speed ($P_{oil,run}$), 55 psig on the prelube pump ($P_{oil,prelube}$), 0 psig (relative) when off. Pressure scales linearly with speed up to 850 rpm, then saturates — a pressure-relief-valve characteristic.

Consequence: a modelled transient dip during startup crossover. Once N exceeds 200 rpm, the running branch takes over at $8.27 \times 10^5 \text{ Pa} \cdot N/850$, which is momentarily **below** the 55 psig prelube target until speed builds. The implemented model exhibits this temporary dip during the transition from the prelube branch to the running-pressure branch; actual machine behaviour depends on the physical lubrication system and has not been validated here. This modelled transient is why the sequencer’s oil-pressure permissive check must wait on a genuine simulated transient rather than a simple threshold-crossing at $t = 0$.

A separate fault-injection path (“slow lube build”) multiplies the effective time constant on this lag from 3 s to 900 s to deliberately delay the prelube permissive past a PLC’s oil-pressure fault timer. The predecessor model used a 90 s time constant for the same fault; this model uses 900 s to more reliably exceed a typical PLC oil-pressure fault timer.

5.12 11. Cooling — fan-count lookup

$$n_{fans} = \text{CMD}_{cooler1} + \text{CMD}_{cooler2} \in \{0, 1, 2\} \quad (16)$$

Three discrete lookups indexed by fan count:

Target	0 fans	1 fan	2 fans
T_{inter}	140 °F	120 °F	105 °F
$T_{ac,target}$	175 °F	130 °F	110 °F
$T_{oil,target}$	200 °F	200 °F	160 °F

Simplification: no heat-exchanger model. No UA (overall heat-transfer coefficient), no air-side flow, no approach temperature. Cooler outlet is a step function of how many fans are running, filtered only by the generic first-order lag of Equation 14. Acceptable because the PLC only ever sees fan status (on/off) and outlet temperature; the physics in between does not affect the logic under test.

5.13 12. Engine speed model

Ramp-rate limited toward a commanded reference, with distinct rates by phase:

Phase	Condition	Rate
Coastdown	not (cat_start and driven_ready and not cat_esd)	first-order, $\tau = 6 \text{ s}$
Cranking	$N < 200 \text{ rpm}$	+70 rpm/s
Acceleration	$N \geq 200 \text{ rpm}$, below reference	+50 rpm/s
Deceleration	$N \geq 200 \text{ rpm}$, above reference	−75 rpm/s

The running reference is $N_{\text{idle}} = 650$ rpm unless “idle/rated” is commanded, in which case $N_{\text{ref}} = 850 + (\text{ao_speed}/100) \times (1000 - 850)$ rpm.

Simplification: no engine torque or load coupling. No governor droop, no load feedback. Speed is a commanded trajectory tracked by rate limits, not a result of torque balance against compressor load. A real engine with a governor would droop under load; this model does not represent that.

6 Symbols, Constants, and Design Point

6.1 Design Point

Converged, 600 s simulated from a pressurised initial condition.

Quantity	Value	Test tolerance
Suction pressure	29.8 psig	± 0.5 psi
ST1 discharge pressure	117.3 psig	± 2 psi
ST2 discharge pressure	377.2 psig	± 5 psi
Final discharge pressure	1149.0 psig	± 10 psi
Stage ratio (each of 3 stages)	2.97	± 0.02
Compressor mass flow	0.945 kg/s	± 0.005
Cylinder 1 discharge temp	245.4 °F	± 2 °F
Cylinder 2 discharge temp	251.7 °F	± 2 °F

Driving command state at the design point: 1000 rpm (100% speed command, idle/rated selected), bypass closed (75% AO), suction valve at 45%, both coolers running, both ESDs open, blowdown closed. All mass flows balance to under 1×10^{-3} kg/s at this point — the model’s primary acceptance criterion (Section 8).

6.2 Constants — Configured Model Parameter Set

Constant	Value	Meaning
R_{sp}	345.5 J/(kg · K)	Specific gas constant
n	1.2693	Polytropic exponent
e_T	0.2122	$(n - 1)/n$
V_{disp}	0.023258 m ³ /rev	Stage 1 swept volume, double-acting
C (clearance)	0.078	Fitted — back-solved to close the design point (Equation 4)
L_{slip}	0.04	Slip/leakage loss fraction
V_s	3.0 m ³	Suction vessel volume
V_d	4.5 m ³	Discharge vessel volume (lumped)
K_{suc}	2.10×10^{-3}	Supply $\rightarrow$ suction flow coefficient
K_{byp}	1.50×10^{-3}	Discharge $\rightarrow$ suction (recycle) flow coefficient
K_{proc}	1.335×10^{-4}	Discharge $\rightarrow$ pipeline flow coefficient

Constant	Value	Meaning
K_{bdv}	4.00×10^{-3}	Suction $\rightarrow$ atmosphere (vent) flow coefficient
P_{src}	60 psig	Source boundary pressure
P_{proc}	1050 psig	Pipeline boundary pressure
T_{suc}	100 °F	Inlet gas temperature
T_{amb}	90 °F	Ambient temperature
$N_{\text{crank term}}$	200 rpm	Crank-terminate / compression-gate threshold
N_{idle}	650 rpm	Idle speed setpoint
$N_{\text{min load}}$	850 rpm	Minimum-load speed setpoint
$N_{\text{max load}}$	1000 rpm	Maximum-load speed setpoint
Speed ramp rates	+50 / -75 / +70 rpm/s	Accel / decel / crank
τ_{coast}	6 s	Coastdown first-order time constant
K_{valve}	20	Valve position-error gain
Valve rates	see the table under Equation 13	Open/close rates, %/s, per valve
$P_{\text{oil,run}}$	120 psig	Running lube oil pressure target
$P_{\text{oil,prelube}}$	55 psig	Prelube pressure target
$P_{\text{oil,fault}}$	25 psig	Forced value under the “low lube oil” fault
$\tau_{\text{oil,p}}$	3 s	Lube oil pressure lag time constant
$\tau_{\text{oil,slow}}$	900 s	Slow-lube-build fault time constant — the predecessor model used 90 s for the same fault
τ_{eoil}	400 s	Engine oil temperature lag time constant
$T_{\text{eoil,run}}$	190 °F	Running engine oil temperature target
Cooling lookup tables	see the table under Equation 16	Fan-count $\rightarrow$ target temperature, 3 tables
$\tau_{T,\text{cyl}}$	45 s	Cylinder discharge temperature lag
$\tau_{T,\text{oil}}$	300 s	Compressor oil temperature lag
$\tau_{T,\text{ac}}$	60 s	Aftercooler outlet temperature lag

Constant	Value	Meaning
Cylinder temperature offset	0, 0, +3, −2 °F	Static per-cylinder calibration offset (Cyls 1–4), cosmetic
Cooler status feedback lag	1.0 s	Cooler run-status feedback lag (contactor pickup/dropout stand-in)
Jacket-water offset	−15 °F	Engine jacket-water temp offset from engine oil temp
Engine oil pressure, running	60 psig	CAT ADEM engine oil pressure at rated speed
Link watchdog timeout	2.0 s	OPC UA link watchdog timeout

6.3 Transmitter Ranges — PLC Instrumentation Interface

Tag	Range
PT_1001 (suction)	0–60 psig
PT_1002 (ST1 discharge)	0–300 psig
PT_1003 (ST2 discharge)	0–1000 psig
PT_1004/PT_1006 (ST3/final discharge)	0–2000 psig
PT_1005 (compressor oil)	0–200 psig
PT_1007 (engine oil)	0–150 psig
ST_1008 (speed)	0–2000 rpm
TT_2001, TT_2004–2013 (temperatures)	0–500 °F
TT_2014 (jacket water)	0–300 °F

Every analog value is clamped to its transmitter range before being written to the PLC — a blocked discharge can drive the internal model well past 2000 psig, but the value written to the PLC is limited to the configured transmitter range, so the PLC sees a saturated 2000. This is consistent with the configured transmitter range-clamping behaviour implemented in the simulator’s instrumentation layer; it is not a claim about how any specific real transmitter model behaves.

7 Assumptions and Limitations

Stated plainly, so nothing here needs to be discovered by inspection later.

Limitation	Why acceptable for this purpose	What it rules out
Compressibility factor $Z = 1$ (ideal gas) throughout	The purpose is PLC logic testing, not a performance guarantee, and no gas composition analysis was supplied	Any use as a performance-prediction tool; density is optimistic by roughly 10–15% at the discharge end for a typical natural gas at design conditions
No choked-flow model — all valve flows use the subsonic orifice form	Blowdown and USD sequences still take real, non-zero time, which is what exercises the PLC's timeout logic	Precise vent-rate or depressurisation-timing validation; the blowdown valve likely vents faster in the model than a real choked vent would
No cylinder thermal mass — discharge temperatures are algebraic targets filtered by a generic lag, not a derived thermal model	The PLC only sees the resulting temperature signal, not the mechanism producing it	Any claim that the 45 s cylinder lag time constant corresponds to a real cylinder's actual thermal time constant
No heat-exchanger model — cooler outlets are lookup tables with no UA, air-side flow, or approach temperature	The PLC only ever sees fan status and outlet temperature	Sizing or performance evaluation of the actual air coolers
No engine torque or load coupling — speed is a commanded trajectory, not a torque-balance result	Sequencing logic tests speed setpoints and ramp timing, not load response	Any claim that engine speed behaviour under real compressor load (governor droop) is represented
No in-cylinder valve dynamics, no pulsation, no rod load	Out of scope for a control-logic test rig by design	Mechanical design, rod-load, or pulsation studies — these belong in a dedicated mechanical analysis
No gas composition analysis supplied — R_{sp} , n are assumed values, not derived from an actual gas analysis	Consistent, repeatable behaviour is what a logic test rig needs, not exact real-gas fidelity	Confidence that discharge temperature or density values would match a specific real gas stream
Clearance fraction $C = 0.078$ was back-solved to close the design point, not sourced from vendor performance data	This is the single largest fitted parameter in the model	Any claim of validated fidelity to the reference machine — this is why the package is framed as a generic three-stage compressor throughout this document

Limitation	Why acceptable for this purpose	What it rules out
No PLC sequencer bundled with this application (unlike its Simulink/CODESYS predecessor)	Sequencing logic under test is supplied by the connected PLC; this simulator does not provide a standalone sequencer	Standalone demonstration of a full startup/shutdown sequence without an external PLC connected

8 Verification Summary

The process model is exercised by **86 automated checks** across seven suites: design-point acceptance (18), transient/dynamic behaviour (27), and five further suites covering fault injection, tag mapping, the OPC UA link, and command-locking behaviour. For the software version documented in this report, the automated verification suite returned 86/86 passing checks. These checks verify the implementation against its configured equations, configured design-point acceptance criteria, interface behaviour, fault mechanisms, tag mappings, and software invariants. They do not constitute validation against measured performance of a physical compressor.

Category	What it checks	Result
Design-point acceptance	All 8 design-point values (Section 6) within tolerance; mass-balance closure $< 1 \times 10^{-3}$ kg/s on both vessels; stage ratios equal to 10^{-6} ; monotonic staging; no negative pressure/NaN/Inf; steady-state drift < 2 psi over 50 s	Pass, 18/18
Transient / dynamic verification	Valve/speed ramp-rate timing; coast-down time constant; load/unload direction; blowdown venting profile; oil-permissive timing (healthy and faulted); fault behaviour recovery; ESD-closed cooling; cooler-loss response	Pass, 27/27
Randomised invariant sweep	400 iterations / 800 s simulated of randomised commands: no NaN, no inverted staging, no negative pressure, no sub-1.0 stage ratio, no cooling-by-compression, no out-of-range valve position	Pass
Timestep insensitivity	Converged final discharge pressure compared across 5/20/50 ms integration steps	Spread < 1.0 psi
Fault injection, tag mapping, OPC UA link, command locking	Remaining five suites	Pass

The physics loop integrates at a fixed 20 ms step using fourth-order Runge-Kutta; the timestep-insensitivity result is the direct evidence this choice does not materially affect the converged values reported in Section 6. The exhaustive per-test narrative is in Appendix C.

Part II — Using the Simulator

9 System Requirements and Deployment Considerations

The simulator is a real-time process simulator with a desktop HMI — it computes and displays the process model described in Part I and exchanges tags with a PLC over OPC UA. It does not perform CFD, FEA, AI inference, 3D rendering, or other high-performance computing tasks, and its computational requirements are modest.

One instance means one copy of the Compressor Simulator application running at a given time — for example, one simulator application running is one instance; two simulator applications running simultaneously are two instances; a machine running several separate simulator instances for several concurrent PLC tests is running multiple instances.

For deployment planning, the following configuration provides a practical baseline for running one Compressor Simulator application at a time. These are practical planning figures, not the result of formal minimum-hardware qualification testing.

	Minimum practical configuration	Recommended configuration
Operating environment	64-bit Windows	64-bit Windows, standard desktop, VM, or Server environment with an interactive desktop session
Processor	Dual-core 64-bit, approximately Intel Core i3 / Pentium Gold class, AMD equivalent, or better	Intel Core i3 class or better — Core i5 or higher is more than sufficient but not required for normal operation
Virtual machine / server CPU	2 vCPU	2 or more vCPU
Memory	4 GB RAM	8 GB RAM or more
Disk space	Approximately 1 GB available	Approximately 1 GB available
Network	Standard Ethernet/network connectivity when OPC UA communication with a PLC is required	Same
Graphics	No dedicated GPU required	No dedicated GPU required

The simulator has modest computational requirements and does not require specialised computing hardware or a dedicated graphics processor for normal operation.

9.1 Server and Virtual-Machine Deployment

The current release is an interactive Windows desktop application, intended to operate within an interactive Windows session. A Windows virtual machine or Windows Server environment with desktop access can therefore be used, subject to normal operating-system and network requirements, including

normal network access to the PLC / OPC UA endpoint. The current release is not a headless Windows service, and headless (no interactive session) deployment has not been tested or validated.

9.2 Multiple Instances

If several simulator applications are run simultaneously, the required CPU and memory should be increased according to the number of active instances, with additional capacity reserved for Windows and any PLC runtime, engineering software, or other workloads hosted on the same machine. No maximum number of simultaneous instances is specified here.

A normal Windows PC or small Windows VM is sufficient for running one Compressor Simulator application. A dedicated GPU or high-end workstation is not required. Larger server resources are mainly relevant when several simulator applications or other workloads are running at the same time.

10 Installation

The application is distributed as a single zipped installer, `CompressorSim-Setup-<version>.zip`. No other files or runtime need to be downloaded separately — the installer bundles everything the app needs except the Microsoft Edge WebView2 Runtime, which it installs automatically on first run if the machine doesn't already have it.

10.1 Extract the Installer

Start with the zip file wherever it was downloaded to — the desktop, in this walkthrough.

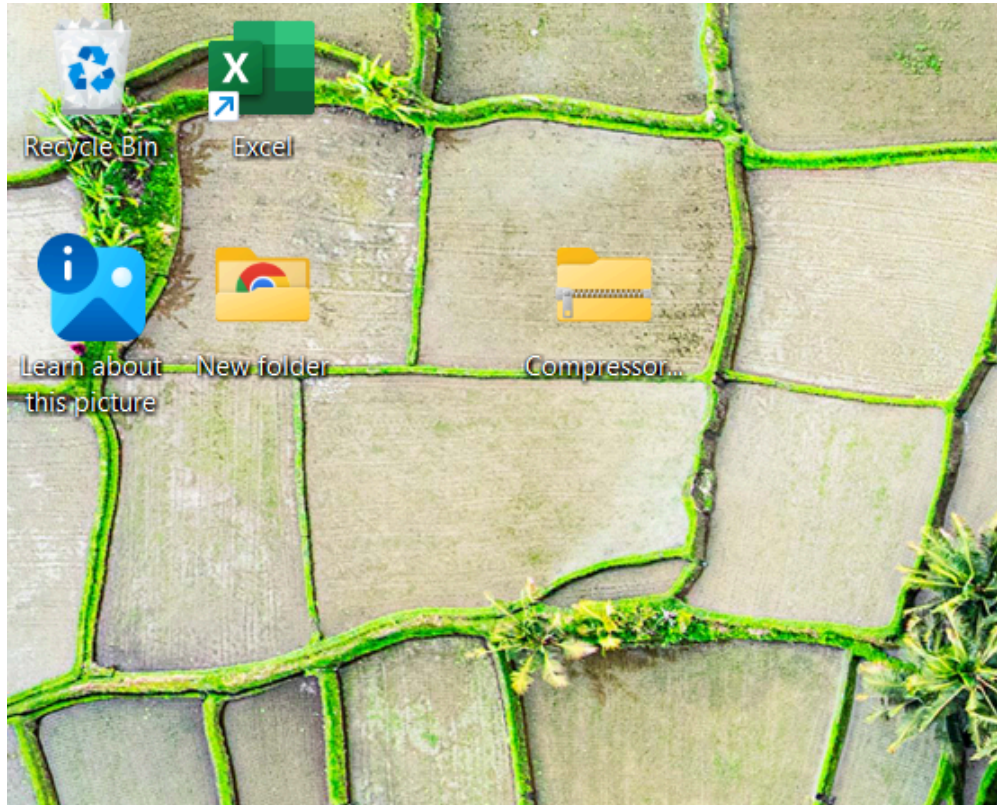


Figure 4. The downloaded installer archive on the desktop, before extraction.

Right-click the zip file and choose **Extract All...** from the context menu.

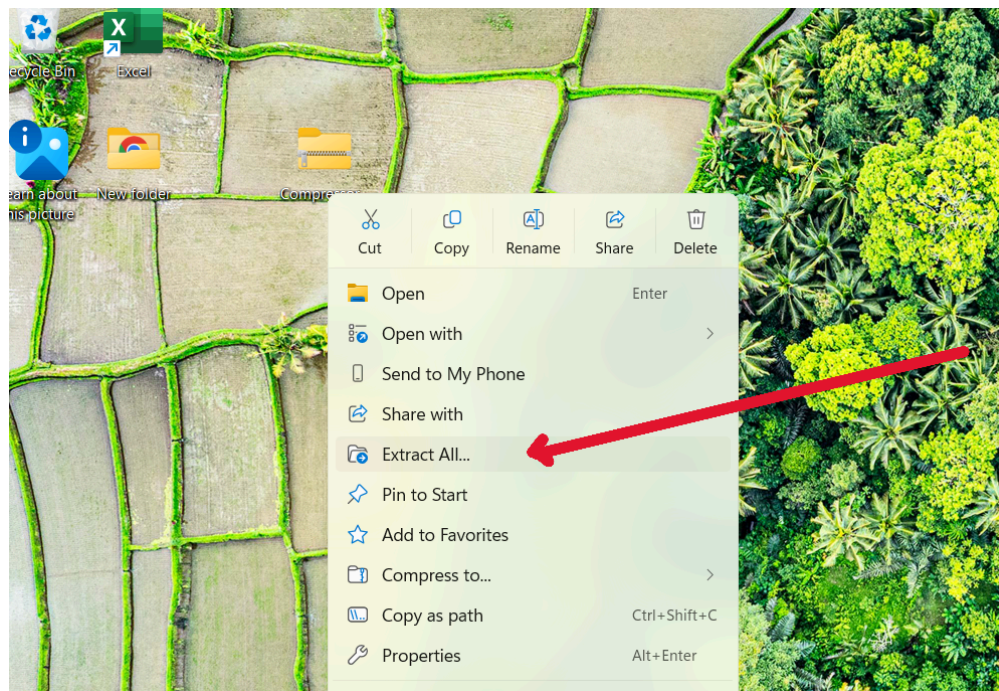


Figure 5. Right-click context menu on the zip file, with Extract All... selected.

This produces a folder next to the zip file with the same name. The installer executable is inside it.

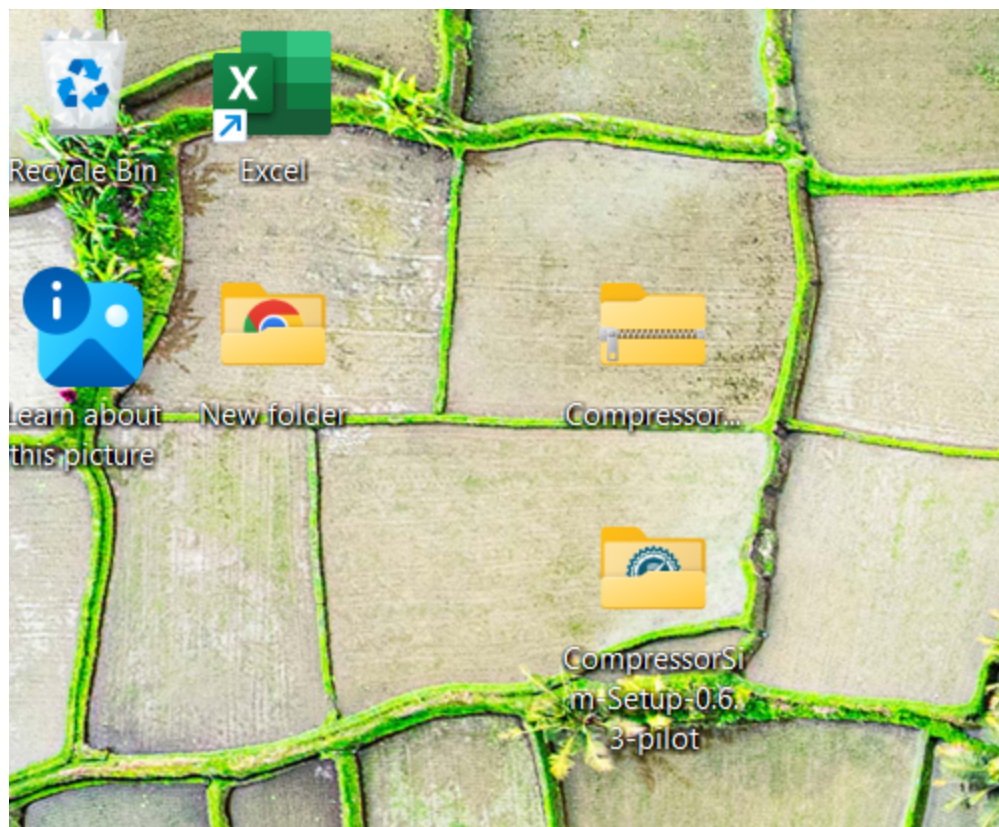


Figure 6. The zip file and the resulting extracted folder shown together on the desktop.

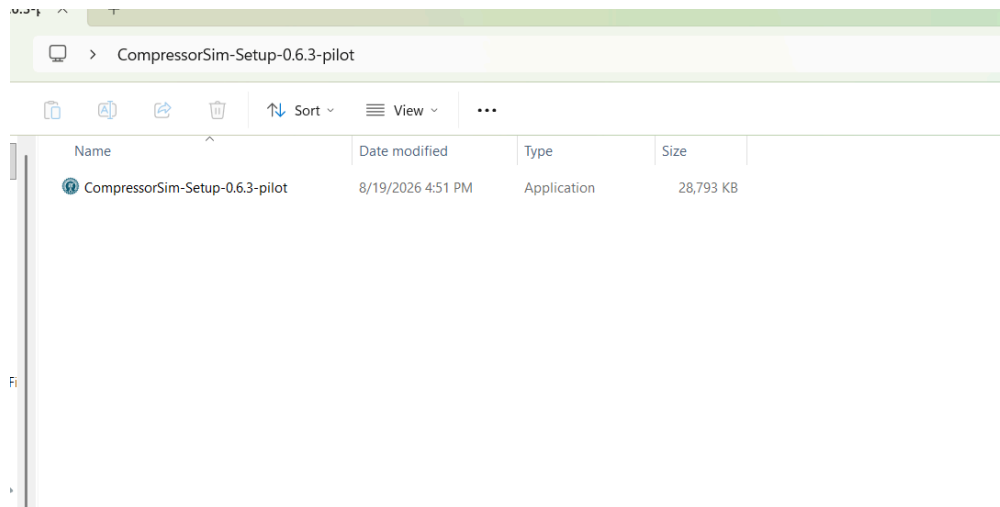


Figure 7. Inside the extracted folder: the installer executable, `CompressorSim-Setup-.exe`.

10.2 Run the Installer

Double-click the installer executable to launch it. Because this build is not code-signed, Windows shows a Microsoft Defender SmartScreen warning the first time it runs — this is expected for any unsigned installer, not a sign something is wrong.

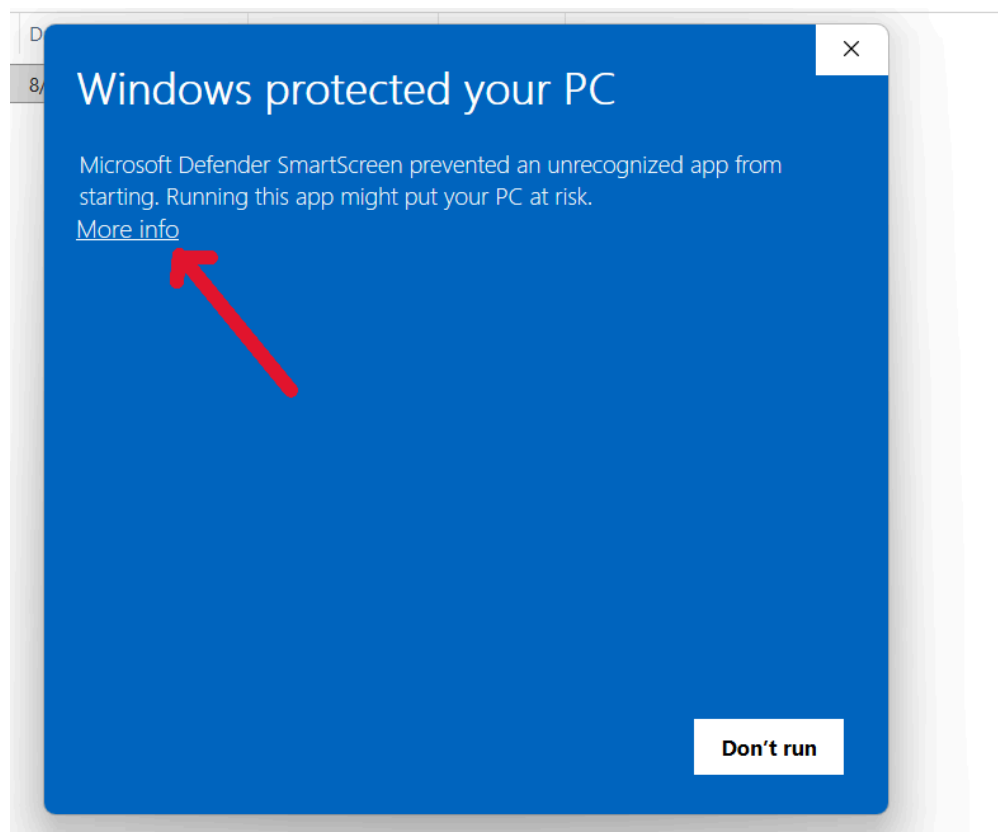


Figure 8. SmartScreen’s initial warning. Click “More info” to reveal the run-anyway option.

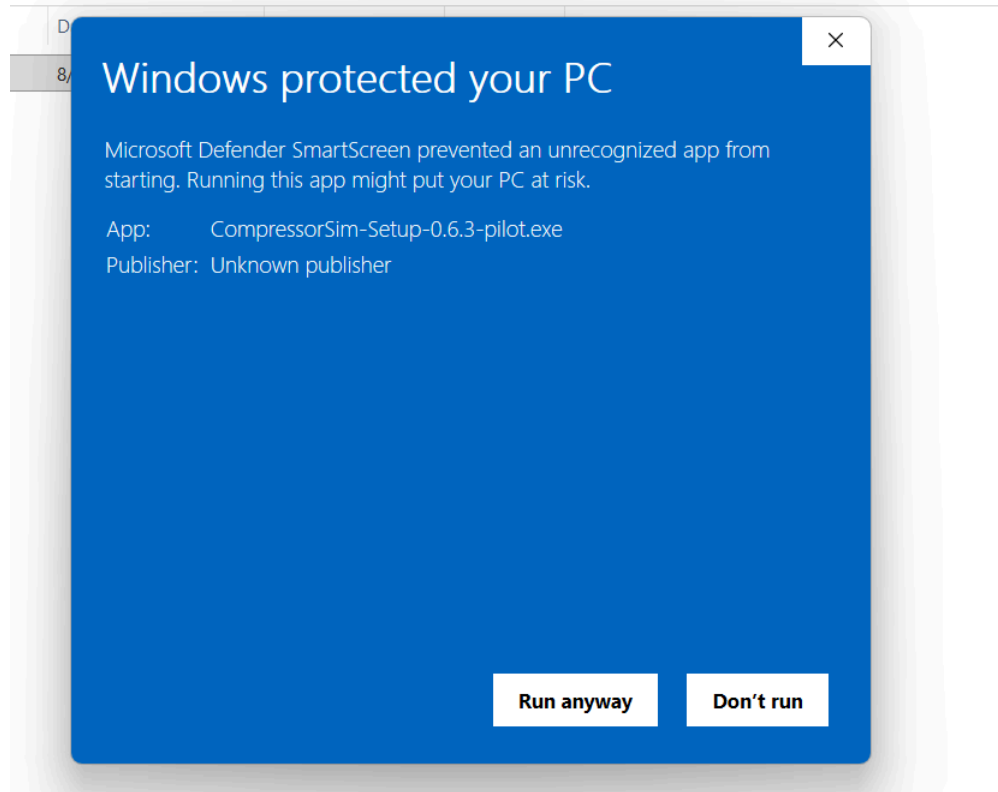


Figure 9. After clicking “More info,” SmartScreen identifies the app and publisher and reveals the “Run anyway” button.

Click **Run anyway**. The Inno Setup installer wizard opens; step through it with the default options (Next → Next → Install → Finish). The install runs per-user and does not require administrator rights.

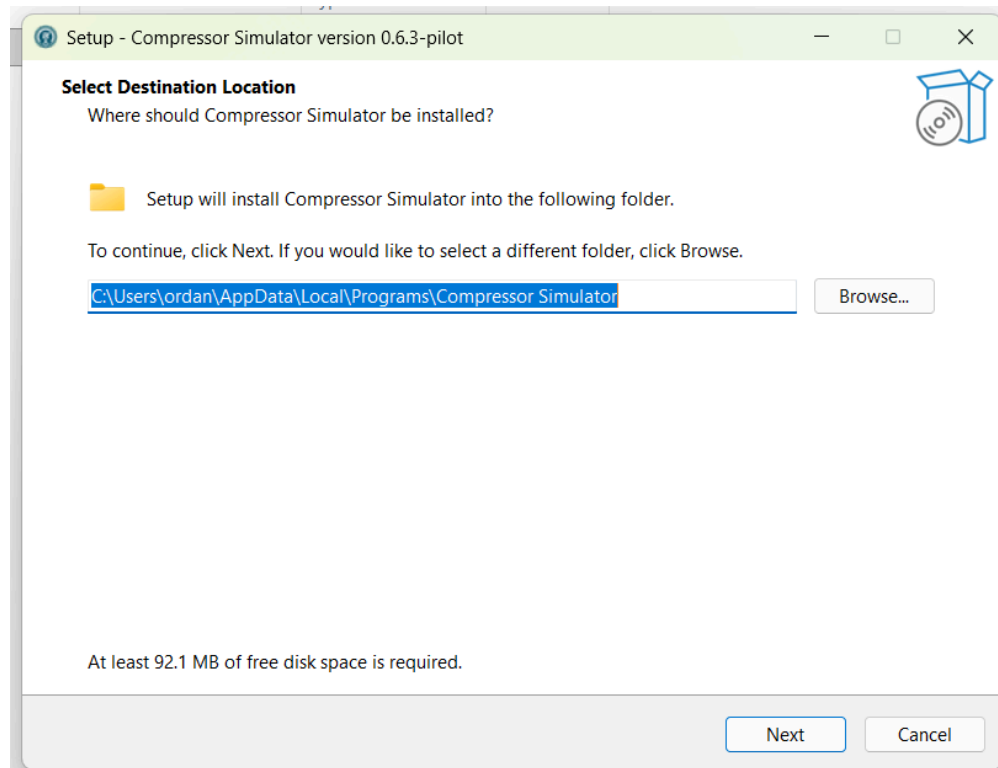


Figure 10. The installer wizard’s destination-folder step. Defaults are appropriate for almost all installs; click *Next* through the remaining pages to finish.

- Leave “Create a desktop shortcut” checked if you want one.
- At least 92.1 MB of free disk space is required, per the installer’s own check.

10.3 First Launch

Once installation finishes, a Compressor Simulator shortcut appears on the desktop (and in the Start menu).

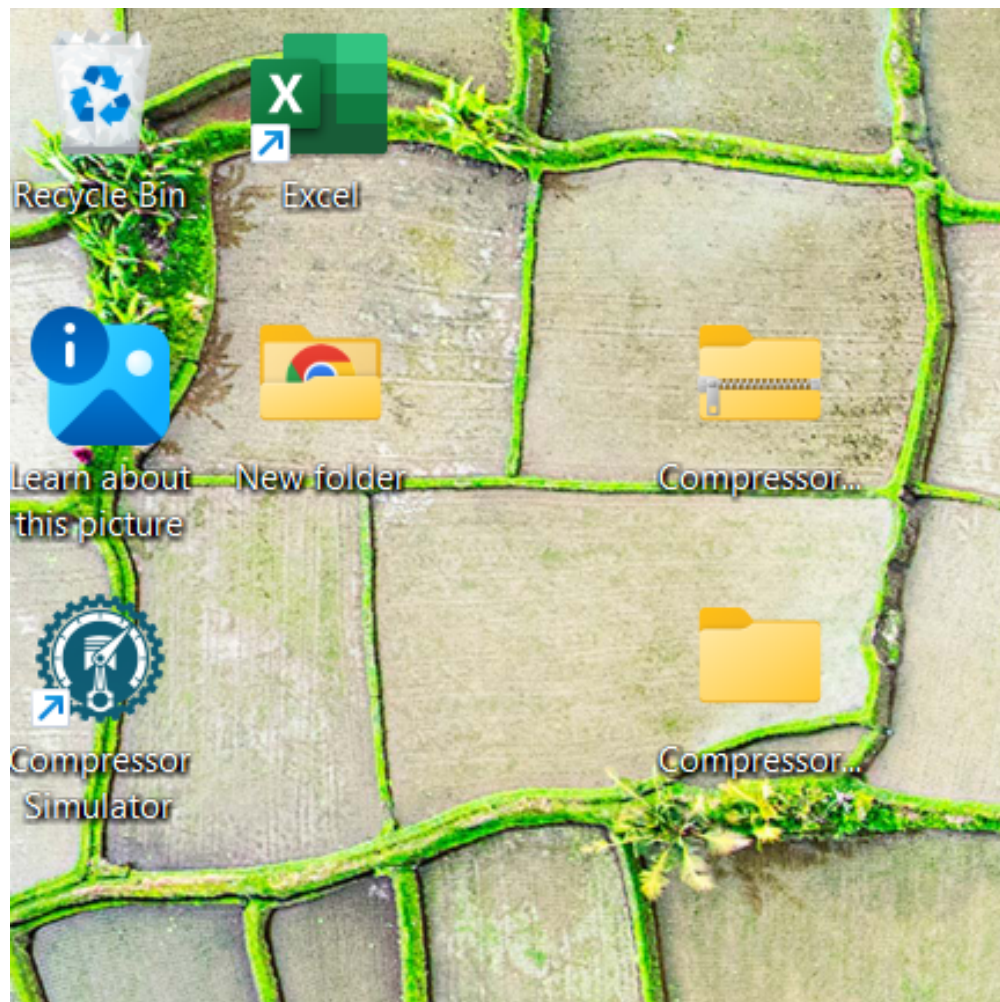


Figure 11. The desktop after installation completes, showing the new Compressor Simulator shortcut alongside the original installer files.

Double-clicking the shortcut opens the application window. The simulation starts running immediately with default boundary conditions and no PLC connected — the P&ID, live instrument readouts, and the Overrides/Faults/Tags tool dock on the right are all visible and interactive before any OPC UA link is made.

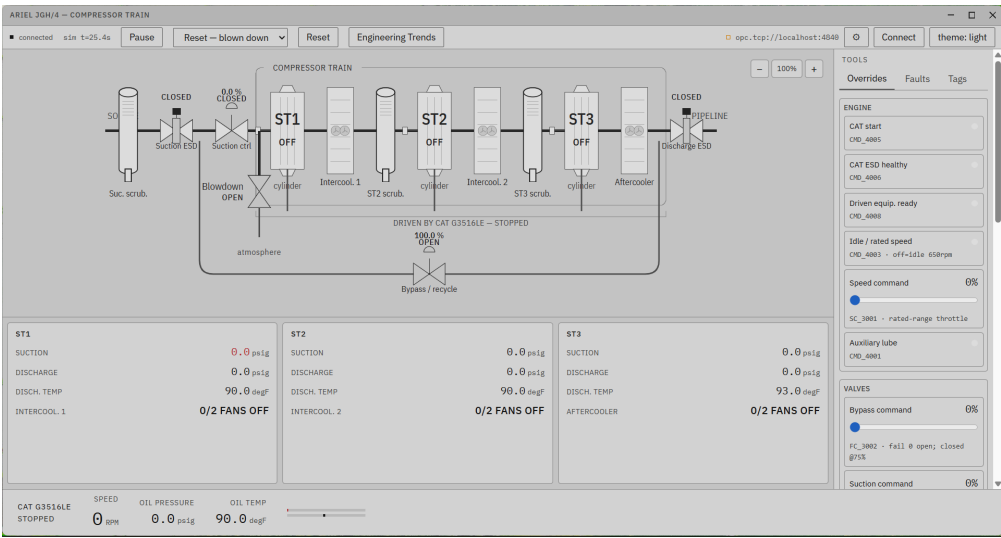


Figure 12. Compressor Simulator on first launch: the ARIEL JGH/4 P&ID with the compressor train stopped and blown down, and the Overrides tool dock open on the right.

VENDOR / MODEL REFERENCES ARE CONTEXT ONLY. Vendor and model references retained in legacy tags, drawings, or interface labels throughout this report — for example the ARIEL P&ID above, and the “CAT”/ADEM-prefixed engine tags used in Section 13 and Appendix A — identify the reference equipment context inherited from the predecessor project only. They do not indicate that this simulator has been validated, endorsed, or certified by those manufacturers.

10.4 Where Things Live Once Installed

App files	%LOCALAPPDATA%\Programs\Compressor Simulator\
Editable configuration	The application configuration file in %LOCALAPPDATA%\CompressorSim\ — not the bundled default installed with the app. This configuration is copied from the bundled default the first time the app runs; edit this copy to change the PLC endpoint or any other parameter. The app never touches its own bundled default.
Logs	%LOCALAPPDATA%\CompressorSim\logs\app.log (rotates, keeps the last 3 files)

Installation is now complete. Connecting this instance to a PLC over OPC UA is covered in Section 11.

11 Connecting to a PLC over OPC UA

With no PLC connected, the simulator runs on its own — the compressor train sits stopped and blown down, and every command tile in the Overrides dock is live and editable (Figure 12 shows this state).

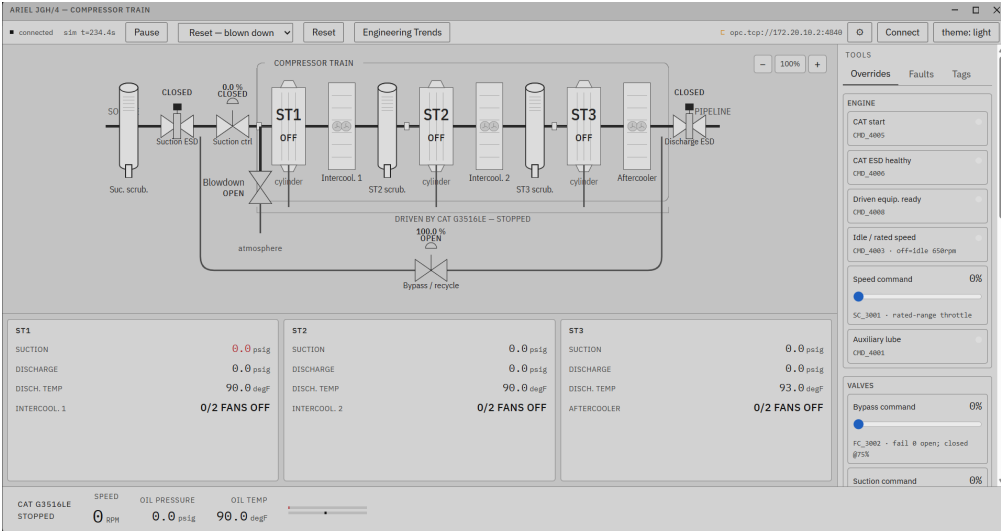


Figure 13. Compressor Simulator with no PLC connected: the compressor train stopped and blown down, all suction/discharge ESD and blowdown valves in their fail-safe position, and the Overrides dock open and editable on the right.

11.1 The OPC UA Endpoint

The OPC UA target address is an editable simulator setting, defined in the application configuration described in Section 10. The configurable connection settings are:

Setting	Purpose
Endpoint	The OPC UA server address to connect to, e.g. <code>opc.tcp://localhost:4840</code>
Symbol-set identifier	Identifies the PLC’s exposed symbol set for tag discovery — needed only for non-standard servers
Browse path prefix	Where in the server’s address space to look for tags — needed only for non-standard servers
Link watchdog timeout	How long to wait without a heartbeat before declaring the link failed — 2.0 s by default
Tag-addressing mode	<code>auto</code> (the default) searches the server’s whole address space for the simulator’s known tag names

For a PLC on the same machine, leave the endpoint at `opc.tcp://localhost:4840`. For a PLC on another device — a networked CODESYS PC or a physical panel — change the host, e.g. `opc.tcp://172.20.10.2:4840`. The default automatic tag-addressing mode needs no symbol-set identifier or browse-path-prefix configuration for a standard CODESYS target; those settings, and the optional node-ID pattern override, exist only for non-standard servers or as a manual override. The app needs to be restarted after any change to the endpoint (or any other connection setting) in the application configuration, since there is no live endpoint-switching control in the current UI.

11.2 Opening the Connect to a PLC Dialog

Click the settings icon next to the connection status in the header to open **Connect to a PLC**. The dialog shows the current connection state, a *Where is the PLC?* choice, and an Advanced section (collapsed by default) for tag-discovery settings.

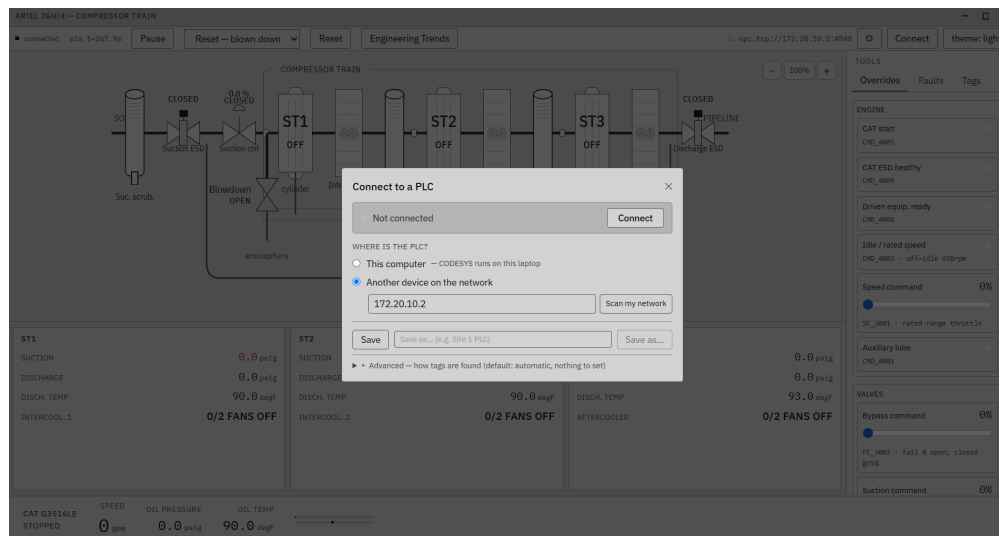


Figure 14. The Connect to a PLC dialog in its default, not-connected state.

11.3 This Computer

Selecting **This computer** targets a CODESYS runtime running on the same machine as the simulator, at `opc.tcp://localhost:4840` — the common case for bench testing.

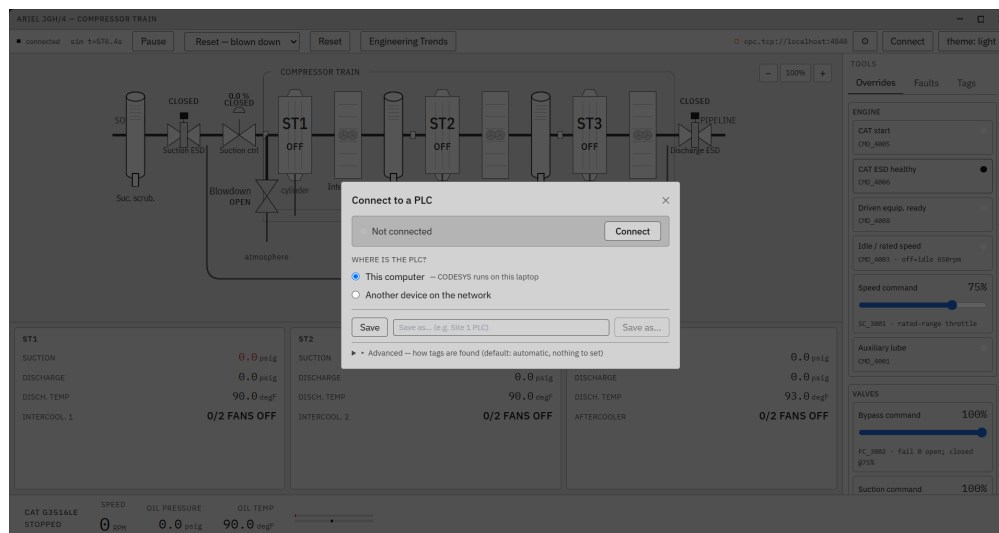


Figure 15. “This computer” selected as the PLC location.

Clicking **Connect** establishes the link. Once connected, the dialog shows the live status and endpoint, and the header’s OPC UA indicator and endpoint text update to match — here the engine has already been started and is running at 75% speed, driven entirely by the PLC.

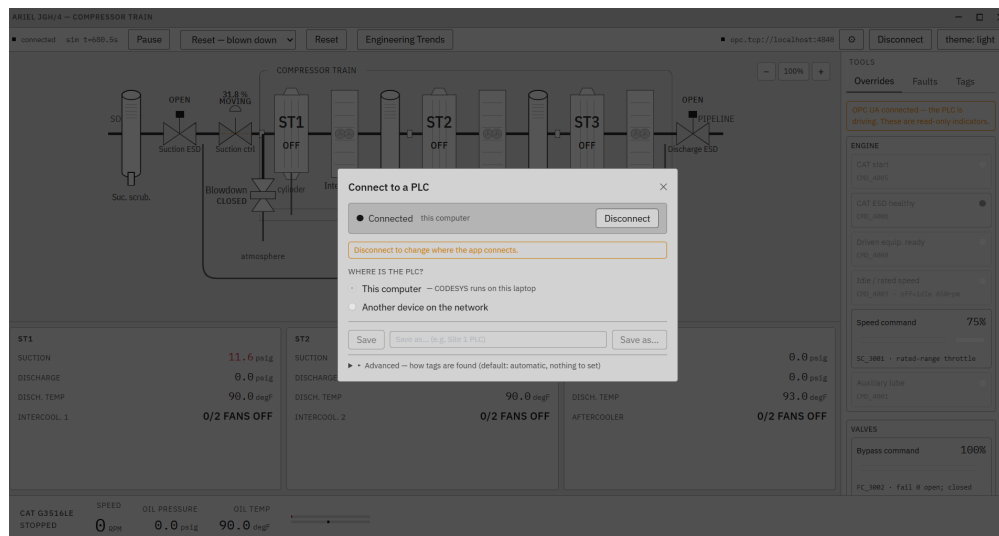


Figure 16. Connected to a local CODESYS runtime. The dialog shows “Connected — this computer”; the header’s connection dot and endpoint reflect the same state.

11.4 Another Device on the Network

For a physical panel or a separate PC, select **Another device on the network** and either type its IP address directly or click **Scan my network** to discover OPC UA servers advertising themselves on the local subnet.

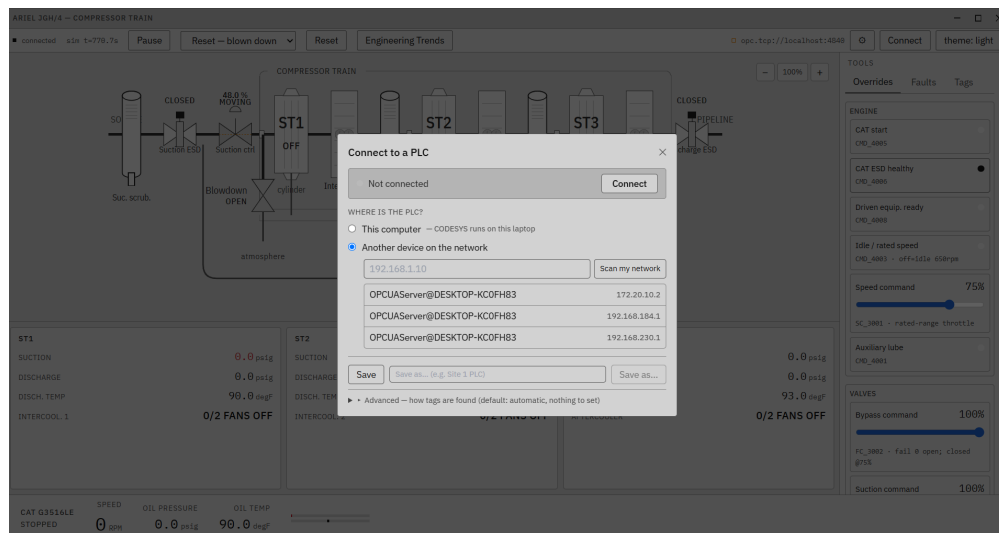


Figure 17. Results of Scan my network: OPC UA servers discovered on the local subnet, each listed with its hostname and IP address.

Selecting a discovered server fills in its address automatically.

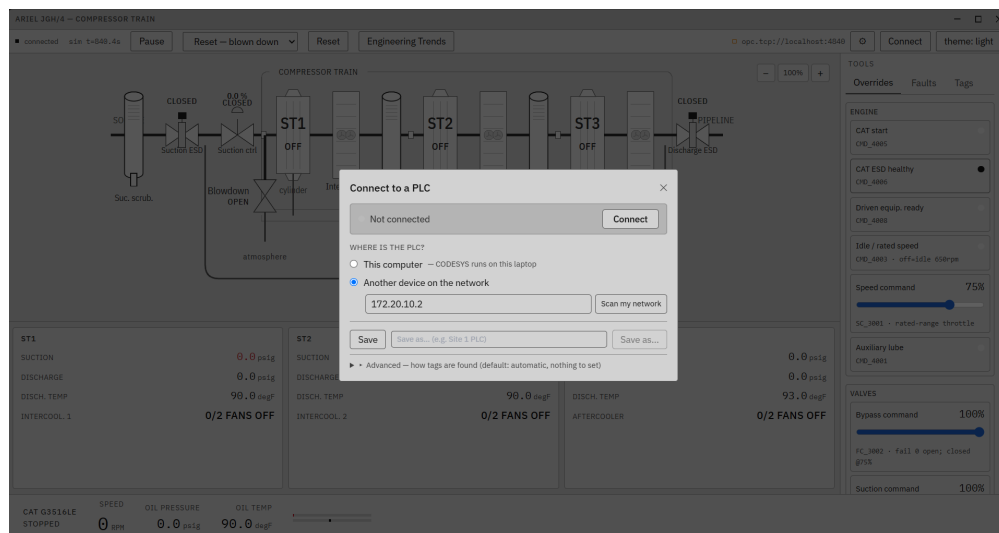


Figure 18. A discovered server's address populated into the connection field after selecting it from the scan results.

Clicking **Connect** links to that remote CODESYS runtime the same way as the local case, just at a routable IP address instead of localhost.

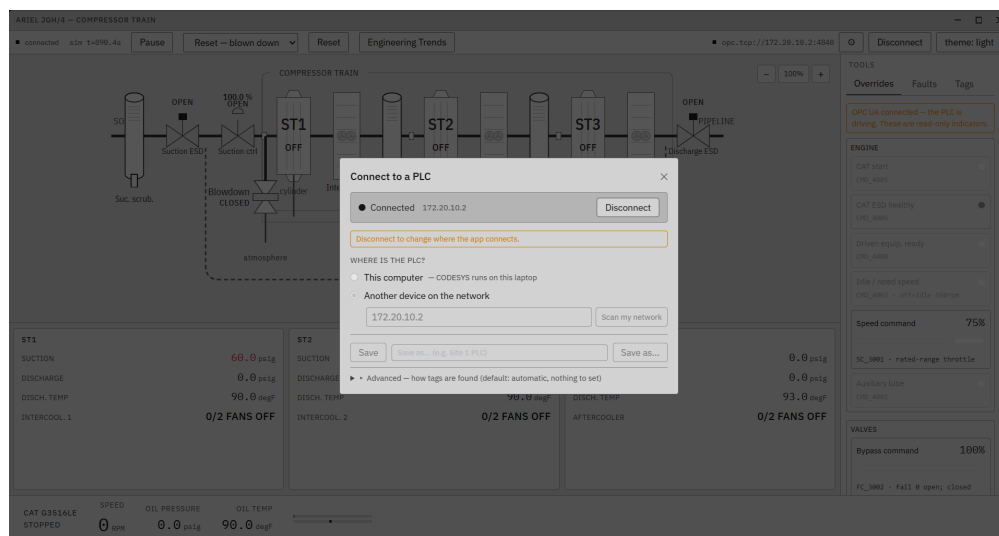


Figure 19. Connected to a remote CODESYS runtime at 172.20.10.2 — the dialog and header both reflect the non-localhost endpoint.

11.5 Confirming the Link Is Up

Once connected, every tile that used to accept manual input in the Overrides dock is grayed out and shows the PLC's own commanded value instead — a banner across the top of the dock states this explicitly.

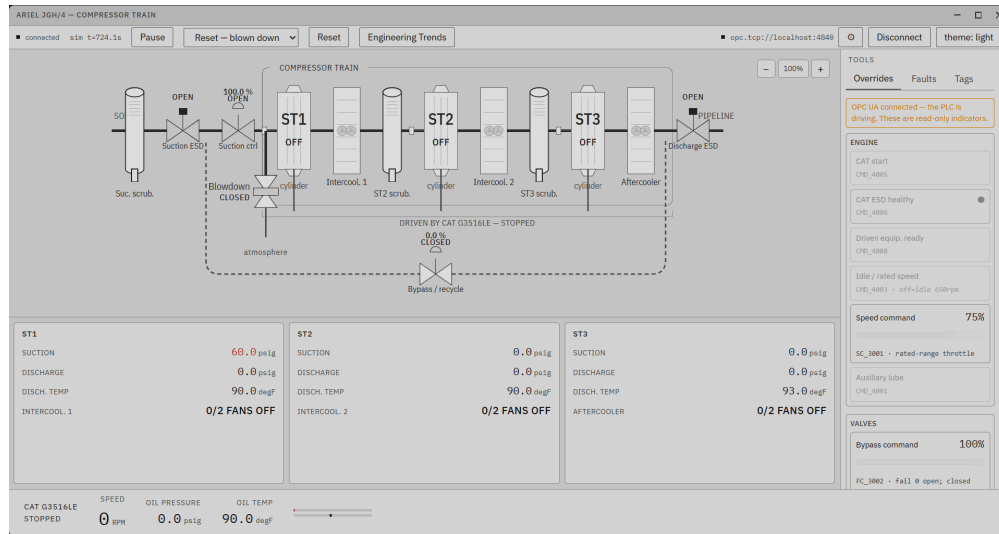


Figure 20. Overrides dock once connected: “OPC UA connected — the PLC is driving. These are read-only indicators,” with every tile reflecting the PLC’s commanded value (Speed command 75%, Bypass command 100%, etc.) instead of accepting input.

On the log side, `app.log` records the connection lifecycle; a successful connect and any link loss are both written there, which is the place to check when the UI status alone is ambiguous (Section 18).

11.6 What Happens on Link Loss

The app increments a watchdog counter every 500 ms; if the peer’s counter (or, on the app’s own side, the OPC UA server’s time source) stops advancing for `watchdog_timeout_s` (2.0 s by default), the link is declared failed and every command tag reverts to its documented fail value (the complete list is Appendix A):

Command	Fail value	Physical result
SC_3001 (speed)	0	Speed command drops to zero
FC_3002 (bypass)	0	Bypass valve opens (fails open)
FC_3003 (suction valve)	0	Suction valve closes (fails closed)
CMD_4004 (blowdown solenoid)	off	Blowdown valve opens (fails open)
All remaining discrete command tags	off	ESDs close, cooler/lube commands drop, start command drops

“All remaining discrete command tags” above means every command tag not listed individually: CMD_4005, CMD_4006, CMD_4008, CMD_4009, CMD_4010, CMD_4001, CMD_4003, CMD_4011, CMD_4012.

This reflects the implemented fail-safe command polarity documented in Section 13, “Command Polarity”: on loss of signal, the ESDs shut and the blowdown opens, venting the unit (Section 4’s mass-accumulation asymmetry governs how that unfolds). Operator pushbuttons (PB_5001, PB_5003, PB_5004, ESD_5002) remain live and unaffected by simulator OPC UA link state. In the simulator architecture these are represented as always-live PLC-facing inputs rather than simulator command outputs, so they are not subject to the fail-value mapping above; whether the real package wires these hard to the PLC independent of any supervisory link is a matter for the applicable project wiring documentation.

11.7 The CODESYS Side: Bringing the Runtime Online

The screenshots above are of the simulator; connecting also requires a CODESYS runtime actually running and serving OPC UA. This is the walkthrough for a CODESYS Control Win V3 runtime on a bench laptop.

1. Launch the **CODESYS Control Win V3 - x64** runtime (search for it from the Start menu if it isn't already running as a tray application).
2. Open the CODESYS IDE project and check **Communication Settings** under the Device node — this shows the gateway and the target device (here, DESKTOP-KC0FH83, reached via local-host:1217), confirming the IDE can see the runtime before attempting to log in.
3. Click **Login** (Alt+F8) on the toolbar to connect the IDE to the running application.

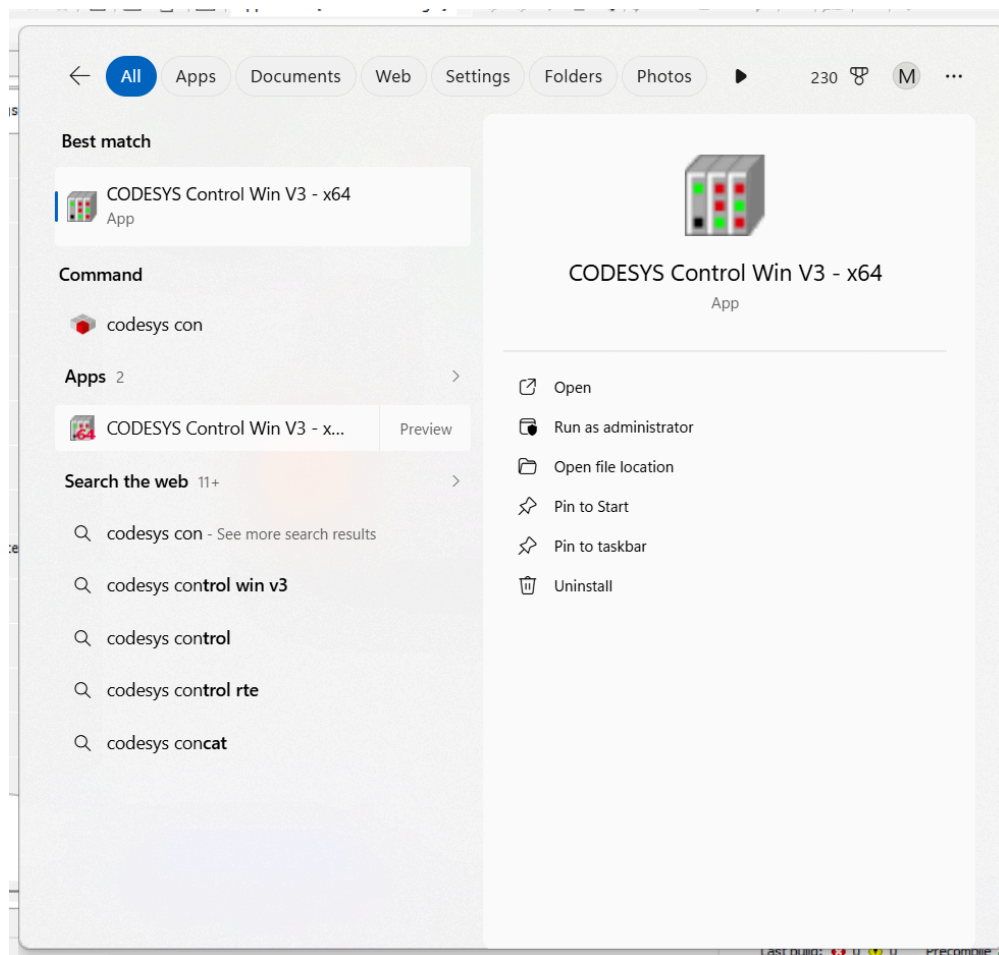


Figure 21. Launching the CODESYS Control Win V3 - x64 runtime from Windows search.

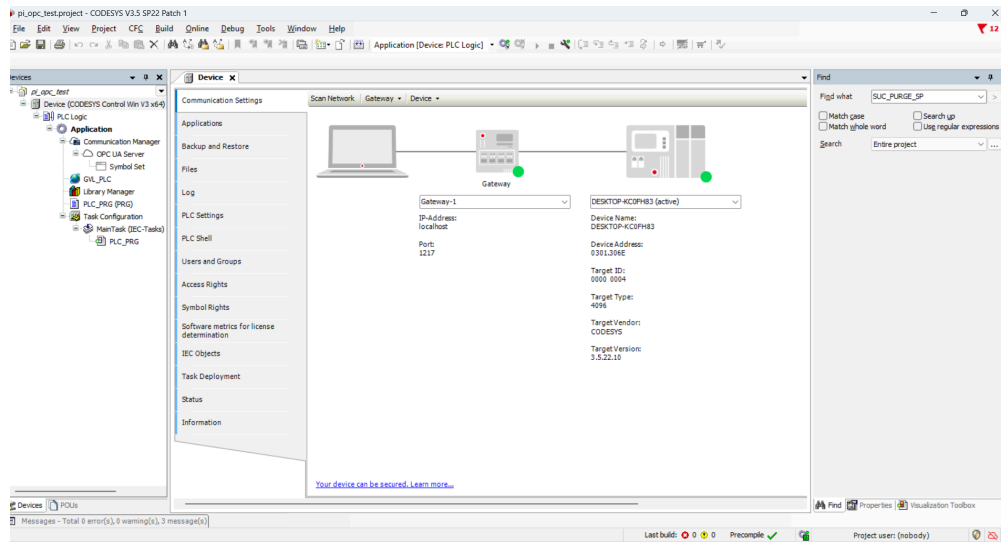


Figure 22. The CODESYS IDE's Communication Settings tab, showing the gateway (Gateway-1, localhost:1217) and the active target device, DESKTOP-KC0FH83.

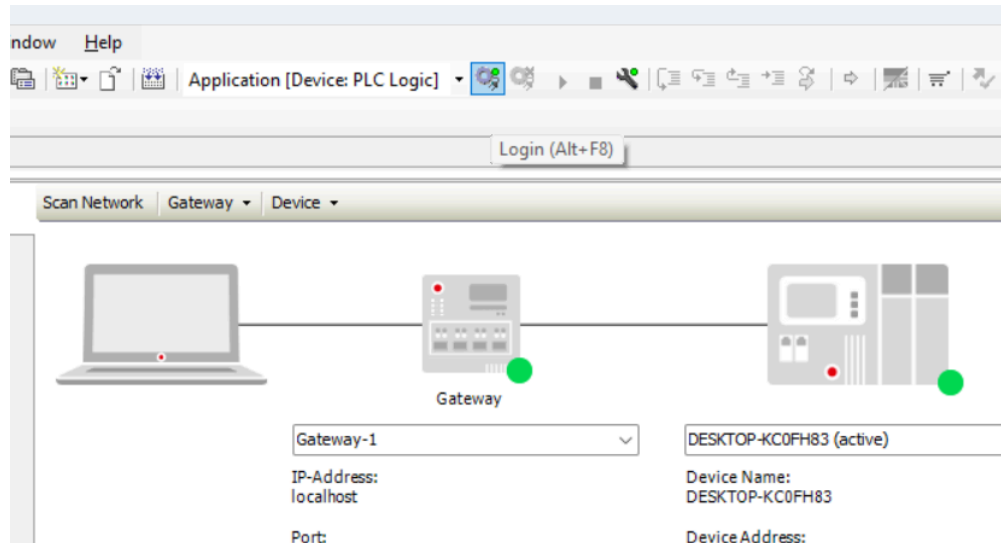


Figure 23. The Login (Alt+F8) toolbar button, which connects the IDE to the running application on the target device.

The runtime's own console confirms its OPC UA server started and which endpoint it's listening on — this is the endpoint the simulator's Connect to a PLC dialog needs to reach.

```

pp>] loaded
2026-08-19T20:44:02.530Z: Cmp=CmpRouter, Class=1, Error=0, Info=1, pszInfo= Setting router <instance>0</instance> address
s to <address>(0001)</address>
2026-08-19T20:44:02.530Z: Cmp=CmpRouter, Class=1, Error=0, Info=1, pszInfo= Setting router <instance>1</instance> address
s to <address>(0001)</address>
2026-08-19T20:44:02.530Z: Cmp=CmpRouter, Class=1, Error=0, Info=1, pszInfo= Setting router <instance>2</instance> address
s to <address>(0002)</address>
2026-08-19T20:44:02.530Z: Cmp=CmpRouter, Class=1, Error=0, Info=1, pszInfo= Setting router <instance>3</instance> address
s to <address>(0001)</address>
2026-08-19T20:44:02.530Z: Cmp=CmpRouter, Class=1, Error=0, Info=1, pszInfo= Setting router <instance>4</instance> address
s to <address>(2ddc:c0a8:b801)</address>
2026-08-19T20:44:02.535Z: Cmp=CmpOPCUAServer, Class=1, Error=0, Info=0, pszInfo= *****
*****
2026-08-19T20:44:02.535Z: Cmp=CmpOPCUAServer, Class=1, Error=0, Info=0, pszInfo= OPC UA Server Started:
Hostname: DESKTOP-KC0FH83, Port: 48
40
2026-08-19T20:44:02.535Z: Cmp=CmpOPCUAServer, Class=1, Error=0, Info=0, pszInfo= URL: opc.tcp://DESKTOP-KC0FH83:4840
2026-08-19T20:44:02.535Z: Cmp=CmpOPCUAServer, Class=1, Error=0, Info=0, pszInfo= Loopbackadapter activated.
All available networkadapters are u
sed.
2026-08-19T20:44:02.535Z: Cmp=CmpOPCUAServer, Class=1, Error=0, Info=0, pszInfo= Multithreading activated. 15 worker
s used.
2026-08-19T20:44:02.535Z: Cmp=CmpOPCUAServer, Class=1, Error=0, Info=0, pszInfo= *****
*****
2026-08-19T20:44:02.537Z: Cmp=CmpApp, Class=1, Error=0, Info=10, pszInfo= Application [<app>Application</app>] started
2026-08-19T20:44:02.538Z: Cmp=CM, Class=1, Error=0, Info=34, pszInfo= CODESYS Control ready
2026-08-19T20:44:02.879Z: Cmp=CmpUserMgr, Class=1, Error=0, Info=0, pszInfo= Create asymmetric key in progress...
2026-08-19T20:44:03.035Z: Cmp=CM, Class=1, Error=0, Info=0, pszInfo= no runtime license - running in demo mode(~2 hours)
2026-08-19T20:44:03.460Z: Cmp=CmpUserMgr, Class=1, Error=0, Info=0, pszInfo= Create asymmetric key done!

```

Figure 24. CODESYS Control Win V3 console log at startup: “OPC UA Server Started,” listening at `opc.tcp://DESKTOP-KC0FH83:4840`.

Logging in for the first time prompts for device-user credentials before the IDE is allowed to view or download to the device.

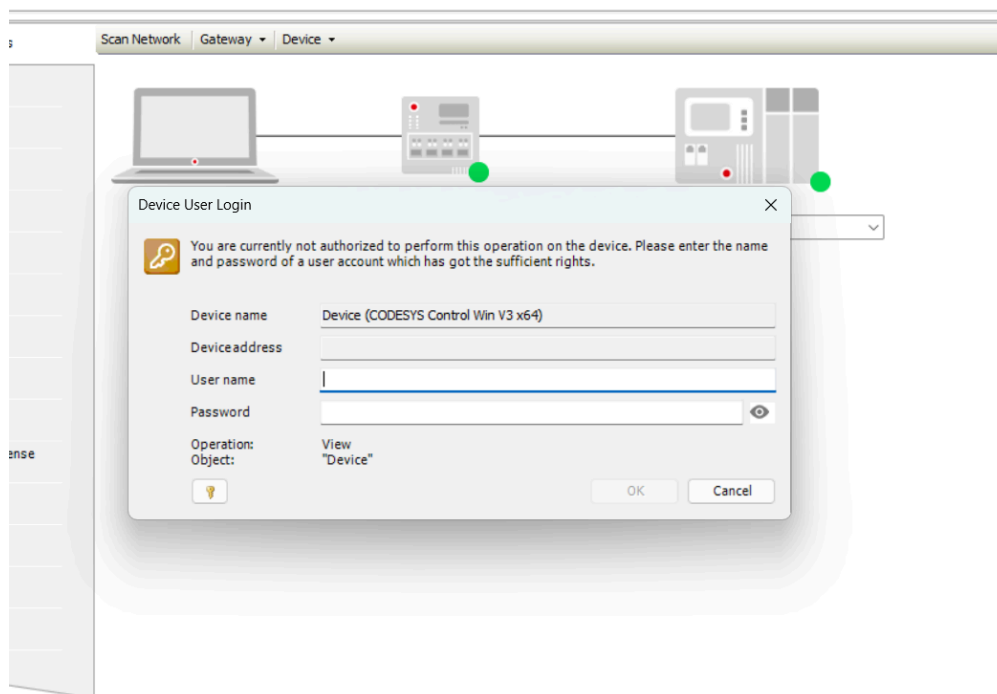


Figure 25. CODESYS’s Device User Login prompt, requesting credentials with sufficient rights to view the Device object.

Once logged in, the Devices tree shows the device as connected and the application as running (both highlighted green), confirming the PLC program is actually executing and not just downloaded.

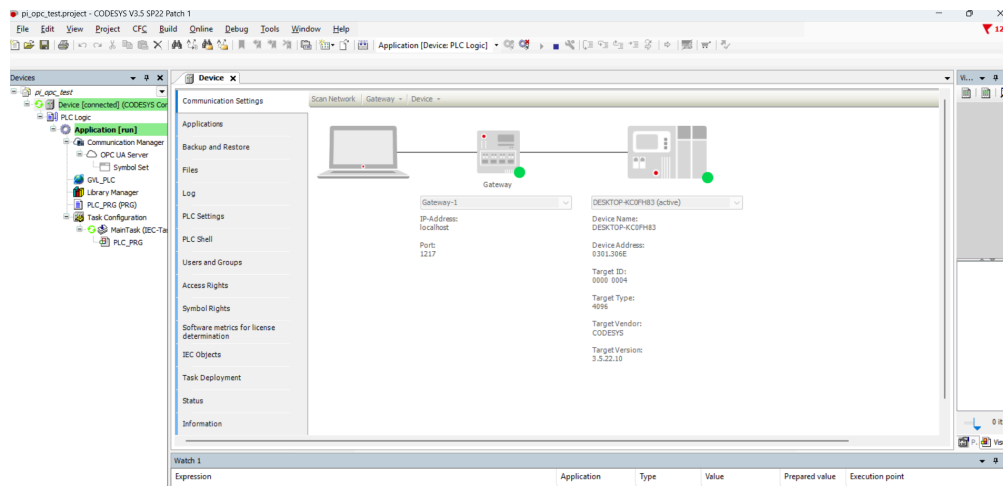


Figure 26. CODESYS IDE online: “Device [connected]” and “Application [run]” both shown in green in the Devices tree.

The full CODESYS engineering-environment procedure — installing the runtime, configuring the project, the complete login sequence — is documented in the predecessor project’s handover report and is not reproduced here.

11.8 Symbol Publishing and Device Security Settings

If the target is a CODESYS soft-PLC, its Symbol Configuration needs to be published for the connection to work — the simulator’s default automatic tag-addressing mode locates tags by browsing the server’s exposed symbol set, and an unpublished symbol set means none of the compressor tags are visible over OPC UA even though the application logic is running correctly. A Download to the PLC regenerates its address space and invalidates any node references the client cached from before the Download; reconnecting (or restarting the simulator) afterward makes it re-discover tags rather than hold stale references. This is a general OPC UA characteristic, not specific to CODESYS, and applies the same way to a physical panel’s OPC UA server after a program change.

The device tree of a CODESYS Control Win project (Device — PLC Logic — Application — Communication Manager — OPC UA Server) is where the Symbol Publishing object controlling which global variables are exposed as OPC UA nodes lives, alongside the Communication Settings tab used earlier to confirm the gateway and target device.

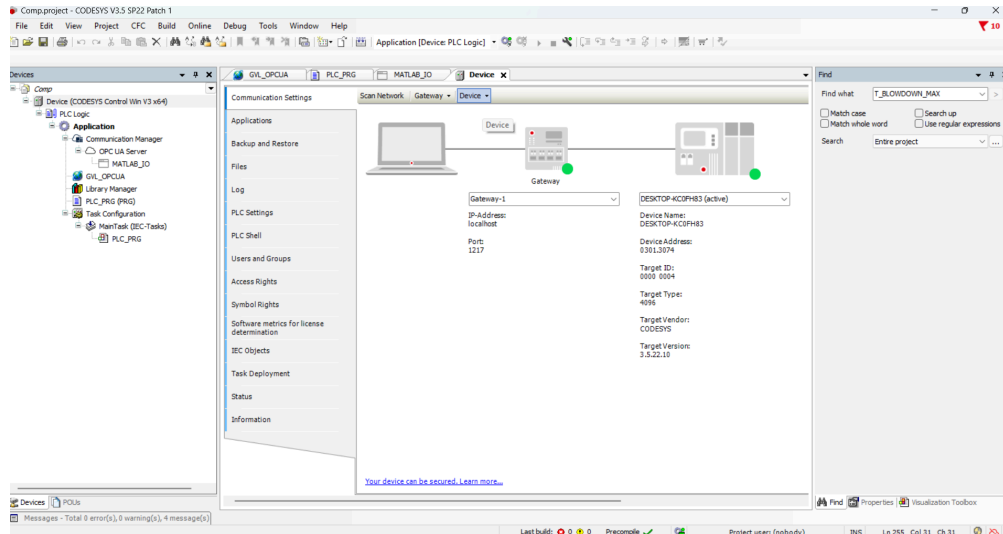


Figure 27. CODESYS Devices tree for a project with symbol publishing configured, and the Communication Settings tab showing the active target device.

Two device-level settings gate every OPC UA client — this application included — and both default to values that **block** connections, so both must be changed explicitly on a fresh CODESYS install:

1. **Device Security Settings** (Device menu — Security Settings...): `CommunicationMode` must be set to **ALL** (the default restricts which channels are permitted and blocks external client connections) and **Activation** must be set to **ACTIVATED** (this also defaults to a connection-blocking state).

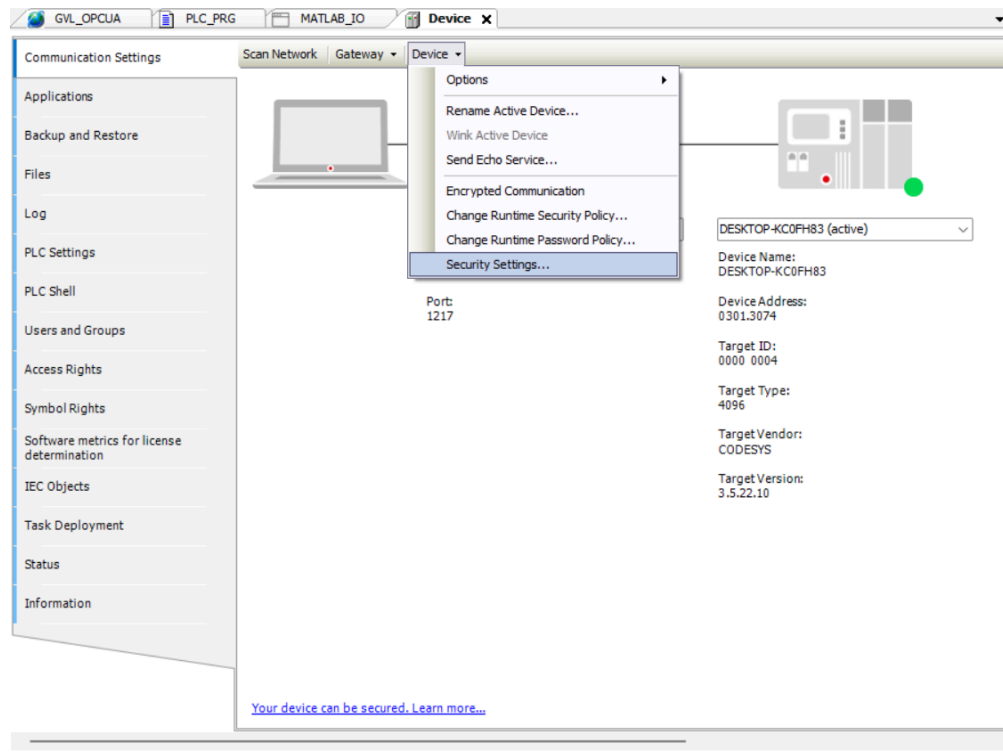


Figure 28. Menu location for Device Security Settings, under the Device dropdown.

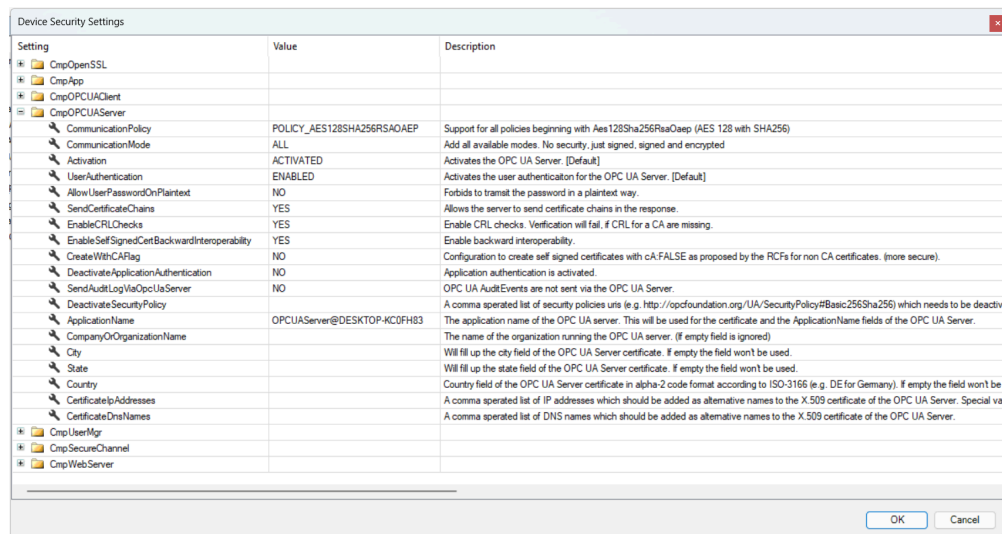


Figure 29. Device Security Settings dialog with the two required non-default values: *CommunicationMode* = *ALL*, *Activation* = *ACTIVATED*, under *CmpOPCUAServer*.

1. **Runtime Security Policy** (Device menu — Change Runtime Security Policy...): encryption set to **Optional**, **Code Signing** = **All**, user management set to **Optional**, and **Allow anonymous login** enabled — anonymous login lets a registered OPC UA client connect without credentials even if user management is otherwise on.

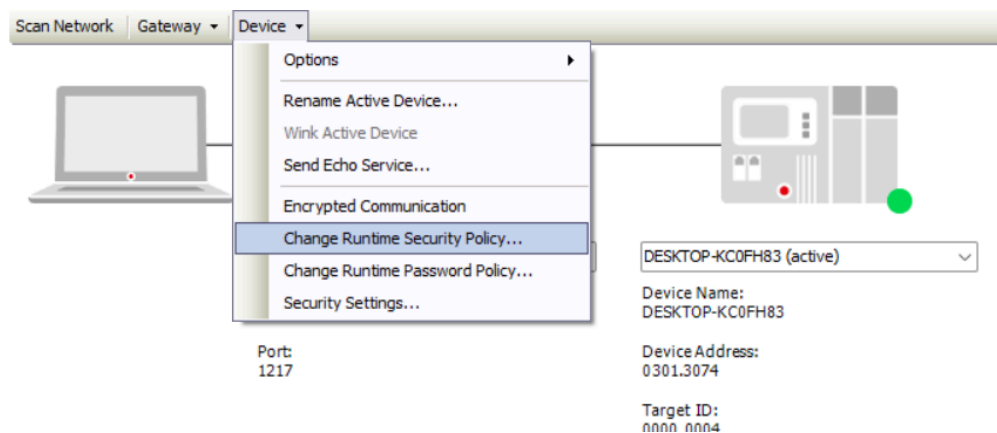


Figure 30. Menu location for the Runtime Security Policy, under the Device dropdown.

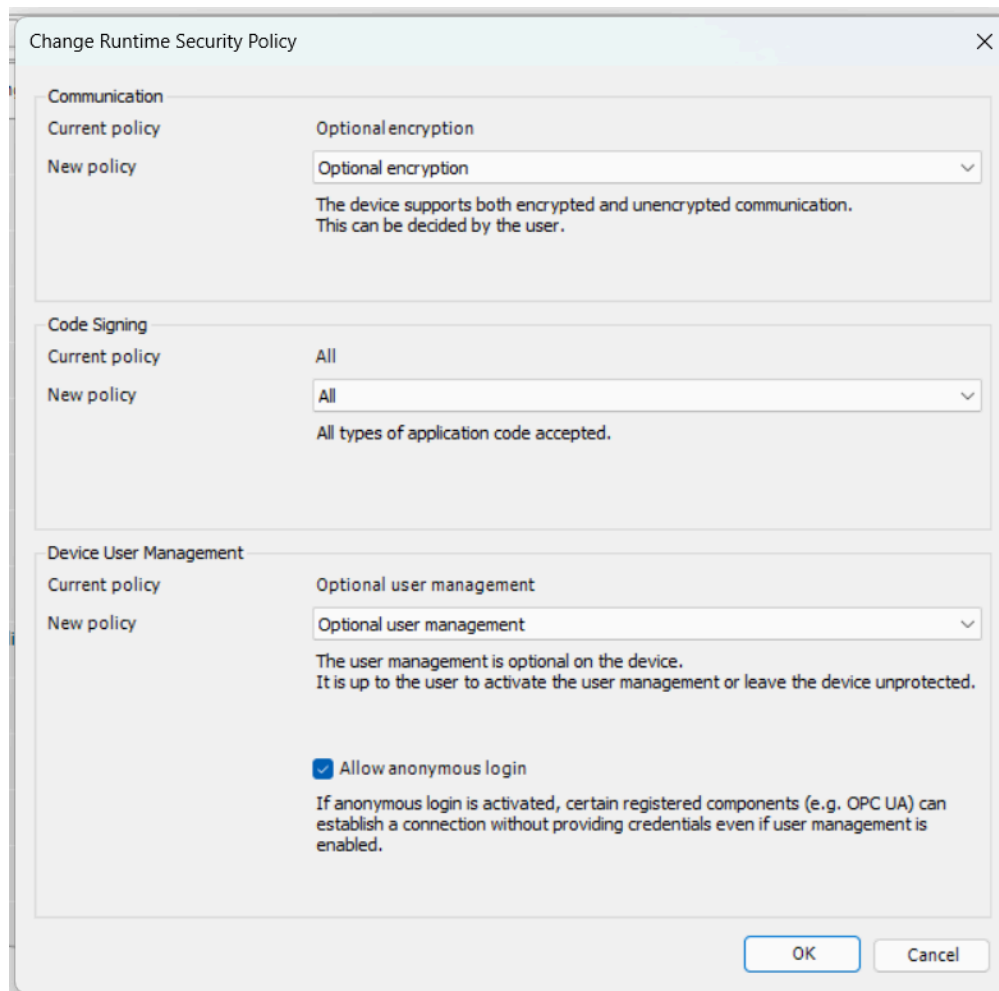


Figure 31. Runtime Security Policy dialog: Optional encryption, Code Signing = All, Optional user management, Allow anonymous login enabled.

Both device-level settings default to values that block connections, so the runtime can appear to start normally while silently refusing every incoming OPC UA client, including this application. If the app cannot connect and `app.log` shows no response from an endpoint that is otherwise reachable, these two dialogs are the first place to check.

11.9 Verifying the Link: Forcing a Tag from the PLC Side

The clearest confirmation that the OPC UA link is actually live — not just “connected” in name — is changing a value in CODESYS and watching it move on the simulator’s HMI. `CMD_4005` (CAT engine start command) in the `GVL_PLC` global variable list is a convenient one to force, since it drives a visible indicator on the P&ID.

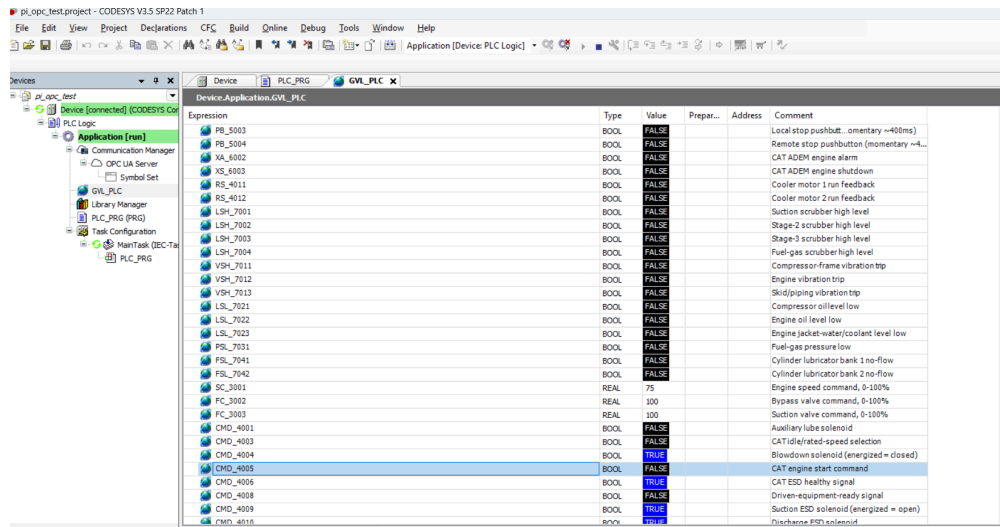


Figure 32. GVL_PLC in the CODESYS IDE with CMD_4005 (“CAT engine start command”) selected, currently FALSE.

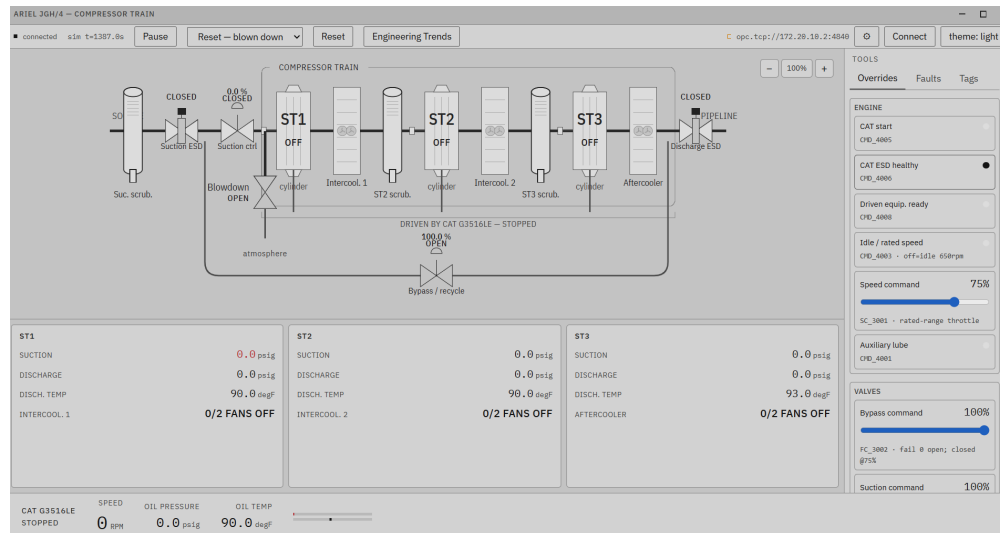


Figure 33. Simulator HMI immediately before the force: CMD_4005 is still FALSE on the PLC side, and the header shows the engine STOPPED.

Forcing CMD_4005 to TRUE in CODESYS (right-click the value → Force Value, or set it directly in a running application) writes the change to the PLC’s own memory — from there the simulator picks it up over the OPC UA link the same way it would from any value the connected PLC program writes to that tag.

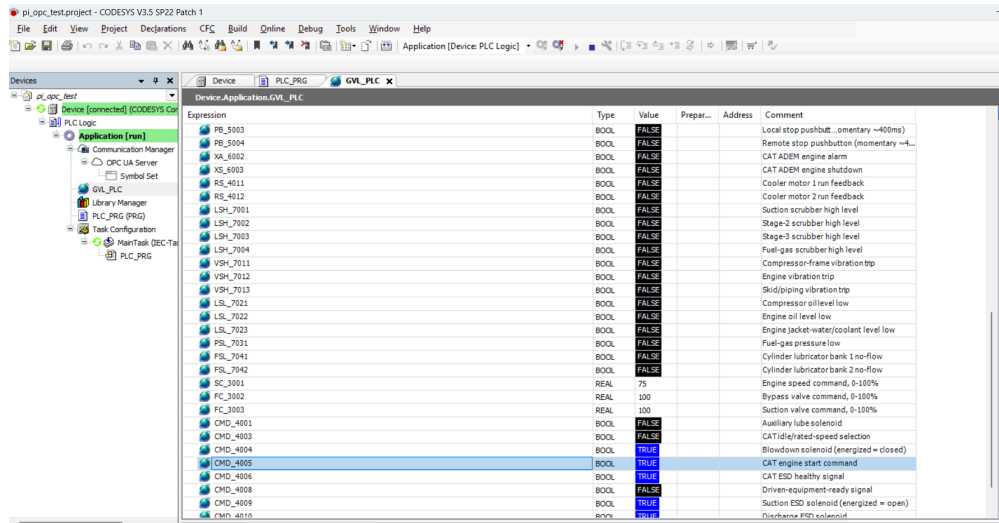


Figure 34. CMD_4005 forced to TRUE in the CODESYS GVL_PLC watch view.

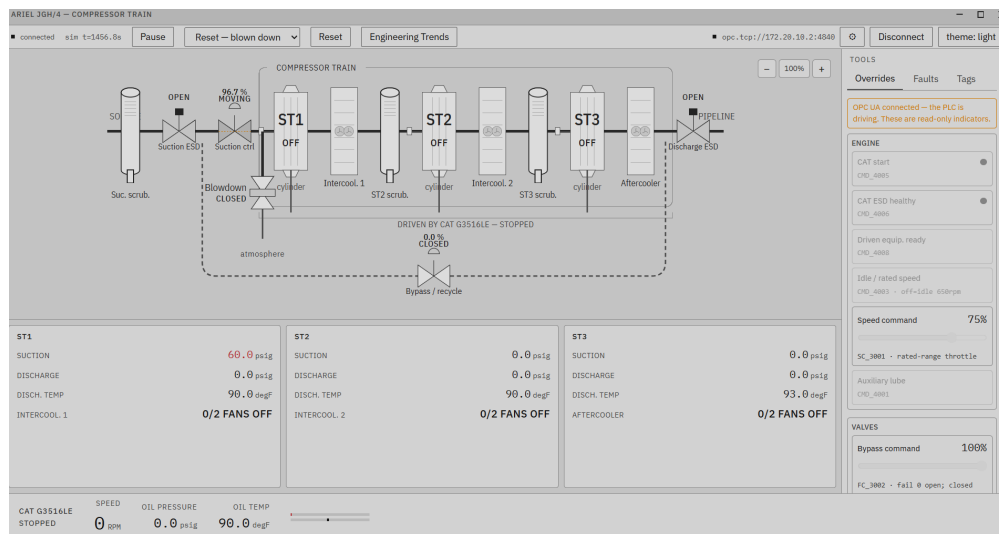


Figure 35. The simulator HMI shortly after the force: CMD_4005 is now TRUE, sourced entirely from the PLC-side change — nothing was clicked in the simulator itself. The engine still reports STOPPED here, since the start command alone doesn't spin the engine without the rest of the sequence (ESD healthy, driven-equipment ready, idle/rated-speed selection) also being asserted.

This is the same mechanism used to test control logic: whatever a real PLC program forces or computes onto its outputs is what the simulator reacts to, tag by tag.

12 PLC Development Roadmap

A recommended order of work for building the PLC application against this simulator, from first communications to a fully tested sequence. Nothing here is enforced by the simulator — it is a suggested path through the work, not a requirement. “Tested” here means exercised against this simulator’s implemented model, not validated against the physical compressor (Section 1).

12.1 Stage 1 — Establish Communications

1. Install the simulator (Section 10).
2. Start (or configure) the CODESYS runtime and confirm it is actually running the application, not just downloaded (Section 11).
3. Publish the CODESYS Symbol Configuration so the compressor tags are visible over OPC UA (Section 11, Symbol Publishing and Device Security Settings).
4. Connect the simulator to the runtime and confirm “Connected” in the header (Section 11).
5. Force one low-impact test command (e.g. `CMD_4005`) and observe the corresponding response in the simulator (Section 11.9).
6. Confirm simulator feedback (a measurement tag) reaches the PLC side, closing the loop in both directions.

This is expanded into a repeatable procedure in Section 17, “First PLC Commissioning Test” — treat that as Stage 1’s acceptance test.

12.2 Stage 2 — Create the I/O Mapping

Map every tag in Appendix A into the PLC project: commands (speed, valve, engine, cooler), process measurements (pressures, temperatures), limit switches, engine-running feedback, oil-pressure-healthy, cooler run feedback, the watchdog counter, and the always-live operator/ECU pushbuttons. Section 13 groups these by function rather than tag number, to make this mapping pass faster.

12.3 Stage 3 — Implement Equipment-Level Logic

Individual output logic for each piece of equipment, independent of sequencing: valve open/close commands, auxiliary lube on/off, engine start/stop, cooler motor commands, speed reference, recycle/bypass positioning. Section 13’s “What Should Happen When I Issue a Command?” section states the expected simulator response for each.

12.4 Stage 4 — Implement Permissives

RECOMMENDED SOFTWARE STRUCTURE, NOT SIMULATOR-ENFORCED.

The simulator does not require, check, or enforce any permissive. Everything in this stage is a recommended engineering practice, not a behaviour the simulator will validate for you.

Typical permissives to consider: communications healthy (watchdog, Section 13); ESD healthy; driven-equipment-ready; correct valve state before a transition; lubrication pressure healthy; and whatever process conditions the project requires before a step proceeds. Do not invent thresholds or timers the project has not specified — mark them **project-specific / to be defined** rather than guessing a realistic-looking number.

12.5 Stage 5 — Implement the Sequence

Develop the startup/run/stop/shutdown state machine. Section 15’s reference sequence and Section 13’s sequence/state reference table are a starting point, not a specification — see “The Predecessor Sequence” note in Section 13.

12.6 Stage 6 — Add Alarm and Trip Evaluation

Evaluate alarms and trips against the simulator’s measurement and fault-affected tags. The simulator never evaluates alarms itself (Section 1); anything coloured on the P&ID or drawn as a dashed line on a trend is a display convenience only, not a live alarm the PLC can read.

12.7 Stage 7 — Fault-Test the PLC

Work through the Faults tab (Section 16) systematically, confirming each fault produces its documented tag-level effect and that the PLC logic under test reacts as intended.

12.8 Stage 8 — Validate Stop, Shutdown, and Recovery Behaviour

Exercise normal stop, unconditional/emergency shutdown, a failed start, lube failure, cooling failure, blocked discharge, instrument faults, an OPC UA link loss, and restart/reset behaviour after each. Section 17’s Acceptance Test Checklist is organised around exactly these cases.

12.9 Recommended PLC Software Architecture

RECOMMENDED ENGINEERING ORGANISATION, NOT A SIMULATOR REQUIREMENT. The decomposition below is one reasonable way to organise a PLC application against this interface. The simulator has no opinion on program structure, POU naming, or coding style — only on the tag values it reads and writes.

```
PLC Application
|
+--- IO Mapping
|
+--- Communication Supervision (watchdog / link health)
|
+--- Equipment Control
|   +--- Engine
|   +--- Auxiliary Lube
|   +--- Suction ESD
|   +--- Discharge ESD
|   +--- Blowdown
|   +--- Suction Control Valve
|   +--- Bypass Valve
|   +--- Coolers
|
+--- Permissives
|
+--- Compressor Sequence
|
+--- Alarm / Trip Evaluation
|
+--- Shutdown Management
```

|

+-+ Diagnostics / HMI

If IEC 61131-3 organisation is useful: a GVL for the OPC UA-mapped tags, one function block per equipment item, an enum-driven state machine for the sequence, and structures for grouping related permissive/alarm bits are all reasonable choices — none of this is dictated by the simulator, and an existing CODESYS project's established style should take precedence over any suggestion here.

13 PLC / Simulator Interface

This chapter is the practical reference for wiring up and coding against the simulator: which tags matter for which subsystem, which ones have counter-intuitive polarity, what to expect in the simulator after issuing a command, what “healthy” looks like at the design point, and how a normal stop differs from an unconditional shutdown. The full, alphabetically-complete tag list with ranges and fail values is Appendix A; this chapter groups the same tags by function instead.

13.1 Command / Feedback Cheat Sheet

13.1.1 Engine

Tag	Direction	Meaning
SC_3001	PLC → Sim	Speed command, 0-100%
CMD_4003	PLC → Sim	Idle / rated speed select
CMD_4005	PLC → Sim	CAT engine start command
CMD_4006	PLC → Sim	CAT ESD, energised = healthy
CMD_4008	PLC → Sim	Driven-equipment-ready
ST_1008	Sim → PLC	Engine speed, 0-2000 rpm
ST_2010	Sim → PLC	Engine running (> 300 rpm)
XA_6002/XS_6003	Sim → PLC	CAT ADEM engine alarm / shutdown status

13.1.2 Lubrication

Tag	Direction	Meaning
CMD_4001	PLC → Sim	Auxiliary lube (prelube pump) solenoid
PT_1005	Sim → PLC	Compressor oil pressure, 0-200 psig
PS_2009	Sim → PLC	Oil pressure healthy (> 10 psig)
PT_1007	Sim → PLC	Engine oil pressure, 0-150 psig

13.1.3 Process Valves

Tag	Direction	Meaning
FC_3002	PLC → Sim	Bypass valve command, 0-100%
FC_3003	PLC → Sim	Suction valve command, 0-100%
CMD_4004	PLC → Sim	Blowdown solenoid, energised = closed
CMD_4009	PLC → Sim	Suction ESD solenoid, energised = open
CMD_4010	PLC → Sim	Discharge ESD solenoid
ZS_2001-2008	Sim → PLC	Valve open/closed limit switches (blowdown, bypass, suction ESD, discharge ESD)

13.1.4 Cooling

Tag	Direction	Meaning
CMD_4011/4012	PLC → Sim	Cooler motor 1 / 2 run command
RS_4011/4012	Sim → PLC	Cooler motor 1 / 2 run feedback
TT_2013	Sim → PLC	Aftercooler temperature
TT_2014	Sim → PLC	Engine jacket-water temperature

13.1.5 Process Measurements

Tag	Meaning
PT_1001	Suction pressure
PT_1002/1003/1004/1006	Stage 1 / 2 / 3 / final discharge pressure
TT_2004-2007	Cylinder 1-4 discharge temperature
TT_2009-2012	Packing temperatures 1-4

13.1.6 Communication

Tag	Meaning
WD_6001	Heartbeat counter, increments every 500 ms — see “Watchdog and Communication Supervision” below

13.2 Important: Command Polarity

IMPORTANT. Do not assume TRUE means “open” or “on” for every tag. Command meaning depends on the specific device and its fail-safe philosophy, and getting this backwards is one of the easiest ways for a new PLC engineer to write logic that looks correct but drives the package the wrong way.

Tag	Energised / TRUE means	Loss of command / de-ener- gised means
CMD_4004 (blowdown solenoid)	Blowdown valve closed	Blowdown valve opens — fails open
CMD_4009 (suction ESD)	Valve open	Valve closes — fails closed
CMD_4010 (discharge ESD)	Valve open (by convention, matching the suction ESD)	Valve closes — fails closed
FC_3002 (bypass command)	Higher % = more closed	0% = valve fully open — fails open
FC_3003 (suction valve command)	Higher % = more open	0% = valve fully closed — fails closed

The pattern to remember: every ESD-type and blowdown element is wired so that **losing signal fails toward the safe, vented, isolated state** — ESDs closed, blowdown open. FC_3002 (bypass) follows

the same fail-open philosophy but is commanded as an analog percentage rather than a discrete, so “0% commanded” and “signal lost” produce the same fully-open result. This mapping is restated where it matters operationally in Section 11, “What Happens on Link Loss,” and is not re-derived each time it recurs below.

13.3 What Should Happen When I Issue a Command?

13.3.1 Auxiliary Lube On (CMD_4001)

Expect: Compressor oil pressure (PT_1005) builds toward the 55 psig prelube target (Section 5, eq. 15). PS_2009 (oil pressure healthy) changes state once the modelled pressure crosses its healthy threshold. The prelube ramp has a real time constant — it does not jump to target instantly.

13.3.2 Engine Start (CMD_4005, subject to its prerequisite commands)

Expect: Speed (ST_1008) progresses through cranking (0→200 rpm at +70 rpm/s), then acceleration toward the idle reference (650 rpm) at +50 rpm/s, then tracks the rated-speed reference once CMD_4003 and SC_3001 move it into the 850-1000 rpm range. See Section 3 and Section 15 for the full permissive chain this depends on.

13.3.3 Close Bypass / Load the Compressor (FC_3002 toward 100%)

Expect: More compressor flow is routed toward discharge/process rather than recycling to suction. Discharge pressures (PT_1002/1003/1004/1006) build stage over stage; the suction/discharge pressure relationship shifts accordingly.

13.3.4 Open Bypass / Unload (FC_3002 toward 0%)

Expect: Discharge gas recycles toward suction. Suction pressure (PT_1001) tends to rise; discharge pressures tend to fall. The engine keeps running throughout — this is the mechanism for unloading without a full stop.

13.3.5 Blowdown (CMD_4004 de-energised, or its permissives never asserted)

Expect: The **suction** volume vents toward atmosphere. Discharge pressure does **not** necessarily collapse at the same time — discharge-side mass can only leave through the bypass valve into suction and then out the vent, so with the bypass closed, discharge pressure can remain high even while suction is fully vented. This is not a model defect; it is the mass-accumulation asymmetry explained in full in Section 4 and is the single most common point of confusion for a first-time reader of this simulator’s behaviour.

13.4 Normal Operating Expectations (“What Good Looks Like”)

The table below restates the design-point values from Section 6 as a fast sanity check while coding or commissioning. These are simulator design-point/reference values for a **generic** package, not a guarantee of any specific real machine’s performance (Section 7).

Quantity	Design-point value
Suction pressure	29.8 psig
Stage 1 discharge pressure	117.3 psig
Stage 2 discharge pressure	377.2 psig
Final discharge pressure	1149.0 psig
Engine speed (rated)	850-1000 rpm
Compressor mass flow	0.945 kg/s

Quantity	Design-point value
Cylinder 1 / 2 discharge temperature	245.4 / 251.7 °F

13.5 Normal Stop vs. Unconditional Shutdown (USD)

Getting this distinction right is central to sequence design, because the simulator produces visibly different pressure outcomes depending on command **ordering** — and enforces none of that ordering itself.

	Normal stop	Unconditional shutdown / USD
Intent	PLC deliberately unloads, isolates, and depressurises in a controlled order	Shutdown actions occur immediately, per the PLC’s own shutdown philosophy
Simulator consequence	If the PLC opens bypass and closes the ESDs before opening blowdown, discharge-side mass has already had a path back to suction, so final pressures settle lower and more evenly	If blowdown opens immediately without that ordering, the suction volume vents fast while discharge pressure can remain high — trapped behind a closed or partially-open bypass

IMPORTANT. The simulator intentionally does not enforce the “correct” ordering of ESD closure, bypass opening, and blowdown opening. Whether a given shutdown sequence leaves the package in a sensible final pressure state is exactly what the PLC is being tested on – see Section 4 for the underlying mass-accumulation mechanism and Section 15 for the reference walkthrough.

13.6 Watchdog and Communication Supervision

WD_6001 increments every 500 ms while the simulator is running and the link is healthy (Section 11). A PLC can supervise this the same way it would supervise any heartbeat: confirm the value is actually changing, not merely present. A stopped or stale WD_6001 is evidence of a simulator-side communication failure. What the PLC should **do** about that — trip, alarm, hold last state, or something else — is a project-specific design decision this document does not prescribe. Section 11, “What Happens on Link Loss,” documents the simulator’s own fail-value behaviour on its side of a dropped link, which is the complementary half of this picture.

13.7 PLC Sequence / State Reference Table

REFERENCE ONLY --- NOT ENFORCED BY SIMULATOR. The predecessor rig’s reference sequence names the phases below (Section 15, Figure 36). The simulator implements none of these states, timers, or transitions — it only responds to whatever commands are asserted at whatever time they are asserted. Any timer, threshold, or permissive not already stated elsewhere in this report is marked **project-specific** below rather than invented.

Phase	Likely command	PLC	Process response to re-serve	Feedback to monitor	Possible failure mode
READY	Permissives asserted, no equipment commanded yet	as-	None — package at rest	ESD/blowdown limit switches, no active faults	Project-specific / to be defined
PURGE	Not modelled by this simulator	by	Not modelled — no purge-gas system in this simulator	Not applicable	Not enforced by simulator
BLOWDOWN	CMD_4004 de-energised (or never energised)	de-ener-	Suction volume vents toward atmosphere (Section 4); discharge pressure may remain elevated depending on bypass state	PT_1001 falling, blowdown limit switch	Project-specific timer for “fully vented”
PRELUBE	CMD_4001 energised	ener-	PT_1005 ramps toward the 55 psig prelube target (Section 5, eq. 15)	PT_1005, PS_2009	Low lube pressure fault, slow lube build fault (Section 16)
CAT_START	CMD_4005 energised, with CMD_4006/ CMD_4008 asserted	en-	ST_1008 ramps through cranking then acceleration toward idle	ST_1008, ST_2010	Engine-fails-to-start fault clamps speed at 550 rpm (Section 16)
WARMUP	Engine held at idle (650 rpm) before loading	held at	Speed steady near idle; oil/cylinder temperatures continue settling	ST_1008, TT_2001	Project-specific warmup duration — not defined by simulator
LOADING	CMD_4003 selected, SC_3001 raised, FC_3003 opened, FC_3002 closed toward the load position	selected,	Stage pressures build per Section 5 eq. 2-3; flow ramps with speed and valve position	PT_1001-1006, ST_1008	Blocked discharge, valve-stuck faults (Section 16)
RUNNING	Steady rated-speed commands maintained	rated-	Design-point values approached (see “Normal Op-	All process measurements	Any Faults-tab condition (Section 16)

Phase	Likely command	PLC	Process response to observe	Feedback monitor	to	Possible failure mode
			erating Expectations” above)			
UNLOADING	FC_3002	opened toward suction	Discharge pressures fall, suction rises, engine keeps running (see “Open Bypass / Unload” above)	PT_1001-1006		Project-specific unload criteria
COOLDOWN	Engine run permissives de-asserted; speed coasts down ($\tau = 6$ s)		ST_1008 decays toward zero per the coastdown time constant	ST_1008		Project-specific cooldown duration before next step
NORMAL_STOP POSTLUBE	CMD_4001 may remain energised after engine stop		PT_1005 holds at the prelube target if lube stays commanded	PT_1005		Project-specific postlube duration
SHUTDOWN / USD	ESDs de-energised, blowdown de-energised immediately, without waiting for the normal-stop ordering		Suction vents fast; discharge pressure state depends entirely on bypass position at the moment of trip (see “Normal Stop vs. USD” above)	PT_1001-1006, ESD/blowdown limit switches		Command ordering error — trapped high discharge pressure is a symptom, not a simulator fault

14 The Operator Interface

The right-hand tool dock exposes three tabs, available whether or not a PLC is connected, plus a full-screen Engineering Trends view opened from the header:

Overrides	Manual control panel; read-only PLC command echo once a PLC is connected.
Faults	Fault injection panel for testing PLC fault handling (@sec-faults).
Tags	Full flat table of every OPC UA tag and its current value.
Engineering Trends	Full-screen strip charts of key tags, opened via a header button.

The app also supports a dark theme (toggled via the “theme” button in the header), shown below with the unit running at rated speed under PLC control.

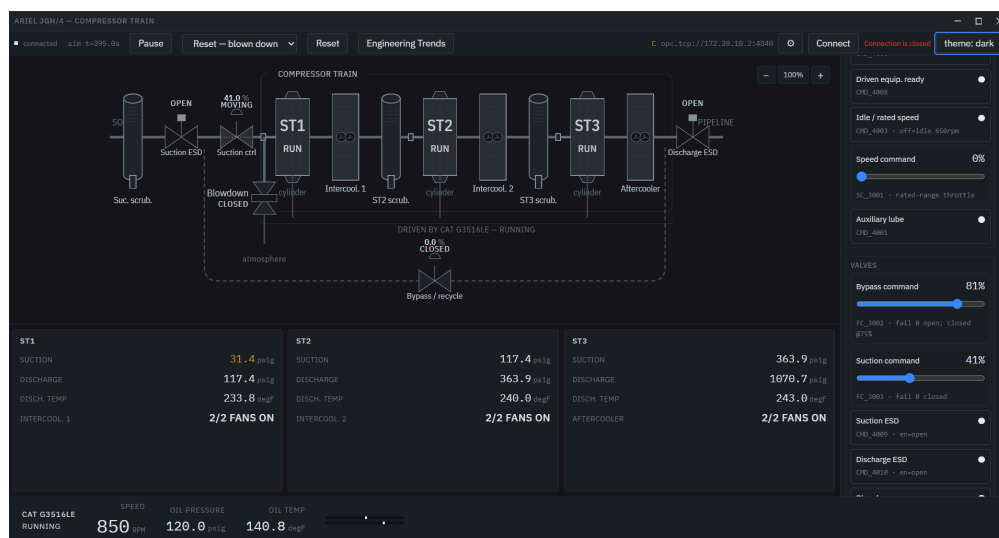


Figure 36. Compressor Simulator in dark theme, unit running under PLC control: all three stages RUN, coolers ON, discharge pressures building stage over stage (ST1 117 psig, ST2 364 psig, ST3 1071 psig), engine at 850 rpm.

14.1 Overrides — Status Feedback and Always-Live Inputs

Beyond the Engine and Valves groups shown earlier (Section 11), the Overrides dock also has a read-only status block reflecting sensor/feedback tags the PLC does not command, and a separate group of operator/ECU pushbuttons that stay live and editable even while a PLC is connected.

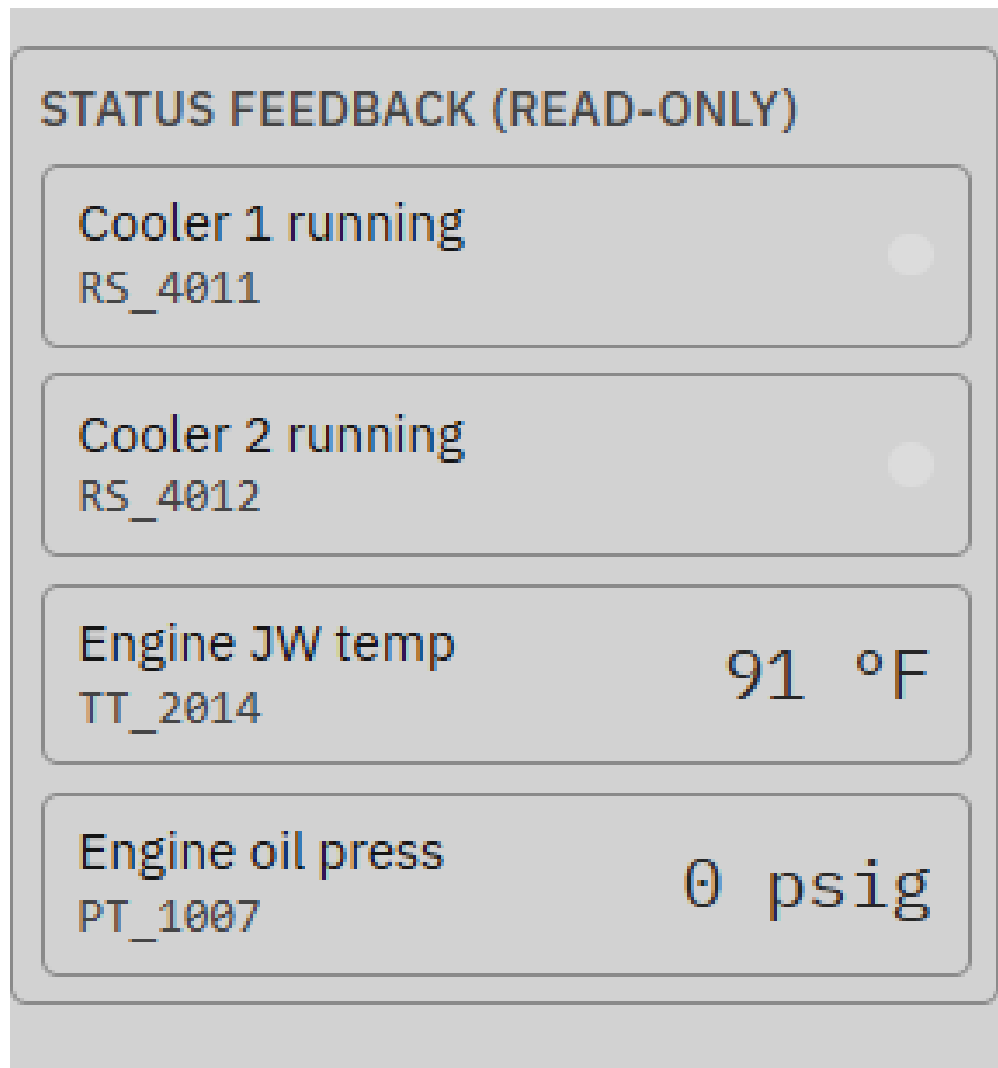


Figure 37. Status feedback (read-only): cooler run feedback (RS_4011, RS_4012), engine jacket-water temperature (TT_2014), and engine oil pressure (PT_1007).

The Tier 1 operator inputs — unit shutdown, local/remote stop, remote ESD, CAT alarm/fail-SD — stay live regardless of PLC connection state. This reflects the implemented design assumption that these inputs are hardwired straight into the PLC’s I/O rather than routed through an upstream system, consistent with the reference project this simulator was adapted from.

OPERATOR / ECU INPUTS (ALWAYS LIVE)

Unit shutdown
PB_5001

Local stop
PB_5003

Remote stop
PB_5004

Remote ESD
ESD_5002

CAT alarm
XA_6002

CAT fail SD
XS_6003

Figure 38. Operator / ECU inputs (always live): unit shutdown (PB_5001), local stop (PB_5003), remote stop (PB_5004), remote ESD (ESD_5002), CAT alarm (XA_6002), CAT fail SD (XS_6003).

14.2 Faults

The Faults tab is grouped by subsystem, with a single **Clear all faults** control at the top. Faults are applied instantly and independently of Reset — clearing one is a separate, explicit action. The complete fault table is Section 16.

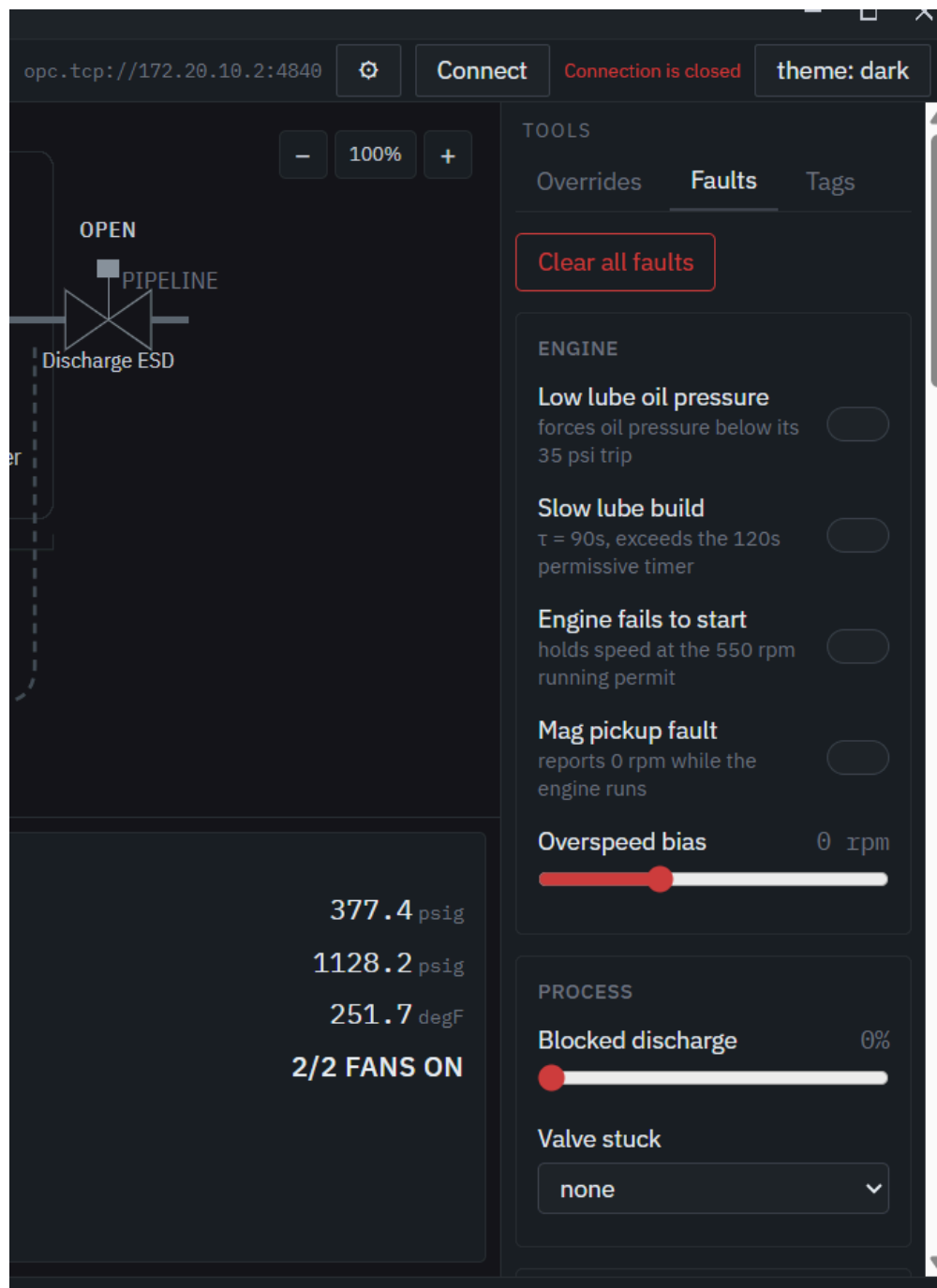


Figure 39. Engine faults — low lube oil pressure, slow lube build, engine fails to start, mag pickup (speed signal) fault, and an overspeed bias slider — followed by the start of the Process group (blocked discharge, valve stuck).

CYLINDER TEMPERATURE BIAS

Cylinder 1

0°F

Cylinder 2

0°F

Cylinder 3

0°F

Cylinder 4

0°F

COOLER MOTORS

Cooler motor 1 trip

Cooler motor 2 trip

SCRUBBER LEVELS (TIER 2)

Suction scrubber

LSH_7001 · level high

ST2 scrubber

LSH_7002 · level high

ST3 scrubber

LSH_7003 · level high

Fuel gas scrubber

LSH_7004 · level high

Figure 40. Cylinder temperature bias per cylinder (1-4), cooler motor trip switches, and Tier 2 scrubber-level switches (suction, ST2, ST3, fuel-gas).

VIBRATION (TIER 2)
Compressor frame
VSH_7011 · vibration trip ☐
Engine
VSH_7012 · vibration trip ☐
Skid / piping
VSH_7013 · vibration trip ☐

OIL / JW LEVELS (TIER 2)
Compressor oil
LSL_7021 · level low ☐
Engine oil
LSL_7022 · level low ☐
Engine JW / coolant
LSL_7023 · level low ☐
Fuel gas pressure low
PSL_7031 ☐

LUBRICATOR NO-FLOW (TIER 2)
Bank 1
FSL_7041 ☐
Bank 2
FSL_7042 ☐

Figure 41. Further Tier 2 discrete faults: vibration trips (compressor frame, engine, skid/piping), oil/jacket-water low-level switches, fuel-gas pressure low, and cylinder lubricator no-flow (banks 1-2).

SIGNAL FREEZE / INVALID		
tag	freeze	invalid
PT_1001	☐	☐
PT_1002	☐	☐
PT_1003	☐	☐
PT_1004	☐	☐
PT_1005	☐	☐
PT_1006	☐	☐
ST_1008	☐	☐
TT_2001	☐	☐
TT_2004	☐	☐
TT_2005	☐	☐
TT_2006	☐	☐
TT_2007	☐	☐
TT_2013	☐	☐
TT_2009	☐	☐
TT_2010	☐	☐
TT_2011	☐	☐
TT_2012	☐	☐

LINK

Link drop
suspends OPC UA writes,
exercises the watchdog

☐

INSTRUMENTATION (SECTION 7
STEP 8)

Signal lag
0.3-2s first-order lag per tag

☐

Signal noise
Gaussian noise per tag

☐

Figure 42. Signal freeze/invalid controls for every analog tag, a Link drop switch (suspends OPC UA writes to exercise the PLC's own watchdog), and the start of the Instrumentation group (signal lag, signal noise).

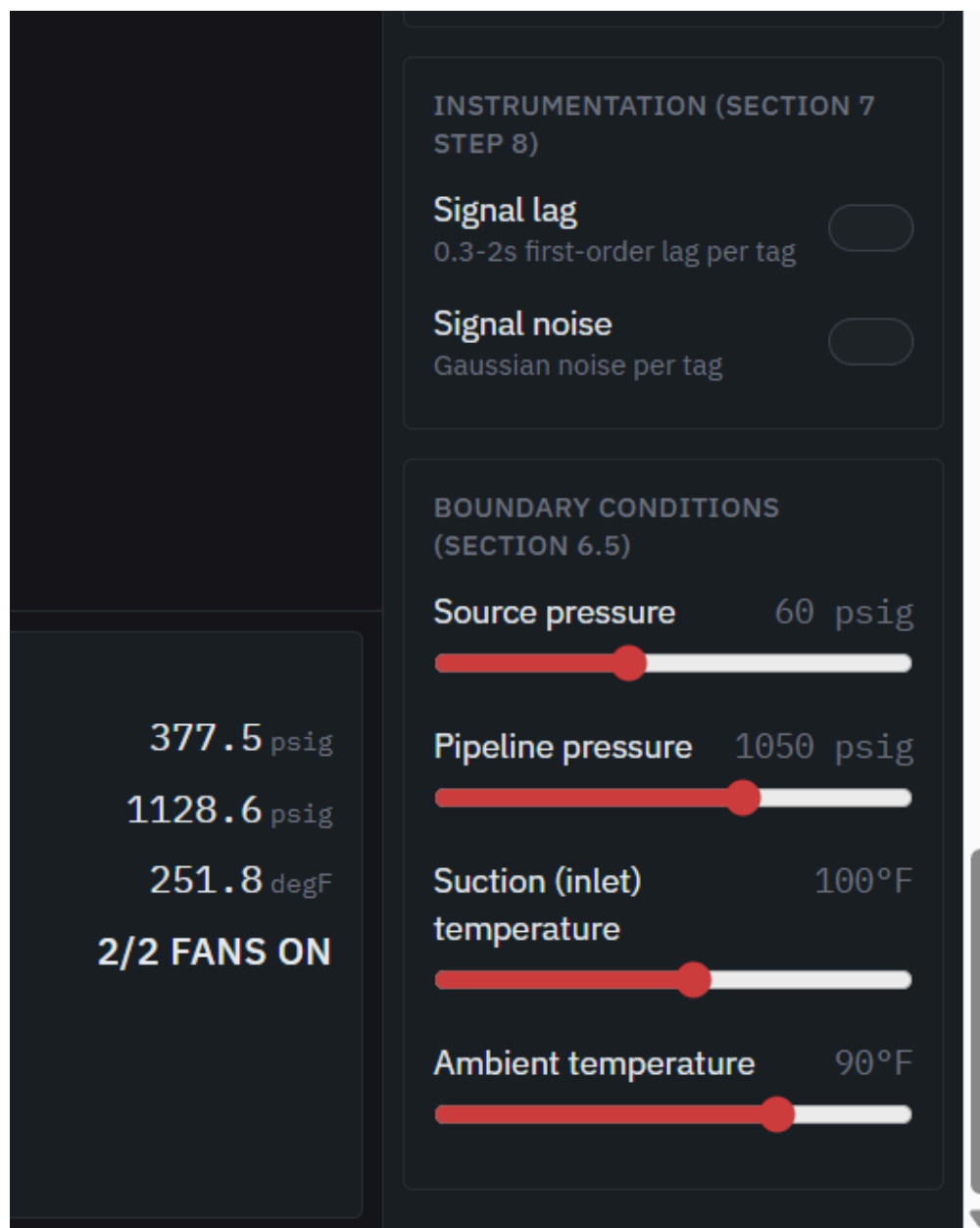
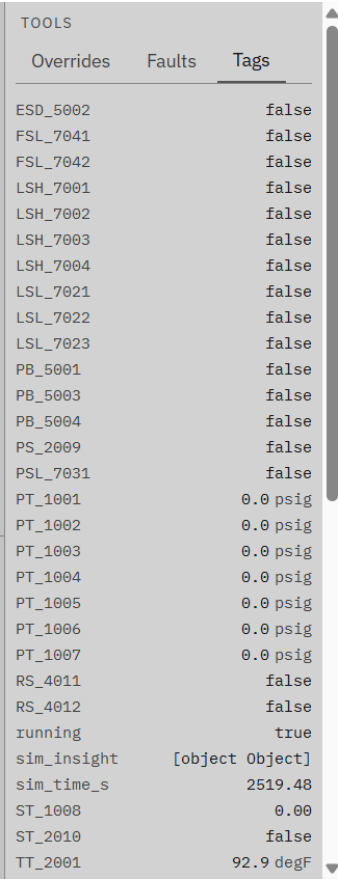


Figure 43. Instrumentation faults (signal lag, signal noise) and the Boundary Conditions panel — source pressure, pipeline pressure, suction temperature, ambient temperature — for changing the process's external conditions.

14.3 Tags

The Tags tab lists every OPC UA tag the simulator exposes and its live current value, useful for confirming exactly what a connected PLC is seeing without cross-referencing the P&ID or Overrides dock tag by tag.



TOOLS		
Overrides	Faults	Tags
		ESD_5002 false
		FSL_7041 false
		FSL_7042 false
		LSH_7001 false
		LSH_7002 false
		LSH_7003 false
		LSH_7004 false
		LSL_7021 false
		LSL_7022 false
		LSL_7023 false
		PB_5001 false
		PB_5003 false
		PB_5004 false
		PS_2009 false
		PSL_7031 false
		PT_1001 0.0 psig
		PT_1002 0.0 psig
		PT_1003 0.0 psig
		PT_1004 0.0 psig
		PT_1005 0.0 psig
		PT_1006 0.0 psig
		PT_1007 0.0 psig
		RS_4011 false
		RS_4012 false
		running true
	[object Object]	sim_insight
		sim_time_s 2519.48
		ST_1008 0.00
		ST_2010 false
		TT_2001 92.9 degF

Figure 44. Tags tab: a flat, alphabetically-ordered list of tag names and live values (discrete switches as true/false, analogs with engineering units).

14.4 Engineering Trends

The full-screen Engineering Trends view (opened via the header’s **Engineering Trends** button) plots up to twelve tags at once against a rolling buffer, sampled at 10 Hz on the simulation’s own time base. Each pen can be toggled independently; the dashed lines mark configured engineering limits for reference and, as the on-screen note states, are not simulator alarms — the simulator itself does not evaluate alarms (Section 1).

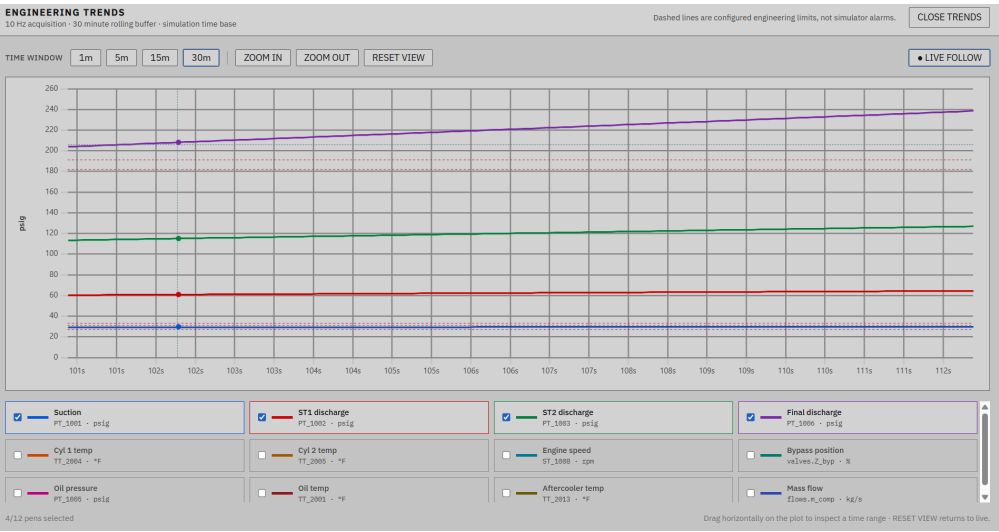


Figure 45. Engineering Trends with four pressure pens selected (suction, ST1 discharge, ST2 discharge, final discharge) over a 30-minute rolling window, ramping up during a startup.

The time-window buttons (1m/5m/15m/30m), zoom controls, and Live Follow toggle let a trend be either watched live or paused and dragged to inspect a specific transient.

15 Running the Simulator

This walkthrough uses the real tags described in Section 14 and Appendix A. Sequencing — deciding what order to assert these commands in, and how long to wait between them — is the job of the connected PLC; this application supplies no sequencer of its own (Section 1). Everything below describes what the operator (through a real PLC, or by hand via the Overrides dock when no PLC is connected) does, and what the simulator does in response. This differs from the earlier Simulink/CODESYS model, which bundled its own reference sequencer for standalone demonstration — without a PLC attached, this application will not run a startup sequence on its own.

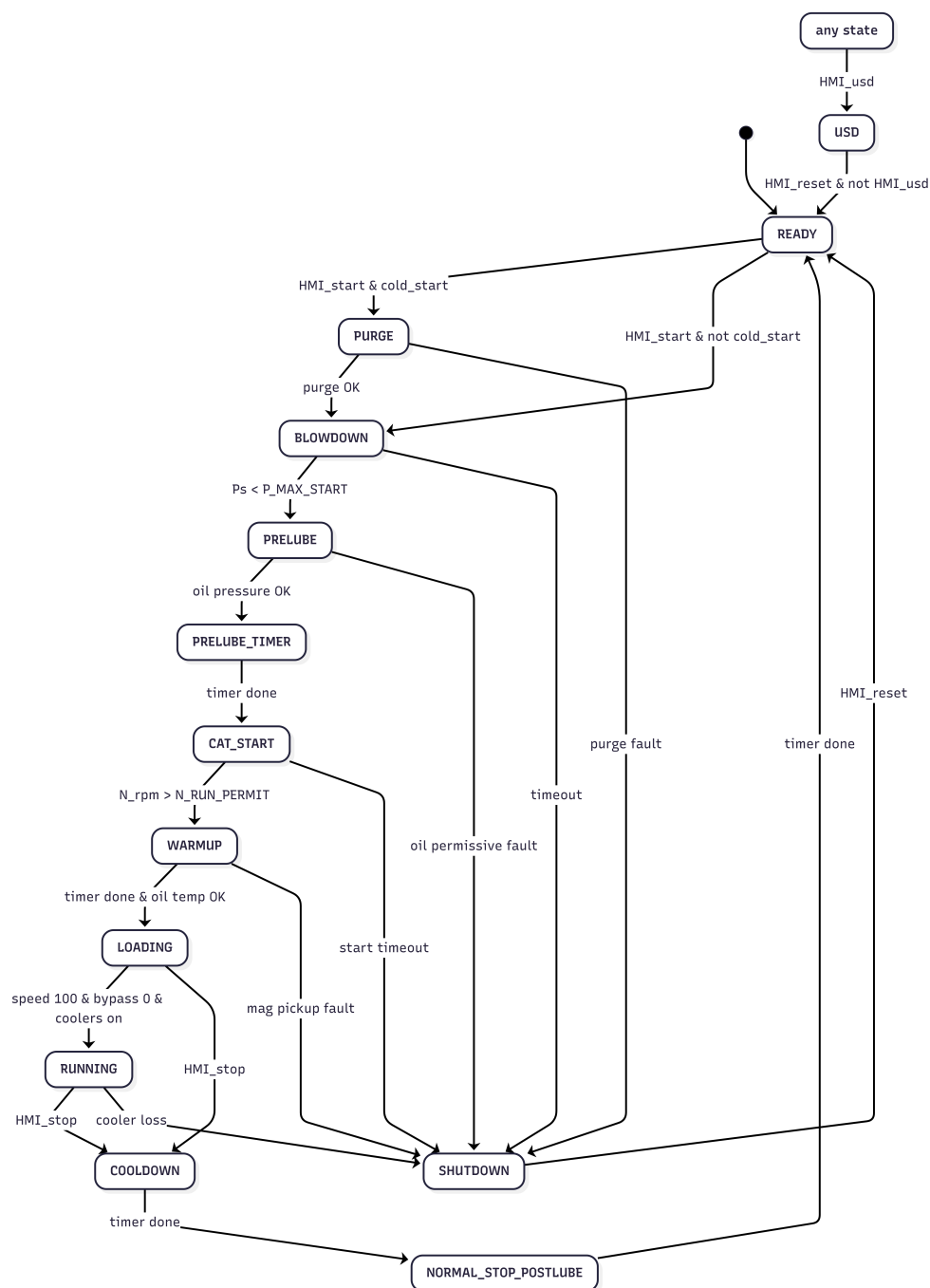


Figure 46. The predecessor Simulink/CODESYS rig’s reference sequencer state diagram (READY – PURGE – BLOWDOWN – PRELUBE – CAT_START – WARMUP – LOADING – RUNNING, with COOLDOWN / NORMAL_STOP_POSTLUBE and SHUTDOWN / USD exit paths). Shown for context only – this application implements none of these states, timers, or transitions itself; every step below is something a connected PLC’s own sequencer (which may follow this same state machine, or a different one) must decide and command.

1. **Launch and confirm the link.** Start the app (Section 10); connect to the target PLC (Section 11) and confirm the header shows “Connected.” With no PLC connected, the same sequence can be driven by hand from the Overrides dock for a demonstration, since every command tile there maps to the same tag the PLC would otherwise drive.

2. **Bring the package from stopped to running.** The PLC (or the operator, via Overrides) asserts the permissive chain: CMD_4009 (suction ESD, energised = open) and CMD_4010 (discharge ESD) open; CMD_4006 (CAT ESD healthy) and CMD_4008 (driven-equipment-ready) are asserted; CMD_4001 (auxiliary lube) commands the prelube pump, and PT_1005/PS_2009 (oil-pressure-healthy) is watched as it builds toward the 55 psig prelube target (Section 5, eq. 15). Once oil pressure is healthy, CMD_4005 (CAT engine start) is asserted: speed ramps through cranking (0→200 rpm at +70 rpm/s), then accelerates toward the idle reference (650 rpm) at +50 rpm/s. Selecting CMD_4003 (idle/rated) and driving SC_3001 up moves the reference into the 850–1000 rpm rated range; FC_3003 (suction valve) opens and FC_3002 (bypass) closes off the 100%-open failed-safe position to route flow toward the process.
3. **Observe the transients.** Watch PT_1001/1002/1003/1006 build stage over stage as the equal-ratio staging (Section 5, eq. 2–3) takes effect, and TT_2004/2005 rise toward their polytropic targets (eq. 11–12) as compression begins — this is the same running state shown in Figure 36. The oil and cooling lags (eq. 14) mean pressures and flow respond quickly (seconds) while oil and cylinder temperatures settle over tens of seconds to minutes, exactly as Section 4’s mass-accumulation framing predicts.
4. **Unload via bypass.** Opening FC_3002 (bypass) recycles discharge gas back to suction without stopping the engine — PT_1001 rises and PT_1002–1006 fall as flow diverts, the same recycle path used to test unload/reload logic without a full shutdown.
5. **Stop.** A normal stop de-asserts the run permissives; the ESDs close and speed coasts down ($\tau = 6$ s, Section 5 eq. 12 phase table). Only once the ESDs are closed does de-asserting CMD_4004 open the blowdown valve, venting the *suction* volume — discharge pressure holds nearly constant through this because it has no path to lose mass except back through the (now-closed or still-recycling) bypass valve, exactly the asymmetry described in Section 4. This is why a normal stop and an unconditional shutdown (blowdown opens immediately, without waiting for the ESDs) leave the package in different states, and why a PLC’s stop sequence needs to get that ordering right independent of anything this simulator enforces — it enforces nothing.

16 Fault Injection

Every fault below is local to the application and never exposed on OPC UA — the PLC under test cannot see or clear its own faults through the link, only through whatever effect the fault has on the process tags it reads. A single **Clear all faults** control resets every fault at once (a separate, explicit action from Reset).

PROJECT-SPECIFIC PASS/FAIL CRITERIA. The right-hand column below states the intended test objective for each fault — what the fault is meant to give the PLC's configured protection logic the opportunity to detect and act on. It is not a required PLC response defined by this document; the actual trip/alarm thresholds, timers, and actions remain project-specific / engineer-defined unless stated elsewhere in this report.

Fault	Operator control	Physical effect	Intended Test Objective / Project-Specific PLC Response
Low lube oil pressure	Toggle	Forces P_oil below its 35 psi trip threshold	Project-specific — intended to exercise the PLC's configured low-lube-oil-pressure protection
Slow lube build	Toggle	Sets the oil-pressure lag time constant to 900 s, pushing the 10 psi start permissive crossing well past a 120 s oil-permissive timer	Project-specific — intended to exercise the PLC's configured Oil Permissive Pressure Fault timer
Engine fails to start	Toggle	Clamps N at 550 rpm, never reaching a running speed	Project-specific — intended to exercise the PLC's configured Engine Failed to Start timer
Mag pickup fault	Toggle	Forces reported ST_1008 to 0 while the engine is actually running	Project-specific — intended to exercise a speed-signal/other-evidence mismatch check, per the PLC's own logic
Overspeed sensor bias	Slider, rpm offset	Adds a continuous offset to reported ST_1008	Project-specific — intended to exercise the PLC's configured overspeed protection

Fault	Operator control	Physical effect	Intended Test Objective / Project-Specific PLC Response
Blocked discharge	Slider, 0-100%	Reduces the effective K_{proc} flow coefficient	Project-specific — intended to exercise the PLC's configured high-discharge-pressure alarm/trip logic
Cylinder temp bias	Slider per cylinder (1-4), °F	Adds an offset to the corresponding TT_2004-2007 reading	Project-specific — intended to exercise the PLC's configured high-cylinder-discharge-temperature alarm/trip logic, per cylinder
Valve stuck	Per-valve select (bypass, suction, suction ESD, discharge ESD, blowdown)	Freezes that valve's position regardless of command	Project-specific — intended to exercise a valve-misalignment check (position feedback vs. command mismatch), per the PLC's own logic
Signal freeze	Per analog tag (set)	Holds the tag's last transmitted value	Project-specific — intended to exercise a stale/non-updating signal check, per the PLC's own staleness logic
Signal invalid	Per analog tag (set)	Drives the tag out of its transmitter range	Project-specific — intended to exercise an out-of-range / bad-quality signal check
Cooler motor trip	Per motor (1, 2, or both)	Drops RS_4011/RS_4012 run feedback while the command stays commanded on	Project-specific — intended to exercise a run-feedback mismatch check, per motor
Link drop	Toggle	Suspends all OPC UA writes from the simulator	Project-specific — intended to exercise the PLC's own watchdog against a stale/frozen link

Fault	Operator control	Physical effect	Intended Test Objective / Project-Specific PLC Response
Tier 2 discrete faults	Per-item toggles	Drive the corresponding scrubber-level / vibration / oil-JW-level / fuel-gas / lubricator tag true	Project-specific — intended to exercise the associated protective function

16.1 Where This Differs from the Predecessor Simulink Rig

A few of these faults are broader in scope than the earlier Simulink model: valve-stuck applies to any of the five valves rather than bypass only, cooler motor trip is per-motor rather than both fans together, signal freeze covers any analog tag rather than discharge pressure alone, signal invalid is new, and cylinder temperature bias is independent per cylinder rather than one shared value. Two numeric values also differ from the predecessor model — the slow-lube-build time constant (90 s in the predecessor model, 900 s here) and the engine-fails-to-start speed clamp (400 rpm in the predecessor model, 550 rpm here) — both chosen to exercise the same PLC permissive timers as the original values did, just with different margins.

17 PLC Validation and Commissioning Tests

17.1 First PLC Commissioning Test

A controlled, deterministic interface check intended to confirm the tested signal path before attempting the full compressor sequence. This is the same tag-forcing mechanism demonstrated in Section 11, “Verifying the Link,” generalised into a repeatable first-commissioning step.

1. Establish the OPC UA connection (Section 11) and confirm “Connected” in the header.
2. Verify `WD_6001` is incrementing and that expected feedback tags (e.g. `PS_2009`, `ST_2010`) are present and at their expected at-rest values.
3. From the PLC, command one low-impact test command — `CMD_4005` is a convenient choice, since it drives a visible indicator on the P&ID and has no destructive effect on its own without the rest of the start permissive chain also being asserted.
4. Observe the corresponding response in the simulator HMI (or the Tags tab, Section 14).
5. Remove the command from the PLC side.
6. Confirm the feedback returns to its prior state.
7. Confirm the PLC is receiving the simulator’s process measurements (e.g. `PT_1001`) and that they read plausible values for the current state (Section 13, “Normal Operating Expectations”).
8. Compare the tag table on both sides — the PLC’s symbol/watch view and the simulator’s Tags tab (Section 14) — to confirm no tag is silently missing or mismatched.

Successful completion provides evidence that the tested OPC UA communication path, addressing, and tag exchange are functioning as intended for the signals checked. It does **not** establish correctness of the PLC’s sequence logic or the complete PLC application — that is a separate validation step, below.

17.2 Acceptance Test Checklist

PROJECT-SPECIFIC PASS/FAIL CRITERIA. This checklist states what to **observe**, not what the “correct” PLC response must be in every case — most of that is a project requirement this document has no authority to define. Where a specific response is not documented elsewhere in this report, the expected-response column says so explicitly rather than inventing one. This checklist is a simulator interface/model test aid for PLC development, not a substitute for the project’s approved FAT/SAT procedure, cause-and-effect documentation, shutdown philosophy, or control narrative.

17.2.1 Communications

Check	Expected observation
Simulator connects to the OPC UA server	Header shows “Connected”; <code>app.log</code> records a successful connect (Section 11)
PLC command changes appear in the simulator	Forcing a command tag from CODESYS is reflected in the HMI within one update cycle (Section 11.9)
Simulator feedback appears in the PLC	Measurement tags update in the PLC’s watch view as the process changes

Check	Expected observation
Watchdog supervision works	WD_6001 increments continuously; a stalled value is detectable by the PLC (Section 13)
Reconnect works after a CODESYS download	Reconnecting (or restarting the simulator) re-discovers tags rather than holding stale node references (Section 11)

17.2.2 Start Sequence

Check	Expected observation
Prelube is commanded	CMD_4001 asserted; PT_1005 ramps toward 55 psig
Oil-pressure permissive is respected	Project-specific / engineer-defined — the simulator only reports PT_1005/PS_2009, it does not decide when a start may proceed
Engine start sequence works	ST_1008 progresses through cranking then acceleration once CMD_4005 and its prerequisites are asserted
Speed reaches idle	ST_1008 approaches 650 rpm
Rated transition works	ST_1008 moves into the 850-1000 rpm range once CMD_4003 and SC_3001 command it
Valves reach expected positions	FC_3002/FC_3003 and their limit switches (ZS_2001-2008) track the commanded position, rate-limited per Section 5 eq. 13
Compressor loads correctly	Stage pressures build per “Close Bypass / Load the Compressor” in Section 13

17.2.3 Running

Check	Expected observation
Stage pressures rise correctly	Monotonic staging matching Section 13’s “Normal Operating Expectations” table
Cooler status works	RS_4011/4012 track CMD_4011/4012; temperatures respond per Section 5 eq. 16
Trends display expected process behaviour	Engineering Trends (Section 14) show smooth transients consistent with the implemented rate limits, first-order lags, and process equations (Section 5), without unintended numerical discontinuities
Unload/reload works	Opening/closing FC_3002 produces the response documented under “Open/Close Bypass” in Section 13

17.2.4 Normal Stop

Check	Expected observation
Machine unloads correctly	Bypass opens before ESDs close, per the PLC’s own sequence

Check	Expected observation
Engine stops	ST_1008 coasts down toward zero, $\tau = 6$ s
Isolation occurs	ESDs close per the PLC's commanded order
Depressurisation behaviour is understood	Final pressure state matches the "Normal Stop vs. USD" explanation in Section 13, not assumed to be fully vented on both sides

17.2.5 Shutdown

Check	Expected observation
Immediate trip handling works	ESDs and blowdown respond to the PLC's shutdown outputs without waiting for a normal-stop sequence
Configured fail-state commands occur	Command tags reach their documented configured fail directions (Section 13, "Command Polarity")
Restart is inhibited until correct reset/recovery	Project-specific / engineer-defined — the simulator does not itself gate restart on any reset philosophy

17.2.6 Fault Injection

Fault family	Expected PLC response
Low lube pressure / slow lube build	Project-specific / engineer-defined
Engine fails to start	Project-specific / engineer-defined
Speed-signal fault (mag pickup) / overspeed	Project-specific / engineer-defined
Blocked discharge	Project-specific / engineer-defined
Stuck valve	Project-specific / engineer-defined
Cylinder-temperature bias	Project-specific / engineer-defined
Cooler trip	Project-specific / engineer-defined
Tier 2 discrete conditions (scrubber level, vibration, oil/JW level, fuel-gas pressure, lubricator no-flow)	Project-specific / engineer-defined
Signal freeze / invalid / lag / noise	Project-specific / engineer-defined
OPC UA link drop	Project-specific / engineer-defined — see "Watchdog and Communication Supervision," Section 13

Each row's simulator-side mechanism is fully documented in Section 16; this table exists only to organise validation, not to prescribe a required PLC response the project has not itself defined.

18 Troubleshooting

App window never opens, or closes immediately. Read `%LOCALAPPDATA%\CompressorSim\logs\app.log` first — the app has no console window, so this file is the only place startup errors go.

“Windows protected your PC” (SmartScreen) on first run. Expected — the installer is not code-signed. Click “More info” → “Run anyway” (Section 10).

Message box: “could not start its window... WebView2 Runtime.” The installer should have silently installed WebView2 already. If this still appears, install it manually from Microsoft’s WebView2 page and relaunch. Windows 11 ships WebView2 built in, so this should only occur on older Windows 10 builds.

“Another instance is already running” but no window is visible. A previous run crashed without releasing its lock in an unusual way (the app checks whether the previous process is actually still alive before refusing to start, so this should self-heal). If it persists, open Task Manager, end any `CompressorSim.exe` process, and relaunch.

OPC UA endpoint unreachable. The Connect to a PLC dialog will not reach “Connected.” Confirm the endpoint in the application configuration (Section 11) is correct, that the target CODESYS runtime or panel is actually running its OPC UA server, and that the machine is reachable on the network (for a remote target, try *Scan my network* to confirm it is visible at all). `app.log` records the underlying connection failure.

Tags not found on an otherwise-reachable server. The app connects but reports missing tags, e.g. *“couldn’t find the compressor tags on this server (N missing, e.g. ...). Is this the right PLC, and is its Symbol Configuration published?”* This means the CODESYS project’s Symbol Configuration has not been published, so the tag names the simulator looks for are not visible in the server’s browsable address space — publish it and reconnect.

Stale behaviour after a PLC program change. If a CODESYS project was re-Downloaded to the runtime after the simulator connected, its OPC UA address space and node references may have changed underneath the existing connection (Section 11). Reconnect (or restart the app) to force it to re-discover tags rather than hold references into the address space as it existed before the Download.

18.1 First-Time PLC Engineer: Likely Early Mistakes

Runtime not actually started. A CODESYS project can be open and downloaded in the IDE without the runtime application actually running (Section 11 shows “Application [run]” in green as the thing to check) — a stopped application still leaves the OPC UA server up, so the connection can succeed while nothing responds to commands.

Symbol Configuration not published. The single most common “connects but no tags” cause (Section 11, Section 18 above) — easy to miss because the connection itself succeeds.

Device security settings left at default. Both `CommunicationMode` and `Activation` default to values that block external OPC UA clients (Section 11, “Symbol Publishing and Device Security Settings”); a fresh CODESYS install needs both changed explicitly.

Simulator connected before the day’s project Download. Reconnect after every Download (Section 11) — an existing connection can hold stale node references silently rather than failing outright.

Command appears “locked.” Once a PLC is connected, the Overrides dock becomes read-only by design (Section 14) — this is not a bug, and is the expected state whenever a PLC is actively driving the simulation.

Wrong command polarity assumed. Writing TRUE to CMD_4009 expecting the suction ESD to close, or driving FC_3002 to 100% expecting the bypass to open, produces the opposite of the intended result — see Section 13, “Important: Command Polarity.”

Blowdown or bypass behaviour misunderstood. Expecting discharge pressure to collapse the moment blowdown opens is the most common first-read misunderstanding of this simulator’s process model — see Section 13, “Blowdown,” and the underlying mechanism in Section 4.

Simulator appears stuck at rest because one required command is missing. The permissive chain in Section 15 has several prerequisites (ESD healthy, driven-equipment-ready, etc.) before speed or valve commands do anything visible — a single missing upstream command can make the whole chain appear unresponsive.

Speed does not increase despite a start command. Usually means one of the engine-start prerequisites (Section 15, step 2) is not actually asserted, not that the command itself failed.

Pressures do not behave as expected. Often traced to the suction/discharge/bypass valve path being in an unintended state — confirm all relevant valve commands and limit switches (Section 13 cheat sheet) before assuming a physics issue.

PLC logic waits indefinitely. Check the Faults tab (Section 16) — an injected fault (e.g. slow lube build) can be the reason a permissive never crosses its threshold, not a logic error.

Heartbeat not actually supervised. Reading WD_6001 without checking that it changes over time gives no protection against a stalled link — see Section 13, “Watchdog and Communication Supervision.”

Part III — Appendices

Appendix A — Signal / Tag List

Full OPC UA tag list, in both directions, as an instrument list rather than a software interface. Direction is stated from the PLC's point of view. Every analog value is range-clamped before being written to the PLC; optional per-signal first-order lag (0.3-2 s) and Gaussian noise are available per tag, off by default (Section 16, Signal Freeze/Invalid).

18.2 Commands In (PLC → Simulator)

Tag	Description	Range	Fail value
SC_3001	Engine speed command	0-100%	0
FC_3002	Bypass valve command	0-100%	0 (valve opens)
FC_3003	Suction valve command	0-100%	0 (valve closes)
CMD_4001	Auxiliary lube solenoid	discrete	off
CMD_4003	CAT idle / rated speed select	discrete	off
CMD_4004	Blowdown solenoid (energised = closed)	discrete	off (valve opens)
CMD_4005	CAT engine start command	discrete	off
CMD_4006	CAT ESD (energised = healthy)	discrete	off
CMD_4008	Driven-equipment-ready	discrete	off
CMD_4009	Suction ESD solenoid (energised = open)	discrete	off
CMD_4010	Discharge ESD solenoid	discrete	off
CMD_4011/4012	Cooler motor 1 / 2 run	discrete	off

18.3 Measurements Out (Simulator → PLC)

Tag	Description	Range
PT_1001	Suction pressure	0-60 psig
PT_1002/1003/1004/1006	Stage 1/2/3/final discharge pressure	0-300 / 0-1000 / 0-2000 / 0-2000 psig
PT_1005	Compressor oil pressure	0-200 psig
PT_1007	Engine oil pressure	0-150 psig
ST_1008	Engine speed	0-2000 rpm
TT_2001	Compressor oil temperature	0-500 °F
TT_2004-2007	Cylinder 1-4 discharge temperature	0-500 °F
TT_2009-2012	Packing temperatures 1-4	0-500 °F
TT_2013	Aftercooler temperature	0-500 °F
TT_2014	Engine jacket-water temperature	0-300 °F

Tag	Description	Range
ZS_2001-2008	Valve open/closed limit switches	discrete
PS_2009	Oil pressure healthy (> 10 psig)	discrete
ST_2010	Engine running (> 300 rpm)	discrete
WD_6001	Heartbeat counter, increments every 500 ms	0-32767
PB_5001/5003/5004, ESD_5002	Operator pushbuttons / remote ESD (always live)	discrete
XA_6002/XS_6003	CAT ADEM engine alarm / shutdown status	discrete
RS_4011/4012	Cooler motor 1 / 2 run feedback	discrete

18.4 Tier 2 Discrete Faults (not in the PN17481 I/O list)

Tag	Description
LSH_7001-7004	Suction / ST2 / ST3 / fuel-gas scrubber level high
VSH_7011-7013	Compressor-frame / engine / skid-piping vibration trip
LSL_7021-7023	Compressor oil / engine oil / engine JW level low
PSL_7031	Fuel-gas pressure low
FSL_7041/7042	Cylinder lubricator bank 1 / 2 no-flow

Alarm and trip setpoints (used only to colour the P&ID and draw trend lines — the simulator never evaluates them) are configured alarm/trip display setpoints in the simulator configuration.

Appendix B — Future Improvements

Improvement	Data or input needed
Real-gas compressibility (Z-factor) in place of the ideal-gas assumption	A gas composition analysis for the actual process stream
A choked-flow model at the blowdown valve	None beyond implementation effort — the physical criterion (critical pressure ratio) is well established
Cylinder thermal mass — replace the algebraic discharge-temperature target with a genuine lag driven by an estimated thermal mass	An estimate of effective cylinder thermal mass, or acceptance of a further-fitted time constant
A UA-based heat-exchanger model to replace the cooler lookup tables	Fan/cooler performance curves and ambient design conditions for the actual air coolers
Engine torque/load coupling with governor droop	The reference engine's governor droop characteristic
Validation of the clearance fraction and valve flow coefficients against real performance curves	Vendor performance data for the reference compressor and engine — currently absent
Rod load and pulsation analysis	Explicitly out of scope for this simulator — belongs in a dedicated mechanical study

Appendix C — Verification Detail

SOFTWARE VERIFICATION, NOT PHYSICAL VALIDATION. The checks below verify the implemented model against the configured design-point acceptance criteria and software invariants. They confirm the simulator does what this report says it does; they are not a comparison against measured performance of a physical compressor (Section 1, “Basis and Validation Status”).

The Design-Point Verification Suite (18 automated checks) checks the implemented model against the configured design-point acceptance criteria, from a pressurised initial condition run for 600 s of simulated time: all eight design-point values (Section 6) against their stated tolerances; mass-balance closure on both vessels under 1×10^{-3} kg/s (the primary acceptance criterion); supply flow equals delivery flow at steady state; all three stage ratios equal to within 10^{-6} relative; pressures rise monotonically through the stages; all pressures remain at or above atmospheric; gas is heated by compression, never cooled; no NaN or Inf appears anywhere in state or algebraic outputs; valve positions stay within 0-100%; all flows are non-negative; and steady-state drift stays under 2 psi over the final 50 s of a 600 s run.

The Transient / Dynamic Verification Suite (27 automated checks) covers, among other checks: cold start stays at atmospheric while stopped; pressurisation toward the source boundary never exceeds it; bypass and blowdown valve open/close timing against their configured rates (within a few percent); speed ramp rates in both directions, including a case-specific note on why the average rate over a full second undercounts the rate-limit during the final approach to setpoint; coastdown reaching approximately 37% of initial speed after one time constant; load/unload direction checks (bypass, speed, suction valve); blowdown venting toward atmosphere and doing so gradually rather than instantly; prelube and oil-permissive-crossing timing, both healthy and under the “slow lube build” fault; fault behaviour for mag-pickup, overspeed bias, blocked discharge, and stuck-valve; recovery to the design point after a fault clears; ESD-closed compression stoppage and post-ESD cylinder cooling; cooler-loss temperature rise; a 400-iteration / 800 s randomised-transient invariant-violation sweep (no NaN, no inverted stage ordering, no negative pressures, no sub-1.0 stage ratio, no cooling-by-compression, no out-of-range valve positions); and a final check that the converged final discharge pressure is insensitive (spread under 1.0 psi) to integration timestep across 5/20/50 ms.

The remaining five suites cover: fault injection (each fault in Section 16 actually produces its documented tag-level effect); tag mapping (the conversion between internal SI process-model units and psig/°F transmitter values, including range clamping); the OPC UA link (connection, disconnection, watchdog timeout, and fail-value application, Section 11); and command-locking behaviour (Overrides tiles correctly becoming read-only once a PLC is connected, and correctly reverting to editable on disconnect).

The process model itself integrates at a fixed 20 ms step using fourth-order Runge-Kutta; the timestep-insensitivity check above is the direct evidence that this choice does not materially affect the converged values reported elsewhere in this document.